Observationally Equivalent Algorithms: What Prices Reveal About Trader Populations, and What Supervisors Need

Pratik Niroula* Prashant Niroula † Sujan Khatiwada‡

12 November 2026

Abstract

Algorithmic traders read prices and, by trading, move them. Populations with the same feedback kernel produce identical prices, profits and equilibria; prices identify nothing finer, and learning composition costs data of order $(\Delta\eta)^{-(4K-2)}$, even near instability, so targeted supervision needs tagged flow. In a pre-registered out-of-sample test on Taiwan's tagged exchange flows (2019–2023), prices improve forecasts of foreign investors' flow as the registered criterion requires, but a pre-specified check attributes the gain to prices revealing past flow rather than to feedback. A pre-registered test of the position-limit prediction on Binance perpetual futures is inconclusive. Core results are machine-checked in Lean 4.

AI-use disclosure. Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, Lean proofs, manuscript drafting and internal critique. Human authors remain responsible for the content. AI checks do not constitute independent review. The authors have verified all mathematical proofs, Lean 4 formalizations and empirical calculations, and take full responsibility for the content of the manuscript.

Relation to other versions. This note states the main results. The online supplement gives every result in full, with proofs, machine-check details and the results omitted here;

*Department of Computer Information Science, Minnesota State University, Mankato, MN 56001, USA. pratik.niroula@mnsu.edu.

†Faculty of Management, Tribhuvan University, Kirtipur, Nepal. prashantniroula20@gmail.com.

‡Department of Electrical, Computer and Robotics Engineering, Minnesota State University, Mankato, MN 56001, USA. sujan.khatiwada@mnsu.edu.

Supplement Sections S1–S7 reproduce the longer core version of 12 November 2026. The full working paper (5 November 2026) calls the trader groups "cohorts".

Keywords: algorithmic trading, feedback trading, sufficient statistics, identification, supervision, disclosure. **JEL:** C32, D83, G14, G18.

Code and proofs. https://github.com/Pratikn03/Fianince.

1 Introduction

Trend-following funds, execution algorithms and AI trading agents read recent prices, decide what to trade and, by trading, move the prices they read. Regulators are now deciding how far to identify them. China's securities regulator issued rules on program trading in 2024, and since July 2025 its exchanges classify accounts as high-frequency by their order rates.[1] India requires every algorithmic order to carry an exchange-issued identifier.[2] Japan has registered high-speed traders since 2018, and its regulator studies their orders by trader and strategy.[3] The EU requires firms to identify their algorithms in order records under MiFID II. In the United States, the SEC withdrew its proposal on firms' use of predictive data analytics in June 2025. Behind these choices is a question this note answers: what does a market's price record reveal about its population of feedback traders, and for which decisions is that enough?

The answer. Prices reveal the market's *feedback kernel*, the response, m periods later, of total feedback flow to a unit move in the price signal. The kernel is all that prices identify, and it is all that anyone trading on public information needs. The composition of the trader population matters only for actions aimed at one group, and learning it from prices is very expensive, even on the eve of a crash. This leaves a supervisor with a problem that the note states precisely at the end: keeping a market safe without knowing who trades in it.

Results.

(1) *Sufficiency* (Theorem 3.1). With a value, noise traders, an insider and a dealer, each following an arbitrary causal rule, two populations with the same kernel give the same prices, order flows, insider profits and dealer inventories, realization by realization. The two games therefore have the same equilibria.

(2) *Identification* (Theorem 3.2). Two populations give identical price responses to every trade sequence exactly when their *merged measures*, the trading weight at each memory rate, coincide. Groups with the same memory cannot be separated however long the record.

(3) *Cost* (Theorem 4.1, Proposition 4.2). Separating K groups whose memories lie within $\Delta\eta$ takes of order $(\Delta\eta)^{-(4K-2)}$ experiments; Prony's method attains this order, and

[1] Shanghai, Shenzhen and Beijing stock exchanges, implementing rules of 3 April 2025 for the CSRC Provisions on Program Trading of May 2024.

[2] SEBI circular SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/0000013, 4 February 2025.

[3] Financial Services Agency, FSA Analytical Notes, 21 November 2024.

passive observation needs at least as much data. The barrier does not lift as the market approaches instability: prices then reveal how close the market is to breaking, not who is trading.

(4) *Supervision* (Section 5). A uniform cut in feedback activity needs only the kernel. Halting the largest group needs composition, at the cost above. Class-tagged flow reports remove the cost. A group that can mimic another group's memory caps what any price-based targeting can achieve.

(5) *Limits* (Section 6). With position limits the sufficient object becomes a measure over memory and leverage. Calm prices reveal only its linear part, the kernel. Probes large enough to reach the limits identify the rest.

(6) *Evidence* (Section 7). We report two pre-registered tests, each run once, with the wording fixed before the results were known.

What is known. Exponential memory is the oldest model of expectations: adaptive expectations (Cagan, 1956; Nerlove, 1958), geometric distributed lags (Koyck, 1954), and Muth (1960)'s proof that exponentially weighted forecasts are optimal for a random walk observed with noise. The idea that an aggregate depends on a distribution of individual coefficients and not on the individuals is also old. Granger (1980) showed it for sums of autoregressions, Forni and Lippi (1997) developed the general theory of aggregation and dynamics, Zaffaroni (2004) and Pesaran and Chudik (2014) the large-economy and large-panel versions, and recovering the distribution is the disaggregation problem (Celov et al., 2010). Results of the form "two structures give the same data, so data cannot choose" have a famous precedent in Sargent (1976), whose observational equivalence of natural and unnatural rate theories settled what reduced-form data could say about monetary rules. Auclert et al. (2024) make the corresponding sufficient-statistic argument for household heterogeneity in macroeconomics. The rate $(\Delta\eta)^{-(4K-2)}$ is the known minimax rate for mixtures with near-colliding components (Heinrich and Kahn, 2018; Wu and Yang, 2020), and the uniqueness step behind the stress result is the uniqueness of networks with one hidden layer (Sussmann, 1992). We do not claim these.

What is new. The market setting, and the policy question it answers: which supervisory actions the observational equivalence class supports, and which need data from outside it. Sufficiency holds path by path inside a strategic market, for every insider and dealer rule, so it transfers to every equilibrium concept. The memory rates are transmission zeros of the closed-loop price response, not poles, which is why passive data are no cheaper than probing

(Supplement Theorem S4.3). The results hold when price impact and persistence come from an inventory equilibrium (Supplement Theorem S3.2). Supervision splits into actions that need the kernel and actions that need composition, with a price on the second. With position limits, calm prices carry exactly no information about how close traders are to their limits.

2 Model

Time is discrete. Let x_t be the pricing error, the gap between the price and the fundamental value. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$. There are K groups of feedback traders. Group i keeps an exponentially weighted average $y_{i,t}$ of past pricing errors and trades $q_i y_{i,t}$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + \varepsilon_t, \qquad y_{i,t} = (1 - \eta_i) y_{i,t-1} + \eta_i x_t, \tag{1}$$

where v_t is a probe, a trade by an observer, and ε_t is news. A desk that tracks a trend with a steady-state Kalman filter (Muth, 1960), or that learns by regularized online gradient descent, updates exactly this way (Supplement Lemma S1.1). Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the trading weight. The *merged measure* $M = \sum_i w_i \delta_{r_i}$ puts weight w_i at rate r_i, merging groups with equal rates, and the *feedback kernel* is its sequence of moments,

$$k_m = \sum_i w_i r_i^m, \qquad m = 0, 1, 2, \ldots$$

Total feedback flow is the convolution of past prices with k. The *feedback budget* $S = a + \kappa \sum_i q_i = a + \kappa \sum_m k_m$ governs stability: with $q_i \geq 0$ the market is stable exactly when $S < 1$.

The observer. The cost results concern a supervisor or exchange that knows (a, κ) and the noise law. It can wait and record prices, probe from a reset state and record the response, or receive class-tagged flow reports. Reset probing is a benchmark for what data can reveal, not a proposed policy.

3 The kernel is sufficient and is what prices identify

For sufficiency we embed the groups in a general market. A value v is drawn, noise traders submit z_t, an insider submits $X_t(v, \mathcal{H}_{t-1})$, and a dealer sets the price $P_t(\mathcal{H}_{t-1}, y_t)$ from the

public history and the total order flow y_t. The groups update on a public signal $s_t = S_t(p_0, \dots, p_t)$ and contribute flow $f_t = \sum_{m<t} k_m s_{t-1-m}$. The maps X, P and S are arbitrary causal rules, which covers Kyle, inventory-averse, learning and misspecified dealers.

Theorem 3.1 (The kernel is sufficient)**.** *If two group configurations have $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on dates $t \le T$. The two games have the same strategy sets and payoffs, and hence the same equilibria under any concept defined from payoffs and the public history. The same holds for any finite population whose flows are causal linear functions of the public signal, with k the sum of the individual kernels. Group-level quantities, such as the weight of the largest group, are not invariant.*

The proof is an induction on dates (Supplement Theorem S2.1 and Proposition S2.2). Its content is the object it isolates, not the induction.

Theorem 3.2 (What prices identify)**.** *In model (1), two economies produce identical price responses to every probe sequence if and only if their merged measures coincide.*

This is Supplement Theorem S2.3(i). The same holds when κ and a are derived from competitive dealers who dislike inventory: the equilibrium price follows (1) path by path with effective impact and persistence, and with the memory rates unchanged (Supplement Theorems S3.1 and S3.2).

4 The price of composition

Theorem 3.2 says that groups with different memories can be told apart in principle. The question is how much data that takes when memories are close.

Theorem 4.1 (The cost of composition)**.** *Fix $K \ge 2$, a center rate r_c, a total weight W and a spacing $\Delta\eta$, with every candidate economy stable. Each reset experiment records the noisy impulse response with Gaussian noise of variance σ^2.*

(i) Lower bound. *There are two K-group economies with rates in $r_c + \Delta\eta[-1, 1]$ and positive weights whose merged measures have the same moments of order $0, \dots, 2K-2$ and whose impulse responses differ by order $(\Delta\eta)^{2K-1}$. Telling them apart with error probability below $1/4$ needs $N \ge c\,\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments.*

(ii) Upper bound. *On any class of separated rescaled rates with weights bounded away from zero, Prony's method applied to the first $2K$ estimated kernel values recovers the weights from $N = C\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments.*

(iii) Waiting is no cheaper. *If prices are instead observed passively, driven by Gaussian news, the information per observation between the pair in (i) is of order* $(\Delta\eta)^{4K-2}$*, so at least of order* $(\Delta\eta)^{-(4K-2)}$ *observations are needed.*

Parts (i)–(iii) are Supplement Theorems S4.1, S4.2 and S4.4. Part (iii) does not weaken as the market approaches instability (Proposition 4.2). For two groups, halving the spread of memories multiplies the data needed by 64; for three groups, by 1,024. The pair in (i) consists of the Gauss and Gauss–Radau quadrature rules, which match moments as far as K nodes allow. Part (iii) uses that the memory rates are zeros of the closed-loop response, so passive data face the same near-cancellation as probes.

Proposition 4.2 (The barrier does not lift near instability)**.** *Hold the pair of Theorem 4.1(i) at a common feedback budget* $S = 1 - \epsilon$*. The passive information per observation between them is at most* $C(\Delta\eta)^{4K-2}$*, with* C *independent of* ϵ*, however close the market is to instability.*

This is Supplement Proposition S4.5, new to this version. One might expect a market near instability to reveal its traders, since the loop amplifies every difference. It reveals instead how close it is to instability: with a common persistence rather than a common budget, the information grows without bound as $\epsilon \to 0$, and all of the growth comes from the budget. The bound is per observation; a finite-sample version uniform in ϵ is not proved. Figure 1 shows both statements.

5 Who needs composition, and what disclosure buys

Proposition 5.1 (Market-wide versus targeted action)**.** *Let* $q_i \geq 0$*, let the market be stable with* (a, κ) *known, and let a supervisor need the feedback budget below a ceiling* $\bar{s}$*, with* $a < \bar{s} < 1$*.*

(i) Market-wide. *Scaling all feedback activity by* θ *meets the ceiling exactly when* $a + \theta(S - a) \leq \bar{s}$*. This depends on the population only through* S*, and so only through the kernel.*

(ii) Targeted. *Halting the largest group does not. For every* $K \geq 2$ *there are two populations whose merged measures have the same moments of order* $0, \ldots, 2K - 2$*, so that their budgets differ only by order* $(\Delta\eta)^{2K-1}$*, while their largest groups differ by a fixed amount, so that for a range of ceilings the halt meets the ceiling in one and not the other. Making this decision from* N *reset records with both errors at most* $1/4$ *needs* $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$*.*

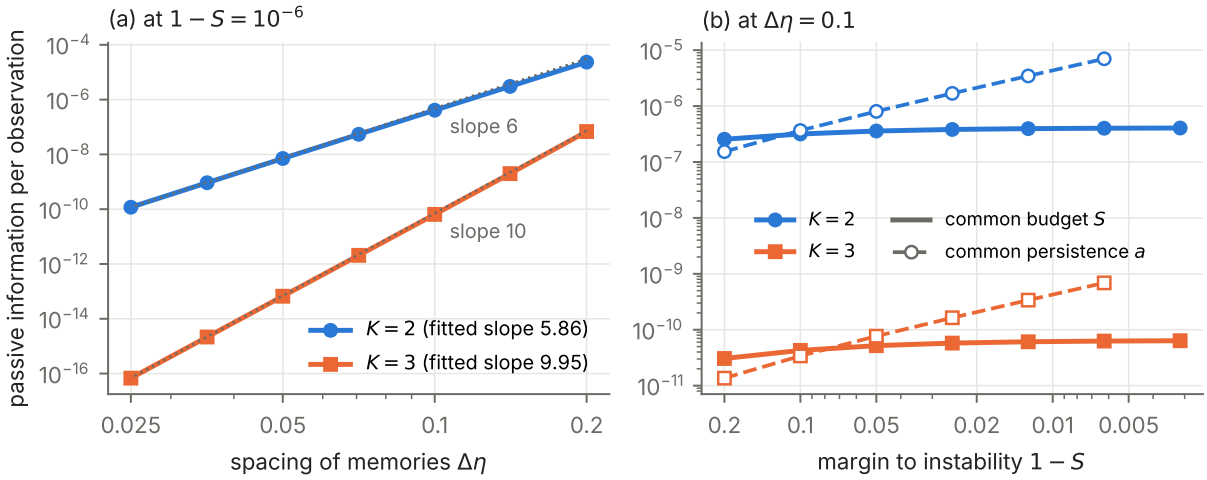


Figure 1: The passive barrier near instability. Both panels show the Gaussian Kullback–Leibler information per observation between the stationary price processes of two populations whose merged measures are the K-node Gauss and Gauss–Radau rules on memories spread over $\Delta\eta$ around 0.5 (total weight 0.3, impact $\kappa = 0.5$). (a) At a common feedback budget one millionth below instability, the information falls like $(\Delta\eta)^{4K-2}$ (dotted lines). (b) Held at a common budget (solid), the information stays bounded as the margin $1 - S$ shrinks; held at a common persistence (dashed), it grows without bound, because the budgets then differ. Computed from the closed-form spectra; scripts in `research_extension/fragility_reveals/`.

Part (i) is Supplement Proposition S5.1(i). Part (ii) is Supplement Proposition S12.4, which is new to this version; its pair comes from a family of quadrature rules whose weights are the values of one function at their nodes (Supplement Lemma S12.3).

Theorem 5.2 (Value of class-tagged disclosure)**.** *Suppose that, in addition to prices, the observer receives each class's flow with Gaussian measurement noise. On the rich class of Theorem 4.1(ii), the number of experiments needed to recover the weights falls from order $(\Delta\eta)^{-(4K-2)}$ to a bound that does not depend on $\Delta\eta$. For the rates it falls to order $(\Delta\eta)^{-2}$.*

This is Supplement Theorem S5.3; Supplement Proposition S5.2 gives the regression that achieves it, one class at a time, with a variance that depends only on that class.

The results so far take the groups' rules as given. A group that expects to be targeted has a reason to change them.

Proposition 5.3 (Hiding caps targeting)**.** *A supervisor observes prices, under any probes and for any record length, and then targets a given group or not. Being targeted costs the group $P > 0$. Suppose the group can adopt, at private cost C_0, a rule that gives the population the kernel of a population that the supervisor targets with probability α. Then in any best*

response the group is targeted with probability at most $\alpha + C_0/P$, whatever the probes, record length and decision rule.

The proof is two lines from Theorem 3.1 (Supplement Proposition S5.4). More data make the group's original rule easier to detect and so make hiding more attractive. The bound concerns one group; the cost of hiding and an equilibrium among several groups are not modelled. A class tag is attached to the account, so changing the trading rule does not remove it.

6 Position limits: calm hides, stress reveals

Real feedback traders have position limits. Let trader i have memory rate r_i, capacity $c_i > 0$ and leverage ℓ_i, and trade $c_i\psi(\ell_i m_{i,t})$, where $m_{i,t}$ is its exponentially weighted signal and ψ is a shape common to all traders. The leading case is the hard limit $\psi(u) = \max(-1, \min(1, u))$: inside its band a trader is linear with weight $c_i\ell_i$; at the edge its position is capped. Write $\nu = \sum_i c_i \delta_{(r_i,\ell_i)}$ for the capacity measure on memory and leverage.

Theorem 6.1 (Two layers)**.** *(i)* Sufficiency. *For any shape, populations with the same ν give the same prices, profits and equilibria. The kernel is one linear functional of ν, $k_m = \int r^m \ell \, d\nu$, and does not determine it.*

(ii) Calm hides. *Under hard limits, two populations with the same kernel give identical price and order-flow paths until some trader first reaches its limit. Any record of the market before date T distinguishes them with total-variation distance at most the probability that a limit is reached before T. With Gaussian news and no probe this is at most $T(n+n')e^{-1/(2\vartheta^2)}$ for the $n+n'$ traders of the two populations, where ϑ is the largest standard deviation of a trader's position relative to its limit.*

(iii) Stress reveals. *If every trader has $r_i \geq 0$ and $\ell_i \neq 0$ and responses are observed exactly, probes of all sizes identify the signed capacity measure $\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i,|\ell_i|)}$, and nothing finer, for the hard limit and for* tanh*. For* tanh*, probes in any neighbourhood of zero suffice. For a cubic shape the result fails.*

Parts (i)–(iii) are Supplement Theorems S6.1, S6.2 and S6.4; for the hard limit and for tanh with probes of all sizes, part (iii) is machine-checked for any number of traders; the small-probe statement is proved on paper. Its last step, for tanh, is the uniqueness of one-hidden-layer networks up to sign flips (Sussmann, 1992), which is why only the signed measure is identified. What the market adds is the reduction: probing recovers the flow map from prices, and two price dates give different memory–leverage pairs different scales.

The economic content is in part (ii). How close traders are to their limits, which governs behaviour in a crash, leaves no trace in calm prices. It can be learned only by pushing traders toward their limits or from disclosure.

7 Scope and conclusion

Evidence. The theory concerns tagged flows and position limits that are rarely observed. We ran two pre-registered tests on the closest public data, each run once, with the wording of the result fixed in advance; Supplement Section S7 reports both in full. In a pre-registered out-of-sample test on Taiwan's tagged exchange flows (2019–2023), prices improve forecasts of foreign investors' flow as the registered criterion requires, but a pre-specified check attributes the gain to prices revealing past flow rather than to feedback. A pre-specified test on Binance perpetual futures was inconclusive (calm difference -0.002, 90% interval $[-0.007, 0.004]$; difference-in-differences 0.041, $p = 0.553$). It neither supports nor contradicts the prediction of Section 6.

The sufficiency results need aggregate feedback flow to be linear in past public signals, or saturating as in Section 6. The identification and cost results need exponential memory; for general completely monotone memory the minimax accuracy is of order $1/\log(N/\sigma^2)$ (full working paper). The passive results assume Gaussian news. Traders who change their rules as they learn are outside the theorems, apart from Proposition 5.3.

The problem this leaves. Together the results pose a problem. A supervisor who must keep a market stable cannot learn who trades in it from prices, at any reasonable cost, in calm conditions or near a crash; the traders can hide if it tries; and the information that would identify them arrives, if at all, only when limits are already being hit. Whether a market can be kept safe without that knowledge, using only what prices do reveal, is the question taken up in the companion paper (Niroula et al., 2026).

What this says about the rules. The results sort the regulatory choices above. A market-wide measure, such as a uniform limit on feedback activity or a pause triggered by the feedback budget, needs no identification and can be set from prices. A measure aimed at particular algorithms cannot be set from prices at any reasonable cost, and the targeted algorithms can make it harder still by imitating others. It needs flow tagged by class or account, which is what the Chinese, Indian, Japanese and European regimes collect. Whether that data make the targeted measures worth their cost is an empirical question that only those tags can answer.

Within these limits the message is simple. Prices tell a supervisor the kernel and nothing more, and the kernel is enough for any action that treats the market as a whole. Actions aimed at one group need composition, which prices supply only at a steep and known price and which a group can hide at a bounded cost. Disclosure tagged by trader class removes that price. The same split recurs with position limits: calm markets reveal the kernel, and only stress or disclosure reveals how close traders are to their limits.

Declarations

Author contributions. Pratik Niroula: conceptualization, methodology, formal analysis, software, investigation, data curation, writing (original draft). Prashant Niroula: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). Sujan Khatiwada: validation (verification of the proofs, the Lean 4 formalizations and the empirical calculations), writing (review and editing). All authors approved the final manuscript.

Funding. This research received no external funding.

Competing interests. The authors declare no competing interests.

Data availability. All data are public. The Taiwan test uses the Taiwan Stock Exchange's daily reports of trading by the three major institutional investors (T86), closing quotes (MI_INDEX), ex-rights and ex-dividend adjustments (TWT49U) and market totals (FMTQIK), from https://www.twse.com.tw. The Binance test uses daily files of five-minute candles and trading metrics (including open interest) for USD-margined perpetual futures from the Binance public data archive (https://data.binance.vision). The registered specifications, amendments, executors, notebooks, result files and test-unlock counters are in the public repository.

Code availability. All code, Lean 4 proofs and result files are public at https://github.com/Pratikn03/Fianince.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. See the AI-use disclosure on the title page.

References

Albertini, F., and E. D. Sontag (1993). For neural networks, function determines form. *Neural Networks* 6, 975–990.

Auclert, A., M. Rognlie, and L. Straub (2024). The intertemporal Keynesian cross. *Journal of Political Economy* 132, 4068–4121.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Barroso, P., R. M. Edelen, and P. Karehnke (2022). Crowding and tail risk in momentum returns. *Journal of Financial and Quantitative Analysis* 57, 1313–1342.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Cagan, P. (1956). The monetary dynamics of hyperinflation. In M. Friedman (Ed.), *Studies in the Quantity Theory of Money*, 25–117. University of Chicago Press.

Celov, D., R. Leipus, and A. Philippe (2010). Asymptotic normality of the mixture density estimator in a disaggregation scheme. *Journal of Nonparametric Statistics* 22, 425–442.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990). Positive feedback investment strategies and destabilizing rational speculation. *Journal of Finance* 45, 379–395.

Dreesen, P., D. T. Westwick, J. Schoukens, and M. Ishteva (2017). Modeling parallel Wiener–Hammerstein systems using tensor decomposition of Volterra kernels. In *Latent Variable Analysis and Signal Separation (LVA/ICA 2017)*, Lecture Notes in Computer Science 10169, 16–25. Springer.

Fefferman, C. (1994). Reconstructing a neural net from its output. *Revista Matemática Iberoamericana* 10, 507–555.

Forni, M., and M. Lippi (1997). *Aggregation and the Microfoundations of Dynamic Macroeconomics*. Oxford University Press.

Granger, C. W. J. (1980). Long memory relationships and the aggregation of dynamic models. *Journal of Econometrics* 14, 227–238.

Grossman, S. J., and M. H. Miller (1988). Liquidity and market structure. *Journal of Finance* 43, 617–633.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture

estimation. *Annals of Statistics* 46, 2844–2870.

Ho, T., and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47–73.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Koyck, L. M. (1954). *Distributed Lags and Investment Analysis*. North-Holland.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Leclère, J., and M. Rosenbaum (2026). Mean-field equilibrium of heterogeneous agents under market impact. arXiv preprint arXiv:2609.03115.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821–830.

Muth, J. F. (1960). Optimal properties of exponentially weighted forecasts. *Journal of the American Statistical Association* 55, 299–306.

Nerlove, M. (1958). Adaptive expectations and cobweb phenomena. *Quarterly Journal of Economics* 72, 227–240.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Pesaran, M. H., and A. Chudik (2014). Aggregation in large dynamic panels. *Journal of Econometrics* 178, 273–285.

Sargent, T. J. (1976). The observational equivalence of natural and unnatural rate theories of macroeconomics. *Journal of Political Economy* 84, 631–640.

Sussmann, H. J. (1992). Uniqueness of the weights for minimal feedforward nets with a given input-output map. *Neural Networks* 5, 589–593.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

Vlačić, V., and H. Bölcskei (2022). Neural network identifiability for a family of sigmoidal nonlinearities. *Constructive Approximation* 55, 173–224.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.

Zaffaroni, P. (2004). Contemporaneous aggregation of linear dynamic models in large economies. *Journal of Econometrics* 120, 75–102.

Online Supplement
Observationally Equivalent Algorithms

Niroula, Niroula and Khatiwada, 12 November 2026

Sections S1–S7 reproduce Sections 2–8 of the longer core version of this paper (12 November 2026): every result in full, with the remarks, machine-check notes and the results that the note omits. The remaining sections give the proofs, further results, machine checks and notation. Theorem numbers in the supplement are those of these sections, not of the note. The note's results correspond to them as follows: Theorem 3.1 to Theorem S2.1 and Proposition S2.2; Theorem 3.2 to Theorem S2.3(i); Theorem 4.1 to Theorems S4.1, S4.2 and S4.4; Proposition 5.1 to Propositions S5.1(i) and S12.4; Theorem 5.2 to Theorem S5.3; Proposition 4.2 to Proposition S4.5; Proposition 5.3 to Proposition S5.4; Theorem 6.1 to Theorems S6.1, S6.2 and S6.4.

S1 Model and motivating examples

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + \varepsilon_t, \tag{2}$$

where v_t is an executed probe quantity and ε_t a price innovation (news). Group i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{3}$$

The ordering is *price first*: the price responds to the probe, then groups update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{4}$$

and the transfer function from probes to prices, $H(z) = \sum_{j \geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z - a) D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{5}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \sum_j h_j = \kappa/(1 - S)$, where $S = a + \kappa \sum_i q_i$ is the feedback budget. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$, where, because groups update on the same period's price, the probe enters every state on impact through

$$b = \kappa\, (1, \eta_1, \ldots, \eta_K)^\top, \tag{6}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{7}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma S1.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0,1). \tag{8}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.*

Why exponential memory, and what is assumed. Exponential memory is the oldest model of expectations in economics: adaptive expectations (Cagan, 1956; Nerlove, 1958) and geometric distributed lags (Koyck, 1954), and Muth (1960) showed that exponentially weighted forecasts are optimal for a random walk observed with noise, which is Lemma S1.1(i). Lemma S1.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same reduced-form state. A third example is a desk that each period trades the amount that maximizes $\hat{v}_t u - \frac{1}{2}(\kappa + \lambda)u^2$, a one-period trade-off between its trend forecast $\hat{v}_t$ and quadratic impact and holding costs. Its trade is $u_t = \hat{v}_t/(\kappa + \lambda)$. If the forecast is updated exponentially, $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta x_t$, the trade follows the same recursion, $u_{t+1} = (1-\eta)u_t + w x_t$ with $w = \eta/(\kappa + \lambda)$. This is a static first-order condition, not a dynamic optimum. In these families the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio, its regularization or its cost parameters. That is the private information a supervisor would like to recover, although real algorithms need not belong to any of these families. Applied to the pricing error, a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). Nothing in the model says such trading is profitable.

The results use six modelling choices: linearity; price-first timing within a period; exponential memory (needed for the identification result, Theorem S2.3, and the cost results of Section S4, but not for sufficiency); algorithms that do not change their rules when observed (Proposition S5.4 relaxes this for one group); known primitives (a, κ); and stated observation contracts. Unknown primitives can only enlarge the set of candidate populations, so the lower bounds remain valid; Corollary S8.2 in the supplement shows that estimating them leaves the upper-bound rate unchanged. Section S3 replaces the primitive (a, κ) by an equilibrium.

The observer. The cost results are stated for one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. It knows (a, κ), the law of the observation noise and, for the K-group results, K. It can *wait*, recording prices driven by natural innovations; *probe*, trading a sequence v_t against the groups from a reset state and recording the response; or *acquire disclosure*, receiving each class's flow. Independent resets are a benchmark that no outside participant can run in a live market. We use them because they make the information content of the data exact. Reset probing is a benchmark for what can be learned, not a proposed policy. Section S4 shows separately that passive observation is no cheaper (Theorem S4.4).

S2 The kernel is a sufficient statistic

This section states the paper's main result. In a market with an insider, noise traders, a dealer and feedback traders, all following arbitrary causal rules, the feedback traders affect every outcome only through their kernel. The notation for the insider follows Kyle (1985), so in this section x_t is the insider's order, not the pricing error.

Theorem S2.1 (The kernel is sufficient)**.** *Consider a market with the following participants and rules.*

- *A value v and noise trades z_t.*
- *An insider who submits $x_t = X_t(v, \mathcal{H}_{t-1})$.*
- *A dealer who sets $p_t = P_t(\mathcal{H}_{t-1}, y_t)$ from the public history $\mathcal{H}_{t-1}$ and the total order flow $y_t = x_t + z_t + f_t$.*
- *Feedback groups whose memory states update on a public price statistic $s_t = S_t(p_0, \dots, p_t)$ by $m_{i,t+1} = r_i m_{i,t} + s_t$ from $m_{i,0} = 0$, with $r_i = 1 - \eta_i$ and aggregate flow $f_t = \sum_i w_i m_{i,t}$, $w_i = q_i \eta_i$.*

The rules X, P and S are arbitrary causal maps. This covers Kyle, inventory-averse, learning and misspecified dealers, and any insider strategy.

(i) $f_t = \sum_{m<t} k_m \, s_{t-1-m}$, *with* $k_m = \sum_i w_i r_i^m$, *the moments of the merged measure M.*

(ii) Two group configurations produce the same aggregate flow on dates $t \le T$ for every signal sequence (s_t) if and only if $k_m = k'_m$ for all $m < T$. On an unbounded horizon this holds if and only if their merged measures coincide. For a fixed signal rule S, the 'if' direction holds for every price path. The 'only if' direction needs a rule that can produce an impulse signal. It fails, for example, for $S \equiv 0$, under which every configuration has zero flow.

(iii) If $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the following coincide on $t \le T$: the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory. The two games therefore have the same strategy sets and the same payoff for every profile. They have the same equilibria, with the same outcomes, under any equilibrium concept defined from these payoffs and the public history.

(iv) Group-level quantities are not invariant. A Gauss and a Radau two-point configuration agree on k_0, k_1, k_2, so they are outcome-equivalent for $T = 3$, while their largest groups carry weights $\frac{1}{2}$ and $\frac{3}{4}$.

Beyond exponential memory. The proof of part (iii) uses only one property of the feedback traders: their aggregate flow is a fixed linear function of past public signals. So the result does not depend on exponential memory.

Proposition S2.2 (Any linear population)**.** *In the market of Theorem S2.1, replace the groups by any finite population of feedback traders $j = 1, \dots, J$ whose flows are causal linear functions of the public signal,*

$$f_{j,t} = \sum_{m<t} \kappa_m^{(j)} \, s_{t-1-m}, \qquad \kappa^{(j)} = (\kappa_0^{(j)}, \kappa_1^{(j)}, \dots) \text{ arbitrary real sequences,}$$

and write $k = \sum_j \kappa^{(j)}$ for the population's kernel.

(i) Two populations produce the same aggregate flow on dates $t \le T$ for every signal sequence if and only if their kernels agree for $m < T$.

(ii) If their kernels agree for $m < T$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on $t \le T$, and the two games have the same equilibria under any equilibrium concept defined from payoffs and the public history.

(iii) Any decomposition of a kernel into individual kernels is consistent with the same outcomes. Prices, flows and all equilibrium outcomes therefore carry no information about the decomposition beyond k.

Proof. (i) Aggregate flow is $f_t = \sum_{m<t} k_m s_{t-1-m}$. If the kernels agree for $m < T$, the flows agree on $t \le T$ for every signal. Conversely, the impulse signal $s_j = \mathbf{1}\{j = 0\}$ gives $f_{m+1} = k_m$. (ii) The strong induction in the proof of Theorem S2.1(iii) (Supplement S9) uses only that f_t is the same function of the signals before t in both markets, which is part (i). (iii) follows from (ii), since outcomes depend on the population only through k. □

S2.1 What prices identify

Theorem S2.1 says that populations with the same kernel cannot be told apart by anything that price-based trading determines. The converse is that prices determine the kernel. In the probing experiment of Section S1, the response of prices to trades is a fixed function of (a, κ, k), and it determines the merged measure.

Theorem S2.3 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{9}$$

Write $a' = a + \kappa M(\{0\})$, so that groups with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$ the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0, 1)|$. A group with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single group with rate $\bar{\eta}$ against two groups with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with all rates in $[\epsilon, 1 - \epsilon]$ for some $\epsilon > 0$. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. Any test with worst-case error at most $\alpha \in (0, 1/4)$ needs*

$$T \;\ge\; \frac{4\sigma_\nu^2 (1 - 2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{10}$$

where $C_{\mathcal{F}}$ depends only on the family, through ϵ, the bound on $|H|$ implied by uniform stability, and the total weight q.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence verified in the full working paper: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark S2.4 (Scope). The result concerns the stated model and observation contract. Groups sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. A group may choose its rate strategically. The observer then still sees only the merged measure of the rules in use; Proposition S5.4 draws the consequence.

S3 Sufficiency and identification in equilibrium

The identification results of Section S2 take the price impact κ and the persistence a of pricing errors as given. This section derives both from dealers who dislike holding inventory. It shows that the equilibrium price is the closed loop of Section S1 with effective parameters and unchanged memory rates. So the kernel stays the sufficient statistic in equilibrium, and the cost of going beyond it (Section S4) keeps its order.

An inventory equilibrium.

- *Fundamental and price.* The fundamental F_t is a public random walk with innovation variance σ_F^2, and the price is $p_t = F_t + x_t$.
- *Groups.* They trade $u_{i,t+1} = r_i u_{i,t} + w_i x_t$, the paper's exponential memory with $w_i = q_i \eta_i$ and $r_i = 1 - \eta_i$.
- *Exogenous flow.* $\xi_t \sim N(0, \sigma_\xi^2)$, i.i.d., stands for noise trades or the observer's probe. Customer flow is $y_t = \sum_i u_{i,t} + \xi_t$.
- *Dealers.* Competitive dealers absorb the flow and lay off a share $1 - \bar{a}$ of their inventory each period, so $H_t = \bar{a} H_{t-1} - y_t$. Each overlapping generation of dealers has CARA risk aversion A and holds H_t to the next date: $H_t = (E_t p_{t+1} - p_t)/(AV)$, with $V = \mathrm{Var}_t(p_{t+1})$.

We study stationary equilibria in which x_t is linear in the dealers' inventory and the groups' next trades. Write $G(D) = \sum_i w_i/(D - r_i) = \int dM(r)/(D - r)$ for the Stieltjes transform of the merged measure.

Theorem S3.1 (Price impact and persistence in an inventory equilibrium)**.** *Consider stationary linear equilibria of the model above.*

(i) Characterization. *The price law*

$$x_t = -\kappa H_t + \kappa \sum_i \frac{u_{i,t+1}}{D - r_i}$$

with $D \neq 0$ and $D \neq r_i$, is an equilibrium exactly when the induced dynamics are stable and

$$D = 1 - \kappa\, G(D), \qquad \kappa(D - \bar{a}) = A\, V\, D, \qquad V = \sigma_F^2 + \frac{\kappa^2 \sigma_\xi^2}{D^2}. \tag{11}$$

An unexpected unit of flow moves the price by κ/D.

(ii) No feedback. *Then $D = 1$ and the price law is exactly the paper's $x_t = \bar{a}\, x_{t-1} + \kappa\, y_t$. The persistence of pricing errors is the dealers' carry share, $a = \bar{a}$, and the impact κ solves $\kappa(1-\bar{a}) = A(\sigma_F^2 + \kappa^2\sigma_\xi^2)$. An equilibrium exists if and only if $\Delta = (1-\bar{a})^2 - 4A^2\sigma_F^2\sigma_\xi^2 \geq 0$. When $\Delta > 0$ there are two, $\kappa_\pm = \big((1-\bar{a}) \pm \sqrt{\Delta}\big)/(2A\sigma_\xi^2)$; the low-impact one $\kappa_- \approx A\sigma_F^2/(1-\bar{a})$ is the price of dealer inventory risk.*

(iii) Invariance. *The coefficients (κ, D, V), and hence the price process, depend on the groups only through G, and so only through the merged measure M. This is Theorem S2.1 in equilibrium form.*

(iv) Small feedback. *Let $\Delta > 0$, fix $\delta > 0$ with $\max_i r_i < 1 - \delta$, and let $\epsilon = \sup_{|D-1|\leq\delta}(|G(D)| + |G'(D)|)$ be small. Then there is a unique equilibrium near $(\kappa_-, 1)$. With $G(1) = \sum_i q_i$, the aggregate feedback intensity,*

$$D = 1 - \kappa_- G(1) + o(\epsilon), \qquad \kappa = \kappa_- + \frac{\kappa_-^2(\bar{a} + 2A\kappa_-\sigma_\xi^2)}{\sqrt{\Delta}}\, G(1) + o(\epsilon).$$

Positive feedback raises the impact κ, and it raises the impact of an unexpected flow shock, κ/D, by more.

Machine check. See Supplement S14.

The identification results hold in equilibrium. With feedback, dealers price the groups' predictable trades in advance, so the response of prices to flow is no longer the reduced form of Section S1 with the primitive (a, κ). The next theorem shows that it is still that closed loop, exactly and path by path, with new parameters and the same memory rates.

Theorem S3.2 (The equilibrium is the paper's closed loop)**.** *Take an equilibrium of Theorem S3.1 with $D \neq \bar{a}$, and set*

$$a^* = \frac{\bar{a}}{D}, \qquad \kappa^* = \frac{\kappa}{D}, \qquad \tilde{w}_i = \frac{D - \bar{a}}{D - r_i}\, w_i.$$

(i) Reduced form, pathwise. *Let $\tilde{u}_{i,t} = \frac{D-\bar{a}}{D-r_i} u_{i,t}$. For every realization, every initial state and every t,*

$$x_t = a^* x_{t-1} + \kappa^*\Big(\xi_t + \sum_i \tilde{u}_{i,t}\Big), \qquad \tilde{u}_{i,t+1} = r_i \tilde{u}_{i,t} + \tilde{w}_i x_t.$$

So the price follows the closed loop (2)–(3) with persistence a^*, impact κ^*, exogenous flow ξ in place of the probe, and groups with the same memory rates and merged weights $\tilde{w}_i$. Its transfer function from flow to price is (5) with $(a^*, \kappa^*, \tilde{w})$. The equilibrium is stable if and only if this closed loop is.

(ii) Zeros. *For $\bar{a} > 0$, distinct rates in $(0, 1)$ and nonzero weights, the memory rates are the transmission zeros of the equilibrium price response, and none cancels (Theorem S4.3(i) with the effective parameters).*

(iii) Boundary. *The effective merged measure is $\tilde{M} = h\, M$, where $h(r) = (D - \bar{a})/(D - r)$ is positive. The effective parameters determine the equilibrium ones: $D = 1 - \kappa^*\tilde{M}([0, 1])/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. So the merged measure remains exactly what equilibrium prices identify (Theorem S2.3).*

(iv) The rates survive. *Let $(a^*, \kappa^*, W, r_c, \Delta\eta)$ satisfy the hypotheses of Theorem S4.1 with $(a, \kappa) = (a^*, \kappa^*)$. Then the two K-group economies of that theorem are the price responses of two inventory equilibria with identical primitives $(A, \sigma_F^2, \sigma_\xi^2, \bar{a})$ and identical (κ, D):*

- *$D = 1 - \kappa^* W/(1 - a^*)$, which exceeds every rate;*
- *$\bar{a} = a^* D$ and $\kappa = \kappa^* D$;*
- *any $\sigma_F^2, \sigma_\xi^2 > 0$, with $A = \kappa^* D(1 - a^*)/(\sigma_F^2 + \kappa^{*2}\sigma_\xi^2)$;*
- *group weights $w_i = \tilde{w}_i(D - r_i)/(D - \bar{a}) > 0$.*

Hence an observer of equilibrium prices who knows the primitives needs $N \geq c\,\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments. Theorem S4.2 applies to the effective parameters, and the true weights follow from the effective ones by a Lipschitz map. So $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ in equilibrium.

In words. Dealers' anticipation does not hide the groups and does not move them. It replaces the persistence and impact by $\bar{a}/D$ and κ/D, and it rescales each group's weight by $(D - \bar{a})/(D - r_i)$. Because that factor rises with r_i, slow-memory groups weigh more in equilibrium prices. The memory rates are untouched.

- Every result of Sections S2 and S4 that concerns the map from order flow to prices therefore holds in equilibrium: identification, sample complexity, passive observation, and probing costs computed from the price path. Its constants are evaluated at the effective parameters.
- The lower-bound pair is not an artefact of the reduced form. Two economies with the same dealers, the same risk aversion and the same impact differ only in their groups, and they are as hard to tell apart as Theorem S4.1 says.
- Part (i), the realization in part (iv) and the effective group recursion are machine-checked (`Paper1EquilibriumReducedForm.lean`).
- One assumption is new: the observer's probe is part of the exogenous flow that dealers price as noise. Dealers who recognized a probe would price it differently.

Further equilibrium results. Four results in Supplement S9 extend this section. None is needed for the main argument.

- *A strategic insider* with short-lived information (Theorem S9.3). The equilibrium is characterized in closed form and nests Kyle's one-period values. Its coefficients depend on the groups only through G, and the observed price is the closed loop driven through a known invertible moving-average filter. So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present. Positive feedback raises total impact, and the insider then trades less and prices reveal less of her news.
- *Fragility* (Theorem S9.2). With positive feedback, a positive-impact equilibrium exists only if dealer risk aversion lies below a threshold. For strong feedback the threshold is inversely proportional to the feedback intensity $G(1)$. With several groups the bound depends on the population only through $G(1)$, again an aggregate.
- *Rents* (Theorem S9.4). A dealer that misjudges the aggregate response by ε hands an insider with a committed schedule a rent of exact order ε, or charges it one. Public data reveal the aggregate response at the ordinary rate. So the information that protects price formation is cheap to learn.

- *A targeted halt in equilibrium* (Theorem S9.5). Two markets that agree in equilibrium to within the cube of the memory spacing can differ at order one after their largest group is halted.

Disclosure keeps its value in equilibrium too (Corollary S9.1).

S4 The price of composition

Prices identify the kernel, and with exponential memory the kernel determines the merged measure. A finite record, however, pins down only finitely many moments of M, each with noise. This section asks how much data it takes to recover the groups. The answer is the known rate for finite mixtures. What the market adds is that the memories must first be extracted from a feedback loop, and that waiting is no cheaper than trading.

S4.1 Many unknown groups

Theorem S4.1 (K-group two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes the noisy taps $h_0, \dots, h_{T-1}$ with $T \geq 2K + 1$ ($T = \infty$ allowed), so that the first tap where the two economies below differ, h_{2K}, is observed; the bound below uses the full response norm, which dominates any finite prefix. There exist two K-group economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \dots, 2K - 2$ coincide while the moment of order $2K - 1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem S4.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which the full working paper matches with an upper bound.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K - 2$ exactly. For $K = 2$ it reproduces the order of Theorem S11.1. Numerically, their low moments agree to machine precision and their moment of order $2K - 1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent. The two rules also differ in their largest weight for every K we checked: for $K = 2, \dots, 200$ the largest Gauss–Radau weight exceeds the largest Gauss weight, by 0.25 at $K = 2$ and by about $0.79/K^2$ for large K (`equilibrium/gauss_radau_maxweight.py`). So the argument given for $K = 2$ in Proposition S5.1 applies to each such K. A proof of this comparison for every K is open. The supervisory result needs less: Lemma S12.3 proves that, for every K, some two rules of this kind have different largest weights.

What counts as an experiment. In Theorem S4.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and the account-level analysis of the full working paper prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section S5).

Theorem S4.2 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j} |s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-group economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments, each observing at least the taps $h_0, \ldots, h_{2K}$ with Gaussian noise of variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem S4.1 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.

Smaller classes and constants. On smaller classes the upper bound need not be tight: if the rates are known, for example, two weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments. For K known clustered rates the requirement is at least of order $\sigma^2(\Delta\eta)^{-2(K-1)}$. A weight perturbation along the $(K-1)$-th finite difference preserves the moments of order $0, \ldots, K-2$ and changes the moment of order $K-1$ by order $(\Delta\eta)^{K-1}$. For $K = 3$, the direction $(1, -2, 1)$ on rates $r_c - \Delta\eta, r_c, r_c + \Delta\eta$ changes the second moment by $2(\Delta\eta)^2$. This is still far below the unknown-rate order $(\Delta\eta)^{-(4K-2)}$. The constants c_0 and C grow quickly with K, through the conditioning of the moment map; the theorem gives the order in $\Delta\eta$ for fixed K, not a usable constant for large K (in the full working paper our double-precision Prony implementation already fails at $K = 4$ on the disclosure grid).

S4.2 Passive data

Theorem S4.3 (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

(i) The zeros of H are the memory rates r_i together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so H is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem S2.3.

(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2|H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), f determines $\sigma_\xi^2\kappa^2$ and the normalized response H/h_0 (recall $h_0 = \kappa$); hence it determines the memory

rates. If in addition κ (or σ_ξ^2) and a are known, it determines H and, by Theorem S2.3, the merged measure.

The result holds for signed weights (mean-reverting groups): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the groups' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem S4.4 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single group at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-group economies of Theorem S11.1, $D = \Theta(\tau^2(\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-group Gauss and Gauss–Radau economies of Theorem S4.1, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of groups. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at the Chernoff-information rate, which is of the same order in the separation. So the orders are the right ones. We do not claim that this attainment is uniform as the separation shrinks. All constants depend only on K, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on δ, $\Delta\eta$ or T. In part (ii) τ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \leq C\,T\,\varepsilon^2$, with ε the leading moment gap (δ^2, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on D and the fixed-pair error exponents are separate statements.

In executed quantity and money. For a *known* pair (one group at rate η against two at $\eta \pm \delta$ with the same total weight), the information needed can be stated in trades. The executed quantity a prober needs is of exact order δ^{-2}: the two price responses first differ at lag two, by $-\kappa^2 q \delta^2$, and a single pulse of size $2\sigma\Phi^{-1}(1-\alpha)/(\kappa^2 q \delta^2)$ attains error α. The capital a counterparty must post to fill such probes is of order δ^{-4}, because the adverse mark of a probe grows with the square of its size. Halving the difference between two memories therefore quadruples the trading needed and multiplies the capital by sixteen (Theorems S10.1 and S10.3 and Lemma S10.5, Supplement S10).

Near instability. Theorem S4.4 is stated on a family that stays uniformly stable. One might hope that the barrier lifts as the market approaches instability, since the loop then amplifies every difference. It does not. What becomes visible is the distance to instability, not the composition.

Proposition S4.5 (The barrier does not lift near instability)**.** *Fix $K \geq 2$ and the K-group Gauss and Gauss–Radau economies A and B of Theorem S4.1, with positive weights, rates in $(0, 1)$ and $\kappa > 0$. For a margin $\epsilon > 0$, choose persistences $a_A(\epsilon)$ and $a_B(\epsilon)$ so that both economies have*

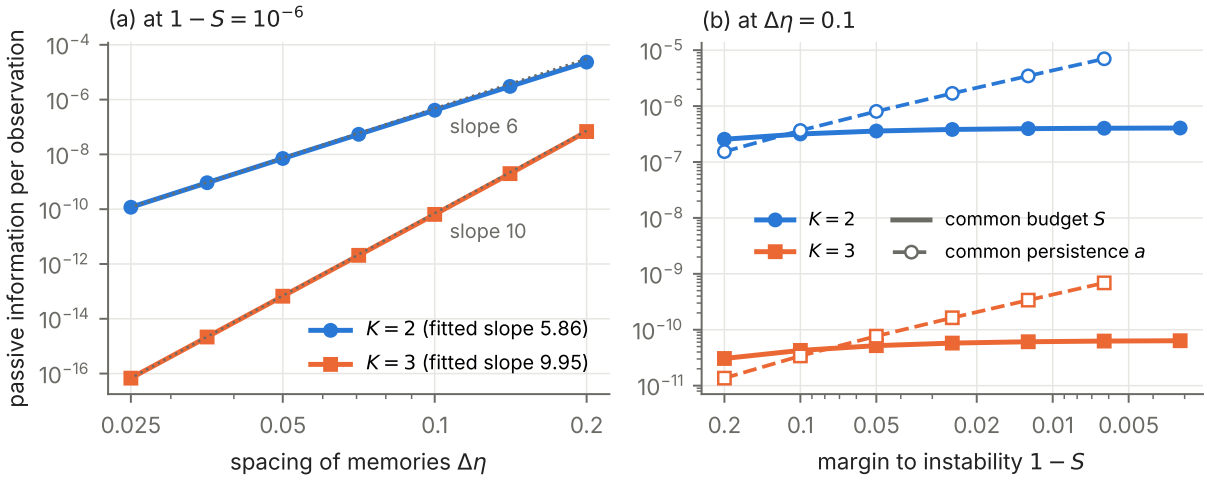


Figure S1: The passive barrier near instability. Both panels show the Gaussian Kullback–Leibler information per observation between the stationary price processes of two populations whose merged measures are the K-node Gauss and Gauss–Radau rules on memories spread over $\Delta\eta$ around 0.5 (total weight 0.3, impact $\kappa = 0.5$). (a) At a common feedback budget one millionth below instability, the information falls like $(\Delta\eta)^{4K-2}$ (dotted lines). (b) Held at a common budget (solid), the information stays bounded as the margin $1 - S$ shrinks; held at a common persistence (dashed), it grows without bound, because the budgets then differ. Computed from the closed-form spectra; scripts in `research_extension/fragility_reveals/`.

feedback budget $S = 1 - \epsilon$, and suppose $a_A(\epsilon), a_B(\epsilon) > 0$ for $\epsilon \le \epsilon_0$. Let $D_\epsilon(B\|A)$ be the Kullback–Leibler divergence rate between the stationary price processes driven by i.i.d. $N(0, \sigma^2)$ innovations. There is C, not depending on ϵ or $\Delta\eta$, such that

$$D_\epsilon(B\|A) \;\le\; C\,(\Delta\eta)^{4K-2} \qquad \text{for all } 0 < \epsilon \le \epsilon_0 \text{ and all small } \Delta\eta.$$

In words. Even as the market approaches a crash, passive prices reveal no more about who is trading than they did in calm conditions, per observation. They reveal more and more about how close the market is to instability: if the two economies share the persistence a instead of the budget, their budgets differ by order $(\Delta\eta)^{2K-1}$, and numerically the divergence rate then grows like ϵ^{-1}. The proof is in Supplement S12.2. Two limits apply. The statement is about the divergence rate per observation; near instability prices are strongly dependent, and a finite-sample bound that is uniform in ϵ is not proved. And probing does gain near instability: the squared response difference grows like ϵ^{-1} (numerical, `research_extension/fragility_reveals`), but probing a market at the edge of instability is not a policy. Figure S1 shows both statements.

S5 Who needs composition, and what disclosure buys

By Theorem S2.1, composition is worth nothing to anyone trading on public information. It can still matter to a supervisor whose action is aimed at one group. This section gives a decision in which it does and the disclosure that removes the cost of learning it. It ends with a bound on what

price-based targeting can achieve when a group can hide.

A decision in which attribution matters. A supervisor must keep the feedback budget $S = a + \kappa \sum_i q_i$ below a prudential ceiling $\bar{s}$. It can either cut all activity uniformly or halt the largest verified class. The target is the feedback budget, not the spectral radius of the state matrix: in an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not a restoration of stability.

Proposition S5.1 (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known a, κ, and $a < \bar{s} < 1$. Write $q_i = w_i/(1 - r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

(i) A uniform scale θ meets the ceiling exactly when $a + \theta(S - a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.

(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem S4.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For N independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require

$$N \ge (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \ge c\sigma^2(\Delta\eta)^{-6}.$$

The record horizon and the response norm must be the same on both sides.

(iii) Tagged-flow estimation under Theorem S5.3, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.

The proof is in Supplement S12. Proposition S12.4 there extends part (ii) to every K, with exponent $4K - 2$.

S5.1 What class-tagged disclosure buys

Proposition S5.2 (Tagged flow removes the barrier)**.** *Let the price x_t be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma^2_\xi \in (0, \infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \ge 0$, $0 \le \lambda_\ell < 1$ the closed-loop modes). Suppose group i has merged weight $w_i > 0$ and the observer receives prices and each group's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance σ^2_f, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma^2_f [(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)], \tag{12}$$

where $A_m = \sum_{s\geq 0} r_i^s \gamma_x(s+m)$, $\Gamma_0 = \left(\begin{smallmatrix}\gamma_x(0) & \gamma_x(1)\\ \gamma_x(1) & \gamma_x(0)\end{smallmatrix}\right)$ *and* $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. *The Jacobian determinant is*

$$\det \Sigma_i = w_i \sum_{\ell,m} c_\ell c_m \frac{1-\lambda_\ell\lambda_m}{1-r_i\lambda_m} \;>\; 0. \tag{13}$$

V_i depends only on group i's own (w_i, r_i), σ_f^2 and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy, $\hat{\beta}_i$ has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least $\gamma_{\min} > 0$ and merged weights at least $w_{\min} > 0$, the limiting covariance V_i is bounded by a constant that does not depend on $|r_i - r_j|$. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family.

Machine check. Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower bound on the determinant of the population instrument matrix that does not depend on the spacing (`p1_tagged_iv_determinant_bound`); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (`p1_iv_boundary_decomposition`). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other groups' rates; the closed form (13) matches the simulated Jacobian (0.1330 against 0.1322 in a two-group economy with 4×10^5 periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem. Proposition S5.2 says that tagging removes the barrier; the next result says by how much.

Theorem S5.3 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem S4.2: rates $r_c + \Delta\eta\, s$ with s in a compact set of γ-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem S4.1, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s\leq t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let N_p and N_t be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \leq C < \infty$, depending on (σ, σ_f) and the class but not on $\Delta\eta$, such that*

(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \leq N_t \leq C$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$;

(ii) to recover the rates to accuracy $c_0\Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \leq N_t \leq C(\Delta\eta)^{-2}$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$.

Proposition S5.2 and Theorem S5.3 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

S5.2 Traders who respond to being watched

The results so far take the groups' rules as given. A group that expects a targeted action has a reason to change its rule. Sufficiency bounds what a supervisor who relies on prices can then achieve.

Proposition S5.4 (Hiding caps targeting)**.** *A supervisor observes prices, under any probes and for a record of any length, and then targets a given group or not. Being targeted costs the group $P > 0$. The group chooses among available rules. Rule ρ has a private cost $C(\rho) \geq 0$ and leads to targeting with probability $\pi(\rho)$. Suppose one available rule ρ_0, with cost C_0, gives the population the kernel of a population that the supervisor targets with probability α.*

(i) $\pi(\rho_0) = \alpha$*, whatever the supervisor's probes, record length and decision rule.*

(ii) Every rule ρ^* that minimizes the group's expected cost $C(\rho) + P\pi(\rho)$ satisfies

$$\pi(\rho^*) \;\leq\; \alpha + \frac{C_0 - C(\rho^*)}{P} \;\leq\; \alpha + \frac{C_0}{P}.$$

(iii) Suppose the group has two rules: its original rule, with cost zero and targeting probability π_T on a record of length T, and ρ_0. It adopts ρ_0 whenever $\pi_T > \alpha + C_0/P$. If $\pi_T \to 1$ and $P(1-\alpha) > C_0$, this holds for all large T, and the targeting probability is then α.

Proof. (i) By Theorem S2.1, populations with the same kernel give the same law of the price record for every probe sequence and every insider and dealer rule. The supervisor's decision is a function of that record and of its own randomization. (ii) $C(\rho^*) + P\pi(\rho^*) \leq C_0 + P\alpha$ by optimality, and $C(\rho^*) \geq 0$. (iii) The original rule costs $P\pi_T$ and ρ_0 costs $C_0 + P\alpha$. □

S6 Beyond the kernel: saturating traders and stress

The kernel is a linear object, and every precedent for it is linear. Real feedback traders have limits: position caps, margin, risk budgets. This section asks what replaces the kernel when trades saturate, and what prices reveal about the part that saturation adds. The answer is a sharper version of the paper's main message. In calm markets, prices reveal the kernel and nothing else. How close traders are to their limits, which is what governs behaviour under stress, is invisible in calm data. It can be learned only by pushing traders toward their limits, or from disclosure.

Saturating traders. In the market of Theorem S2.1, let trader i have memory rate $r_i \in [0, 1)$, capacity $c_i > 0$ and leverage $\ell_i \in \mathbb{R}$. Its memory state is $m_{i,t} = \sum_{j<t} r_i^{t-1-j} s_j$, as before, and its flow is

$$f_{i,t} = c_i \, \psi(\ell_i m_{i,t}),$$

for a policy shape ψ common to all traders. A shape is *band-linear* if $\psi(u) = u$ for $|u| \leq 1$. The hard limit $\psi(u) = \max(-1, \min(1, u))$ is the leading case: inside its band trader i trades $c_i \ell_i m_{i,t}$, and at the band edge its position is capped at c_i. Inside the bands the population is the linear one of Section S1, with merged weights $w_i = c_i \ell_i$ and kernel $k_m = \sum_i c_i \ell_i r_i^m$. Write

$$\nu = \sum_i c_i \, \delta_{(r_i, \ell_i)}$$

for the capacity-weighted measure on memory and leverage.

Theorem S6.1 (What replaces the kernel)**.** *For any shape ψ, aggregate feedback flow is $f_t = \sum_i c_i \psi(\ell_i m_{i,t}) = \int \psi(\ell\, m_t(r))\, \nu(dr, d\ell)$, with $m_t(r) = \sum_{j<t} r^{t-1-j} s_j$, so it depends on the population only through ν. If two populations have the same ν, that is $\sum_i c_i g(r_i, \ell_i) = \sum_j c'_j g(r'_j, \ell'_j)$ for every function g, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and profit and the dealer's inventory coincide at every date. The two games have the same equilibria under any equilibrium concept defined from payoffs and the public history. The kernel is the first leverage moment of ν at each memory, $k_m = \int r^m \ell\, d\nu$, and in general it does not determine ν.*

The proofs of Theorems S6.1–S6.2 and Proposition S6.3 are in Supplement S13.

Machine check. `Paper1SaturatingSufficiency.lean` proves the pathwise statement for every shape ψ and every causal insider, dealer and signal rule (`p1_sat_sufficiency`, through the generic `p1_flow_market_invariance`). The passage to equilibria is on paper, as in Theorem S2.1.

Theorem S6.2 (No information without stress)**.** *Let ψ and ψ' be band-linear. Take two populations P and P' with the same kernel, $\sum_i c_i \ell_i r_i^m = \sum_j c'_j \ell'_j r'^m_j$ for every m, but possibly different capacities and leverages.*

(i) Pathwise. *Run both markets with the same strategy profile and the same realization. Let τ be the first date at which some trader of P or of P', with its state computed from the signals of the P-market, leaves its band ($|\ell m_t| > 1$). Then the two markets have identical price and order-flow paths on every date $t < \tau$. On those dates both also coincide with the* linear *market with kernel k, so τ is the first exceedance date of the linear market.*

(ii) Records. *Let O_T be any record that is a function of the paths on dates $t < T$, plus noise independent of them. The laws of O_T under P and P' satisfy*

$$\mathrm{TV}(\mathcal{L}_P(O_T), \mathcal{L}_{P'}(O_T)) \le \Pr_{\mathrm{lin}}(\tau < T),$$

the probability that some trader of $P \cup P'$ leaves its band before T in the linear market.

(iii) Calm passive data. *In the market of Section S1 with innovations $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$ i.i.d., no probe, a start from rest, and a linear market with $S_{\mathrm{abs}} = a + \kappa \sum_m |k_m| < 1$, let $s_r^2 = \sup_t \mathrm{Var}_{\mathrm{lin}}(m_t(r))$ and $\vartheta = \max_g |\ell_g| s_{r_g}$ over the $n + n'$ traders of $P \cup P'$. Then*

$$\mathrm{TV} \le T\,(n + n')\, e^{-1/(2\vartheta^2)}, \qquad s_r \le \frac{\sigma_\varepsilon}{(1 - r)(1 - S_{\mathrm{abs}})},$$

and any test that tells P from P' with both error probabilities at most $\alpha < 1/2$ needs $T \ge (1 - 2\alpha)\, e^{1/(2\vartheta^2)}/(n + n')$ observations.

(iv) Probes. *Without process noise, if the linear market's response to a deterministic probe keeps every trader of $P \cup P'$ inside its band, the two records have the same law, for any number of repetitions and any observation noise. For a single pulse of size A, no information is obtained below*

$$A^* = \min_g \Big(|\ell_g| \sup_t \Big| \sum_{j<t} r_g^{t-1-j} h_j \Big| \Big)^{-1},$$

with h the linear impulse response.

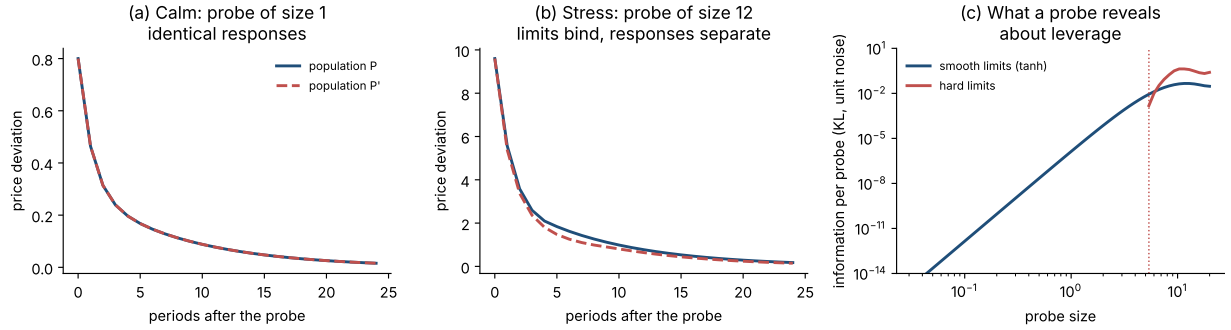


Figure S2: The two layers. Populations $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$ at memory 0.7 have the same kernel ($a = 0.5$, $\kappa = 0.8$). (a) Under hard limits, a calm probe gives identical responses. (b) A stress probe pushes the most levered trader to its limit and the responses separate. (c) Information per probe about the difference (Kullback–Leibler divergence over 30 taps, unit noise): under smooth limits it grows like the sixth power of the probe size (fitted slope 6.00); under hard limits it is exactly zero below the size at which the first trader reaches its limit (5.38, dotted).

Machine check. The pathwise part (i), including the statement that the market follows the linear market until the first exceedance, is machine-checked for every pair of band-linear shapes and every causal insider, dealer and signal rule (`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`). Parts (ii)–(iv) are on paper.

An example, and what differs at stress. At memory $r = 0.7$, take $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$. Both have merged weight 0.1, so they have the same kernel and are equivalent until stress. Under a hard limit, their total capacities, the largest aggregate flow they can deliver, are 1 and 1.25 (`p1_sat_capacity_differs`). In the market of Section S1 with $a = 0.5$ and $\kappa = 0.8$, single pulses up to size 4 give exactly identical responses. Larger pulses push the most levered trader of P' to its limit, and the responses separate by 0.072 at size 6 and 0.364 at size 12. Under Gaussian innovations, the simulated probability that some trader leaves its band within 1,000 dates lies below the bound of part (iii) at every noise level tested. It is zero up to $\sigma_\varepsilon = 0.3$, and 0.0175, 0.0875 and 0.63 at $\sigma_\varepsilon = 0.35$, 0.4 and 0.5. The scripts are `research_extension/nonlinear_scoping/sim_wiener.py` and `sim_passive_clip.py`. Figure S2 shows the two layers.

Proposition S6.3 (Smooth saturation: information grows like the sixth power of the probe)**.** *Let $\psi = \tanh$. In the probing experiment of Section S1, with reset state, no process noise, a single pulse of size A and taps observed with independent $N(0, \sigma^2)$ noise, compare two populations P and P' with the same kernel and with $\rho_P = a + \kappa \sum_i c_i |\ell_i| / (1 - r_i) < 1$, and likewise $\rho_{P'} < 1$. Let $\hat{S} = \max(\rho_P, \rho_{P'})$, $\Lambda_P = \frac{1}{3} \sum_i c_i |\ell_i|^3 / (1 - r_i)^3$, and $\Delta M_3 = \sum_i c_i \ell_i^3 - \sum_j c'_j \ell'^3_j$.*

(i) The two price responses satisfy $\|x^P - x^{P'}\|_2 \le C_D |A|^3$ with $C_D = \kappa^4 (\Lambda_P + \Lambda_{P'}) / (1 - \hat{S})^2$. Hence N experiments that tell P from P' with both errors at most $\alpha < 1/2$ need

$$N \ge \frac{4\sigma^2 (1 - 2\alpha)^2}{C_D^2 \, A^6}.$$

(ii) The first tap after the pulse differs by $x_1^P - x_1^{P'} = -\frac{1}{3} \kappa^4 \Delta M_3 A^3 + O(A^5)$. If $\Delta M_3 \neq 0$, then

for all small enough $|A|$ the averaged first tap tells P from P' with error at most α once $N \geq \left(12\,\sigma\,\Phi^{-1}(1-\alpha)\right)^2/(\kappa^8 \Delta M_3^2 A^6)$. So the order A^{-6} is attained. The cubic coefficient of the first tap identifies $\sum_i c_i \ell_i^3$, the third leverage moment of ν.

S6.1 What stress reveals

Theorem S6.2 says what calm data cannot show. The next result says what stress does show: everything that matters. Write

$$\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i, |\ell_i|)}$$

for the signed capacity at each memory and each leverage level. A trend follower and a contrarian with the same memory and the same leverage magnitude offset exactly, since $c\,\psi(-\ell m) = -c\,\psi(\ell m)$ for odd ψ.

The result holds for any shape with one property. An odd shape ψ has *independent dilations* if rescaled copies of it at distinct scales are linearly independent: for any distinct $s_1, \ldots, s_n > 0$ and any weights $w_1, \ldots, w_n$, $\sum_k w_k \psi(s_k u) = 0$ for every u implies $w_1 = \cdots = w_n = 0$.

Theorem S6.4 (Stress identifies the leverage distribution)**.** *Let ψ be odd with independent dilations, and let every trader have $r_i \geq 0$ and $\ell_i \neq 0$. In the probing experiment of Section S1, with reset state and exactly observed responses, two populations give identical price responses to every probe sequence if and only if $\tilde{\mu}_P = \tilde{\mu}_{P'}$. Probe responses therefore identify, within each memory class, the net capacity at every leverage level. The kernel $k_m = \int \lambda r^m\, \tilde{\mu}(dr, d\lambda)$ is one linear functional of this measure.*

The property holds or fails as follows.

(i) The hard limit has independent dilations. For it the result needs probes of all sizes: probes that keep every memory state inside every band identify only the kernel (Theorem S6.2(iv)).

(ii) tanh *has independent dilations. Moreover, identical responses to all probes in an arbitrarily small neighbourhood of zero already imply $\tilde{\mu}_P = \tilde{\mu}_{P'}$.*

(iii) A cubic shape $\psi(u) = u - \gamma u^3$ does not: the scales $1, 2, 3$ with weights $5, -4, 1$ give $\sum_k w_k \psi(s_k u) = 0$ for every u. For such a shape, probes identify only the first and third leverage moments within each memory.

Machine check. The general statement of Theorem S6.4 is machine-checked for any number of traders (`p1_thm_stress_identification` in `Paper1StressIdentificationGeneral.lean`). So are the first sentence of part (i) (`p1_clip_indep_dilations`), the first sentence of part (ii) (`p1_tanh_indep_dilations`) and the counterexample in part (iii) (`p1_cubic_not_indep_dilations`). The theorem for the two shapes, with no hypothesis left on ψ, is `p1_thm_stress_identification_hard_limit` and `p1_thm_stress_identification_tanh`. Two things are proved on paper only: the small-probe statement in part (ii), and the last sentence of part (iii). In the formal statement, "identical responses" reads: every pair of a probe and a price path of the first market is such a pair of the second.

The proof (Supplement S13) uses three facts. First, probes can steer the price along any path, so identical responses mean identical maps from past prices to flow. Second, two dates of prices turn that map into a sum of scaled copies of ψ, one per memory–leverage pair, with distinct scales for a suitable direction. Third, independent dilations separate the copies. For the hard limit, each copy adds a kink where its pair saturates, and the copy with the largest scale has the first kink. For

tanh, each copy approaches its limit at its own exponential rate, and the copy with the smallest scale has the slowest tail. The kink argument is constructive. In the example of Figure S2, read along one direction, the flow map has kinks at 3.702 and 14.813, and their slope jumps return net capacities 0.24989 and 0.99988, against true values 0.25 and 1. Locating the second group needs a memory state of almost 15, about four times the stress level at which the first trader saturates. Knowing the kernel and the first group fixes only the second group's total $c\ell$, not its leverage.

Theorem S6.4 turns Section S6 into a two-layer theory. Prices in calm markets identify the kernel. Prices under stress identify the full signed measure $\tilde{\mu}$, which by Theorem S6.1 is all that any price-based outcome depends on. Under hard limits, no information beyond the kernel arrives until the first group saturates. Under smooth limits, the order-n term of the response scales with the probe size to the power $2n+1$. We expect its information to scale with the power $2(2n+1)$, but only the case $n=1$ is proved (Proposition S6.3).

S6.2 Deleveraging traders and dealers who anticipate

Proposition S6.5 (Deleveraging traders)**.** *Let each trader follow any causal rule, for example deleveraging gradually, liquidating, or switching strategy after a loss, provided it trades linearly, $w_i m_{i,t}$, as long as its state has stayed inside its band on every date up to t. Then Theorem S6.2 holds verbatim for two such populations with the same kernel. Up to the first date at which some trader leaves its band, the paths coincide and equal the linear market's path, and parts (ii)–(iv) follow.*

The proof of the pathwise part is the induction of Theorem S6.2(i). It is machine-checked for arbitrary causal rules outside the bands (`p1_delever_stress_invariance`). Deleveraging that starts before a hard limit only moves the band edge inward. Calm data remain silent until the first trader starts to cut its position.

After the first breach: an open question. What happens after a breach is not covered by these results. A natural conjecture is a threshold. If traders who breach their limits liquidate, their sales move the price, and the price move pushes other traders' states toward their own limits. One expects a cascade when the liquidation triggered per unit of price move, summed over the traders close to their limits, exceeds the market's capacity to absorb it, and no cascade otherwise. We have no proof of such a threshold and do not state one.

Proposition S6.6 (Who sets prices matters)**.** *(i) Theorems S6.2 and S6.4 and Proposition S6.5 hold for any causal dealer rule that is the same in the two markets. This covers every rule that depends on the population only through its kernel, for example a dealer who knows only the kernel, or one who prices current flow.*

(ii) A dealer who knows the population and quotes on the expected next-period feedback flow uses more than the kernel. Suppose the next memory states are Gaussian, $m_g^+ \sim N(\bar{m}_g, s_g^2)$, with every trader inside its band in expectation, $|\ell_g \bar{m}_g| < 1$, and hard limits. Then the expected next-period flows under two populations with the same kernel differ by at most

$$\sum_{g \in P \cup P'} c_g \sqrt{2/\pi}\, \sigma_g\, e^{-\delta_g^2/(2\sigma_g^2)}, \qquad \sigma_g = |\ell_g| s_g, \quad \delta_g = 1 - |\ell_g \bar{m}_g|.$$

The proof is in Supplement S13.

What the supervisor can and cannot learn. The two results split the population into a part prices reveal in calm markets, the kernel, and a part they reveal only under stress: the distribution of leverage within each memory, weighted by capacity. The second part decides how the market behaves when traders hit their limits, through the total capacity and through how many traders are near their limits. Calm-market price data are therefore the wrong instrument for it, however long the record and however sophisticated the estimator. The options are stress probes, whose cost and risk Proposition S6.3 prices, actual stress episodes, or disclosure of capacities: position limits, margins and risk budgets. This is a different disclosure from the class-tagged flow of Section S5. Tagged flow reveals each group's memory and weight. Capacity disclosure reveals how far each group is from its limit.

What is open. Three questions remain open. The first is rates for the higher orders of Theorem S6.4: Proposition S6.3 gives the first nonlinear moment, and the general order is not proved. The second is a full equilibrium with forward-looking dealers and saturating or deleveraging traders, including what happens after the first breach, in a fire sale. The third is learning agents whose rules change continuously, not only at a band edge. Related ideas appear in the literature on funding constraints and liquidity spirals (Brunnermeier and Pedersen, 2009) and in evidence that crowding predicts tail risk (Barroso et al., 2022). The identification of sums of nonlinear branches is a known problem in system identification (Dreesen et al., 2017). What is new here is the sufficient measure inside a strategic market, the exact absence of calm-market information about it, and the reduction of its recovery under stress to a known uniqueness problem.

S7 Scope and limits

Linearity. Sufficiency (Theorem S2.1, Proposition S2.2) needs aggregate feedback flow to be a linear function of past public signals, plus independent idiosyncratic shocks if there are any. With group-specific Gaussian demand shocks, a second aggregate enters, the spectral density of the aggregate shock process, and the equivalence class shrinks accordingly (Supplement S9). The passive results of Section S4 are stated for Gaussian innovations. With non-Gaussian innovations, higher-order moments of prices could carry further information; we do not analyse that case. Saturating rules are treated in Section S6. Other nonlinear rules, and traders who change their rules as they learn, are outside the theorems. The full working paper reports simulations of stylized nonlinear rules and of online learners trained in the closed loop. In those simulations, the costs of Section S4 hold when the competing populations' feedback weights match. Populations whose learned weights differ are told apart by those weights instead. Hard thresholds hide groups from small probes altogether. These are simulations, not theorems.

Exponential memory. The identification and cost results need memory to be a finite mixture of exponentials. For general completely monotone memory, the full working paper proves that the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$. Halving the error then squares the data needed, instead of multiplying it by a constant.

The observer. The lower bounds hold for an observer who knows (a, κ) and the noise law, so they hold a fortiori without that knowledge. The upper bounds assume reset experiments or a known initial state. Strategic responses to being observed are modelled only in Proposition S5.4, and there for one group. Everywhere else the groups do not change their rules when probed.

Evidence. The full working paper reports pre-specified tests on two sets of trader-class reports: the CFTC's weekly Commitments of Traders in twenty futures markets and Binance's class ratios in 30 perpetual markets. Of 23 recorded outcomes, 11 pass. The class reports constrain the persistence of class positions far more tightly than prices do. They do not identify learning memory, because that persistence can also reflect turnover, reporting aggregation or inventory retention. A pre-registered out-of-sample test of the feedback premise on Taiwan's tagged exchange flows (spec P1T, holdout 2019–2023; the full working paper reports every diagnostic) gives the following. $D = 0.0426$ $[0.0206, 0.0643]$, $p = 0.000$, meets the registered supportive criterion. The pre-specified reverse-causality check gives $D = -0.0065$ $[-0.0316, 0.0165]$, which does not meet it. The registered verdict is 'supportive'. Read with its pre-specified diagnostic, the test does not provide evidence of feedback beyond prices revealing past flow. We do not claim that tagged flow in Taiwan responds to price deviations. The registered test is supportive, but its pre-specified diagnostic attributes the gain to reverse causality.

A second test, of the two-layer prediction of Section S6, is registered on Binance perpetual futures (spec P7S, amendment A1). Its primary sample is retrospective (July 2024 to October 2026), specified in November 2026 after the sample period, and a prospective window from November 2026 serves as a replication. The primary test was run once on 11 November 2026. A pre-specified test on Binance perpetual futures was inconclusive (calm difference -0.002, 90% interval $[-0.007, 0.004]$; difference-in-differences 0.041, $p = 0.553$). It neither supports nor contradicts the prediction of Section S6. The prospective replication had not been run when this version was written.

Open problems. Four questions remain open: a long-lived strategic insider who times her trades against the groups' memory; rates for the higher orders of the identification of saturating traders, and a full equilibrium with forward-looking dealers and deleveraging traders after a breach; sufficiency for learning agents whose rules change; and a supervisory loss function estimated from real interventions, which would turn the information costs here into a value of disclosure.

S8 Proofs of the main results

A lemma used in the proofs.

Lemma S8.1 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_{\ell} \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_{\ell} \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, even though J is non-normal. In price/memory coordinates the price-only gain is $\sup_t \|e_x^\top J^t\|_2 = 1$.

Lemma S8.1. Omit zero-weight groups. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots), \qquad u = (\sqrt{a}, \sqrt{b_1 \eta_1}, \ldots)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J' e_x = \sqrt{a}\, u$, so $h_j = (\kappa/a) e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0, 1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is

completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Lemma S1.1. (i) For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2)+\sigma_\zeta^2$, i.e. $P^2-\sigma_\zeta^2 P-\sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = (\sigma_\zeta^2+\sqrt{\sigma_\zeta^4+4\sigma_\zeta^2\sigma_\nu^2})/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (8), and the update $\hat{v}_{t+1} = \hat{v}_t+\eta(r_t-\hat{v}_t)$ has the stated form. (ii) $\nabla_\theta \frac{1}{2}H(x_t-\theta)^2 = -H(x_t-\theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t+\alpha H(x_t-\theta_t) = (1-\eta)\theta_t+\eta q x_t$ with $\eta = \gamma+\alpha H$ and $q = \alpha H/\eta$.

Theorem S2.3. With zero initial states, $y_{i,t} = \sum_{s\le t}\eta_i r_i^{t-s}x_s$, so $\sum_i q_i y_{i,t} = \sum_{s\le t} k_{t-s}x_s$ and every response is determined by (a,κ,k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa z D(z)/[(z-a)D(z)-\kappa N(z)]$ with $D = \prod(z-r_i)$ and $N(z) = \sum_i w_i z\prod_{j\ne i}(z-r_j)$, as in (5); at a retained r_i the denominator equals $-\kappa N(r_i) = -\kappa w_i r_i\prod_{j\ne i}(r_i-r_j) \ne 0$ (memory rates are positive because $\eta_i < 1$), so no root of D cancels. The numerator $\kappa z D(z)$ has no roots other than 0 and the r_i, so these are the only possible cancellations. The remaining numerator factor z cancels if and only if the denominator vanishes at 0; since $N(0) = 0$ the denominator there equals $-aD(0)$, which is nonzero for $a > 0$ and zero for $a = 0$. Hence the McMillan degree, which is the dimension of a minimal realization, is $1+|\mathrm{supp}(M)\cap(0,1)|$ for $a' \ne 0$ and $|\mathrm{supp}(M)\cap(0,1)|$ for $a' = 0$ (an atom at rate 0, a group with $\eta = 1$, contributes the constant $M(\{0\})$ to G and no pole, so it shifts a to $a' = a+\kappa M(\{0\})$ and the argument above applies with a') (numerically, the rank of the Hankel matrix of $(h_j)_{j\ge1}$ is $K+1$ and K respectively for $K = 1,2,3$). For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z-1+\bar{\eta})\big((z-1+\bar{\eta})^2-\delta^2\big)}.$$

The denominator is the product of the three linear factors $z-r$ with $r \in \{1-\bar{\eta},\, 1-\bar{\eta}\pm\delta\}$, all in $[\epsilon, 1-\epsilon]$ by assumption; on $|z| = 1$ each satisfies $|z-r| \ge 1-r \ge \epsilon$, so $|G_\delta - G_0| \le 2q\delta^2/\epsilon^3$ there. Uniform stability gives $M_H = \sup|H_\theta| < \infty$, and the identity $H_\delta - H_0 = z^{-1}H_\delta H_0(G_\delta - G_0)$, whose factor z^{-1} has modulus one on the circle, gives an ℓ_2 gain at most $C_{\mathcal{F}}\delta^2$. The mean difference of a power-constrained probe has squared norm at most $C_{\mathcal{F}}^2\delta^4 PT$; the equal-noise KL is at most $C_{\mathcal{F}}^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1-2\alpha$ gives (10).

Theorem S4.1. Let (s^A,ω^A) and (s^B,ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1,1]$, normalized to unit mass. Both are positive and exact for polynomials of degree $2K-2$, and their moments of degree $2K-1$ differ by $g_K \ne 0$. To see that $g_K \ne 0$, let the Radau nodes be $-1, s_2, \dots, s_K$ and put $P(s) = (s+1)\prod_{j\ge2}(s-s_j)^2$. This is monic of degree $2K-1$ and vanishes at every Radau node, so the Radau rule gives it 0. Gauss is exact to degree $2K-1$, so it gives $\int P\,d\mu > 0$, since $P \ge 0$ on $[-1,1]$ and vanishes only at the nodes. The two rules agree on every lower-degree term, so they differ on s^{2K-1}. Put $r = r_c+\Delta\eta\, s$ and weights $W\omega$, and let $\mu_j^X = W\sum_i \omega_i^X (s_i^X)^j$ for $X \in \{A,B\}$. Then $|\mu_j^X| \le W$, and $\mu_j^A = \mu_j^B$ for $j \le 2K-2$.

Kernel gap. By the binomial theorem, $k_m^X = \sum_j \binom{m}{j} r_c^{m-j}(\Delta\eta)^j \mu_j^X$. So, with $n = 2K-1$, $\Delta k_m = k_m^A - k_m^B = \sum_{j\ge n}\binom{m}{j} r_c^{m-j}(\Delta\eta)^j(\mu_j^A-\mu_j^B)$. Since $\binom{m}{j} \le \binom{m}{n}\binom{m-n}{j-n}$ for $j \ge n$, this gives $|\Delta k_m| \le 2W\binom{m}{n}(\Delta\eta)^n(r_c+\Delta\eta)^{m-n}$. Summing with $\sum_m \binom{m}{n}x^{m-n} = (1-x)^{-(n+1)}$ gives $\|\Delta k\|_1 \le 2W(\Delta\eta)^{2K-1}/(1-r_c-\Delta\eta)^{2K}$ over all lags.

Response gap. Let $\varsigma = a + \kappa W/(1 - r_c - \Delta\eta) < 1$, the theorem's hypothesis. Each kernel has $\|k^X\|_1 \le W/(1 - r_c - \Delta\eta)$. From $h_0 = \kappa$ and $h_{t+1} = a h_t + \kappa \sum_j k_j h_{t-j}$, every partial sum of $|h^B|$ is at most $\kappa/(1-\varsigma)$. The difference $\Delta h = h^A - h^B$ has $\Delta h_0 = 0$ and $\Delta h_{t+1} = a\Delta h_t + \kappa \sum_j (k_j^A \Delta h_{t-j} + \Delta k_j h_{t-j}^B)$, so every partial sum of $|\Delta h|$ is at most $\kappa \|\Delta k\|_1 \|h^B\|_1/(1-\varsigma)$. Hence, for every horizon,

$$\|h^A - h^B\|_2 \le \|h^A - h^B\|_1 \le C_U (\Delta\eta)^{2K-1}, \qquad C_U = \frac{2\kappa^2 W}{(1 - r_c - \Delta\eta)^{2K}(1-\varsigma)^2}.$$

In the other direction, $\Delta k_j = 0$ for $j < 2K-1$, so by induction $\Delta h_t = 0$ for $t \le 2K-1$, and $\Delta h_{2K} = \kappa\, \Delta k_{2K-1} h_0^B = \kappa^2 W g_K (\Delta\eta)^{2K-1}$. When the taps $h_0, \dots, h_{T-1}$ are observed with $T \ge 2K+1$, this gives $\|h^A - h^B\|_2 \ge \kappa^2 W |g_K| (\Delta\eta)^{2K-1}$. With $T \le 2K$ the observed responses coincide. The impossibility bound holds for every T, since it uses only the upper order.

Testing. N reset experiments have divergence $N\|h^A - h^B\|_2^2/(2\sigma^2)$. By Pinsker's inequality, a test whose two error probabilities are at most α needs divergence at least $2(1-2\alpha)^2$. So $N \ge 4(1-2\alpha)^2 \sigma^2 (\Delta\eta)^{-(4K-2)}/C_U^2$, and the constant does not depend on T. For the W_1 version, the two merged measures are the images of the fixed normalized rules under $s \mapsto r_c + \Delta\eta\, s$ scaled to mass W, so their distance is $W\Delta\eta$ times the distance $2c_K > 0$ between the rules, and the same two-point argument applies with the metric loss.

Theorem S4.4. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi} \int_{-\pi}^{\pi} (\varrho - 1 - \log \varrho)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = z^{-1} H_1 \Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(z^{-1} H_1 \Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with w_k the group weights (any factor of η_k is absorbed in ϕ), where ϕ is analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1 - \eta$ stays inside the disk). Expanding ϕ around $\bar\eta$, $\Delta G = \sum_{j \ge 0} \frac{\partial_\eta^j \phi(z, \bar\eta)}{j!}\, \Delta\tilde M_j$, where $\tilde M_j = \sum_k w_k (\eta_k - \bar\eta)^j$. For the pair, $\Delta\tilde M_0 = \Delta\tilde M_1 = 0$ and $\Delta\tilde M_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde M_j = 0$ for $j \le 2$ and $\Delta\tilde M_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (21); for the K-group Gauss and Gauss–Radau economies with rates $\bar\eta + \Delta\eta\, s$ and weights $W\omega$, both rules integrate polynomials of degree $2K-2$ exactly, so $\Delta\tilde M_j = W(\Delta\eta)^j \sum_k (\omega_k^A (s_k^A)^j - \omega_k^B (s_k^B)^j)$ vanishes for $j \le 2K-2$ and equals $\pm W g_K (\Delta\eta)^{2K-1} \ne 0$ at $j = 2K-1$. Higher terms are smaller by a factor of the spacing: $|\Delta\tilde M_j| \le 2W(\Delta\eta)^j$, and analyticity of ϕ in η on a disk of radius $\varrho > 0$ around $\bar\eta$, uniformly on the circle, gives $|\partial_\eta^j \phi|/j! \le C\varrho^{-j}$, so for $\Delta\eta < \varrho/2$ the tail beyond the first nonzero term is $O((\Delta\eta)^{j_0+1})$. Hence $\sup_\omega |u| = O(\delta^2)$, $O(\tau(\Delta\eta)^3)$, respectively $O((\Delta\eta)^{2K-1})$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$, $\tau(\Delta\eta)^3$ or $(\Delta\eta)^{2K-1}$ and $c(\omega)$ proportional to $2\,\mathrm{Re}\big(z^{-1} H_0\, \partial_\eta^{j_0} \phi(\cdot, \bar\eta)\big)/j_0!$, $j_0 = 2$, 3 or $2K-1$. The function c does not vanish identically: $f = z^{-1} H_0 \partial_\eta^j \phi$ is analytic outside the disk including at infinity, where it vanishes, and is not identically zero; an analytic function on the exterior whose real part vanishes on the circle is an imaginary constant, which here would have to be zero. For $|u| \le \frac{1}{2}$, $u^2/4 \le u - \log(1+u) \le u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i)–(iii). The sample requirements need a finite-T bound that is uniform in the separation, which we obtain directly. The first T observations are $N(0, \Sigma_j)$ with Σ_j the $T \times T$ Toeplitz matrices of the spectral densities $f_j = \sigma^2 |H_j(e^{i\omega})|^2$. On the compact, uniformly stable family the transfer functions have no zeros or poles on the circle (Theorem S4.3), so $0 < \underline{f} \le f_j \le \bar f < \infty$ with constants that do not depend on the separation. A Toeplitz matrix has eigenvalues within the range of its symbol, so $\lambda_{\min}(\Sigma_0) \ge \underline{f}$ and $\|\Sigma_1 - \Sigma_0\| \le \sup_\omega |f_1 - f_0| = \sup_\omega f_0 |u| \le \bar f \sup_\omega |u|$. With $A = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2}$,

whose eigenvalues lie in $[\underline{f}/\bar{f}, \bar{f}/\underline{f}]$, the inequality $\lambda - 1 - \log\lambda \le (\lambda-1)^2/\lambda$ gives

$$D_{\mathrm{KL}}(P_1^T \| P_0^T) = \tfrac{1}{2}\sum_k (\lambda_k - 1 - \log\lambda_k) \le \frac{\bar{f}}{2\underline{f}}\,\|A - I\|_F^2 \le \frac{\bar{f}}{2\underline{f}}\,T\Big(\frac{\bar{f}}{\underline{f}}\Big)^2 \sup_\omega |u|^2 \le C\,T\epsilon^2,$$

with C independent of T and of the separation, because $\sup_\omega |u| = O(\epsilon)$ uniformly. The Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6) then gives the stated lower bounds on T for every T, not only asymptotically. For a fixed pair, $T^{-1}D_{\mathrm{KL}}(P_1^T\|P_0^T) \to D$ and Stein's lemma (one error fixed) and the Chernoff bound (both errors vanishing) give the error exponents of the likelihood-ratio test, of the same order in the separation; that statement is pointwise in the pair.

Theorem S4.3. In the delay variable $\lambda = z^{-1}$, $H = \kappa\prod_i(1 - r_i\lambda)/[(1-a\lambda)\prod_i(1-r_i\lambda) - \kappa\lambda\sum_i w_i\prod_{j\neq i}(1-r_j\lambda)]$, which is (5) divided through by z^{K+1}; its nonstructural zeros are the memory rates, which do not cancel under the nondegenerate assumptions of Theorem S4.3. In the z-plane there is also the structural timing zero at zero: the numerator is proportional to $z\prod_i(z-r_i)$. Stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem S2.3 then gives the merged measure.

Proposition S5.2. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ holds exactly and involves only group i, so $\tilde{u}_{i,t} = r_i\tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with the MA(1) error $e_{i,t}$. Prices are independent of the measurement noise, so $\mathbb{E}[Z_t e_{i,t}] = 0$ and, by the ergodic theorem for the stationary linear price process, the IV estimator is consistent when $\Sigma_i = \mathbb{E}[Z_t(\tilde{u}_{i,t-1}, x_t)]$ is nonsingular. Since $u_{i,t-1} = w_i\sum_{s\ge0} r_i^s x_{t-1-s}$, $\mathbb{E}[x_{t-1}u_{i,t-1}] = w_i A_0$ and $\mathbb{E}[x_t u_{i,t-1}] = w_i A_1$, which gives Σ_i. The price is a nonnegative mixture of geometric responses to its innovation (Lemma S8.1), so $\gamma_x(s) = \sum_\ell c_\ell\lambda_\ell^s$ with $c_\ell = \sigma_\xi^2\kappa^{-2}\omega_\ell\sum_m\omega_m/(1-\lambda_\ell\lambda_m) \ge 0$; substituting, $\det\Sigma_i = w_i(\gamma_x(0)A_0 - \gamma_x(1)A_1) = w_i\sum_{\ell,m} c_\ell c_m(1-\lambda_\ell\lambda_m)/(1-r_i\lambda_m)$, each term positive. By summation by parts, $\sum_{t=1}^T Z_t e_{i,t} = \sum_{t=1}^T(Z_t - r_i Z_{t+1})\varepsilon_{i,t} + O_P(1)$. Since $(\varepsilon_{i,t})$ is i.i.d. with mean zero and variance σ_f^2 and independent of the whole stationary price process, the summands $m_t = (Z_t - r_i Z_{t+1})\varepsilon_{i,t}$ form a stationary ergodic martingale difference sequence with respect to $\mathcal{F}_t = \sigma((x_s)_{s\in\mathbb{Z}}, \varepsilon_{i,s} : s \le t)$, with $\mathbb{E}[m_t m_t^\top] = \sigma_f^2[(1+r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top)] = \Omega_i$, so the Billingsley–Ibragimov martingale central limit theorem gives (12). On a uniformly stable family the modes and weights lie in a compact set with $\lambda_\ell \le \bar{\lambda} < 1$, so with $w_i \ge w_{\min} > 0$ we get $\det\Sigma_i \ge w_{\min}(1-\bar{\lambda}^2)\gamma_x(0)^2 > 0$ and $\|\Omega_i\|$ is bounded, uniformly in the spacing.

Theorem S4.2.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = a h_{t-1} + \kappa\sum_m k_m h_{t-1-m}$ (plus κ at $t=0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_1, \dots, h_{2K}$. Because $h_0 = \kappa$ is a known primitive, each k_m is a polynomial in $h_1, \dots, h_{m+1}$ with coefficients depending on (a, κ) (the only division is by $\kappa h_0 = \kappa^2$), so on the bounded set of responses within sup-distance 1 of the class this map is L_h-Lipschitz, with L_h depending on the class but not on $\Delta\eta$. Let E be the event $\max_{1\le t\le 2K}|\bar{h}_t - h_t| \le \sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$, where $\bar{h}_t$ averages the N responses; a Gaussian tail bound and a union bound over the $2K$ estimated taps give $P(E) \ge 1-\beta$. On E, and provided this radius is at most 1, the estimated kernel values satisfy $\max_m|\hat{k}_m - k_m| \le L_h\sigma N^{-1/2}\sqrt{2\log(4K/\beta)}$. Centring at r_c is a fixed triangular

linear map, $\mu_m = \sum_{j \le m} \binom{m}{j} (-r_c)^{m-j} k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s, \omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j} (s_i - s_j)^4$, bounded away from zero on the compact class. On ordered, γ-separated nodes F is injective, because a measure with K distinct atoms is determined by its first $2K$ moments. A C^1 map that is injective on a compact set Q and has a nonsingular Jacobian there is injective on an open neighbourhood U of Q: otherwise there are $x_n \neq y_n$ with $F(x_n) = F(y_n)$ and distance to Q tending to zero, whose limits $x, y \in Q$ satisfy $F(x) = F(y)$, so $x = y$, contradicting local injectivity at x. Applying the quantitative inverse function theorem uniformly on Q, there are $\rho_0 > 0$ and $L < \infty$, uniform over the class, such that $F(U)$ contains the ball of radius ρ_0 around every point of $F(Q)$ and the single inverse $F^{-1} : F(U) \to U$ is L-Lipschitz on each such ball. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K-1$. Since $\Delta\eta < 1$ and the centring map is fixed, on E we get $\|\hat{t} - t\|_\infty \le C_1 \sqrt{\log(4K/\beta)}\, \epsilon_N$ with C_1 equal to $\sqrt{2} L_h$ times the norm of the centring map; and because $\sigma N^{-1/2} \sqrt{2 \log(4K/\beta)} \le \sqrt{2}\, \epsilon_N \sqrt{\log(4K/\beta)}$, the radius condition holds whenever $c_0 \le 1/\sqrt{2}$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. For an input in such a ball, $x = F^{-1}(\hat{t})$ has K distinct nodes and nonzero weights, so the $K \times K$ Hankel matrix of $\hat{t}$ is nonsingular and Prony's method returns exactly x: the Prony polynomial is determined uniquely by the Hankel system, and since the K real nodes of x reproduce $\hat{t}$ it equals $\prod_i (z - \hat{s}_i)$, so its roots are real and distinct and no complex pair can appear; Prony's method computes the algebraic inverse of the moment map up to the order of the atoms, so, listing its output in increasing order of rate (which the error bound below keeps unambiguous once $C\sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$), its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = L C_1$ and $c_0 = \min\{\min(\rho_0, \gamma/(2L))/C_1,\ 1/\sqrt{2}\}$, which also keeps $C\sqrt{\log(4K/\beta)}\, \epsilon_N < \gamma/2$. No separate root-perturbation bound is needed: on the ball where F^{-1} is L-Lipschitz, Prony's roots and weights are the coordinates of $F^{-1}(\hat{t})$, so the conditioning of the root step is part of L (for an explicit step-by-step analysis of Prony's method with clustered exponents see Katz et al., 2023). The matching statement requires the class to contain the Gauss and Gauss–Radau pair of Theorem S4.1 (ordered nodes with separation at least the smaller of their two minimum gaps, and weights within $[\omega_{\min}, \omega_{\max}]$ after scaling to total weight W); for any such class the orders match, with constants that depend on the class. □

Theorem S5.3.

Proof. $N_p \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ by Theorems S4.1 and S4.2. *Upper bounds.* Average the N price responses to estimate h with error of order $\sigma N^{-1/2}$ per tap. Given h, class i's mean flow is a two-parameter family $F(w_i, r_i; h)$ whose Jacobian columns $\sum_s r_i^{t-s} h_s$ and $w_i \sum_s (t-s) r_i^{t-s-1} h_s$ are linearly independent, with a Gram determinant bounded below uniformly on the class because $w_i \ge \omega_{\min}$, the rates lie in a compact subset of $(0, 1)$ and $h_0 = \kappa > 0$; the bound does not involve the other classes or $\Delta\eta$. Least squares on class i's flows with the plug-in h therefore estimates (w_i, r_i) with error $O_P(N^{-1/2})$, uniformly on the class, which gives the upper bounds in (i) and in (ii) for accuracy $\epsilon\Delta\eta$. For (i) there is also a finite-sample version: in the first period the tagged flow of class i is $w_i \kappa$, so the mean of the N first-period records gives $P(|\hat{w}_i - w_i| \ge \epsilon) \le \sigma_f^2 / (N \kappa^2 \epsilon^2)$ by Chebyshev's inequality, for every configuration (this step is machine-checked). *Lower bounds.* At a configuration in the interior of the class, perturb one class's weight by ϵ, or its rate by $\epsilon\Delta\eta$; for small ϵ both perturbed configurations remain in the class because the weight interval is nondegenerate and the rate set has nonempty interior. The mean differences of both the price and the flow records are of order ϵ, respectively $\epsilon\Delta\eta$, so the Kullback–Leibler divergence of N experiments is at most $CN\epsilon^2$, respectively $CN\epsilon^2(\Delta\eta)^2$, and Le Cam's two-point lemma gives the lower bounds. □

Unknown primitives.

Corollary S8.2 (Unknown primitives cost nothing in the rate)**.** *In the setting of Theorem S4.2, suppose in addition that every memory rate is at least $\varepsilon > 0$ (that is, $r_c - \Delta\eta \geq \varepsilon$), and that the fundamental value p^*, the impact $\kappa \in [\underline{\kappa}, \overline{\kappa}] \subset (0, \infty)$ and the persistence $a \in [0, \bar{a}] \subset [0, 1)$ are unknown. Put $\tilde{k}_0 = k_0 + a/\kappa$ and $\tilde{k}_m = k_m$ for $m \geq 1$.*

(i) The response depends on (a, κ, M) only through κ and $\tilde{k}$: $h_0 = \kappa$ and $h_{t+1} = \kappa \sum_{m=0}^{t} \tilde{k}_m h_{t-m}$. The persistence a acts exactly like an extra group with memory rate zero and merged weight a/κ.

(ii) The taps $h_0, \ldots, h_{2K+1}$ identify (κ, a, M): $\kappa = h_0$; the recursion gives $\tilde{k}_0, \ldots, \tilde{k}_{2K}$; the values $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of $\mu' = \sum_i w_i r_i \, \delta_{r_i}$; and $a = \kappa(\tilde{k}_0 - \sum_i w_i)$.

(iii) Subtract the pre-probe price to remove p^* (doubling the noise variance), estimate $\hat{\kappa} = \hat{h}_0$, deconvolve with $\hat{\kappa}$ alone, apply Prony's method to the estimated moments of μ', and set $\hat{w}_i = \widehat{w_i r_i}/\hat{r}_i$ and $\hat{a} = \hat{\kappa}(\hat{\tilde{k}}_0 - \sum_i \hat{w}_i)$. There are constants $c_0, C > 0$ depending on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, \varepsilon, \underline{\kappa}, \overline{\kappa}, \bar{a}, r_c)$ but not on $\Delta\eta$ such that, whenever $\epsilon_N \leq c_0$,

$$\max_i |\hat{w}_i - w_i| + |\hat{a} - a| + |\hat{\kappa} - \kappa| \;\leq\; C\sqrt{\log(8K/\beta)}\,\epsilon_N$$

with probability at least $1 - \beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice, the same order as with known primitives.

Proof of Corollary S8.2. (i) Substitute $k_0 = \tilde{k}_0 - a/\kappa$ in $h_{t+1} = a h_t + \kappa \sum_{m \leq t} k_m h_{t-m}$: the term $a h_t$ cancels against $\kappa(a/\kappa)h_t$. (ii) Solving the recursion of (i) for $\tilde{k}_t$ gives $\tilde{k}_t = (h_{t+1} - \kappa \sum_{m<t} \tilde{k}_m h_{t-m})/\kappa^2$, a polynomial in $h_1, \ldots, h_{t+1}$ divided by a power of $h_0 = \kappa$. For $m \geq 1$, $\tilde{k}_m = \sum_i w_i r_i^m = \sum_i (w_i r_i) r_i^{m-1}$, so $\tilde{k}_1, \ldots, \tilde{k}_{2K}$ are the moments of orders $0, \ldots, 2K-1$ of μ', which has K distinct atoms with positive weights; these moments determine μ' (Theorem S2.3). Since $r_i \geq \varepsilon > 0$, $w_i = (w_i r_i)/r_i$, and then $a = \kappa(\tilde{k}_0 - \sum_i w_i)$. (iii) On the event that every estimated tap is within $\sigma N^{-1/2}\sqrt{2\log(8K/\beta)}$ of its mean (probability at least $1 - \beta/2$ by the Gaussian tail and a union bound over $2K + 2$ taps, with the doubled variance absorbed into the constant), $\hat{\kappa} \geq \underline{\kappa}/2$ once $\epsilon_N \leq c_0$, so the map from taps to $\hat{\tilde{k}}$ is Lipschitz with a constant that depends on $(K, \underline{\kappa}, \overline{\kappa}, \bar{a}, W)$ and not on $\Delta\eta$. The weights of μ' lie in $[\varepsilon\omega_{\min}, \omega_{\max}]$ and its rates are those of the class of Theorem S4.2; the proof of that theorem uses only a $\Delta\eta$-free Lipschitz deconvolution, the rescaling by $(\Delta\eta)^{-m}$ and weights bounded below, so it applies verbatim to μ' and gives $\max_i |\widehat{w_i r_i} - w_i r_i| \leq C_1\sqrt{\log(8K/\beta)}\,\epsilon_N$ and rate errors smaller by a factor $\Delta\eta$, with probability at least $1 - \beta/2$. The maps $(v, r) \mapsto v/r$ on $r \geq \varepsilon/2$ and $(\kappa, \tilde{k}_0, w) \mapsto \kappa(\tilde{k}_0 - \sum_i w_i)$ are Lipschitz on the bounded set reached, and $|\hat{\kappa} - \kappa|$ and $|\hat{\tilde{k}}_0 - \tilde{k}_0|$ are of order $\sigma N^{-1/2} \leq \epsilon_N$. Combining gives the bound. □

Proposition S2.2. The proof is given after the statement in the main text.

S9 Equilibrium: further results and proofs

Corollary S9.1 (Disclosure in equilibrium)**.** *Take a positive-impact equilibrium of Theorem S3.1 with positive weights and $D > \max_i r_i$.*

(i) The groups' flows obey $u_{i,t+1} = r_i u_{i,t} + w_i x_t$ with their true *weights. The price is the stable closed loop of Theorem S3.2, driven by ξ, with positive effective weights. So Proposition S5.2 applies: tagged flow identifies each group's (r_i, w_i), with a limiting covariance that has no singularity as rates collide.*

(ii) On classes in which both the true and the effective weights lie in compact positive intervals, Theorem S5.3 holds in equilibrium. Tagging cuts the experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ for weights and $(\Delta\eta)^{-(4K-4)}$ for rates.

In words. What makes disclosure valuable survives the dealers' optimization. Prices still hide the groups at the same polynomial cost, and each group's own flow still reveals that group's memory and weight directly.

Theorem S9.2 (Fragility: where liquidity disappears)**.** *Let there be one feedback group with weight $w > 0$ and memory rate $r \in (0,1)$, with $\bar{a} \in [0,1)$ and $\sigma_F, \sigma_\xi > 0$. Call an equilibrium of Theorem S3.1 with $\kappa > 0$ and $D > 0$ a positive-impact equilibrium. For $D \in (m,1)$ with $m = \max(\bar{a}, r)$, let*

$$\kappa_w(D) = \frac{(1-D)(D-r)}{w}, \qquad \Psi_w(D) = \frac{\kappa_w(D)\,(D-\bar{a})\,D}{\sigma_F^2 D^2 + \kappa_w(D)^2 \sigma_\xi^2}, \qquad A^*(w) = \max_{D\in(m,1)} \Psi_w(D).$$

(i) Threshold.

- *The positive-impact equilibria are exactly the pairs $(\kappa_w(D), D)$ with $D \in (m,1)$ and $\Psi_w(D) = A$, and every one of them is stable.*
- *A positive-impact equilibrium exists if and only if $A \le A^*(w)$.*
- *For $A < A^*(w)$ there are at least two; for $A > A^*(w)$ there are none.*

(ii) Monotonicity. *$A^*(w)$ is strictly decreasing in w. As $w \downarrow 0$ it tends to $A_0^* = (1-\bar{a})/(2\sigma_F\sigma_\xi)$, the threshold of Theorem S3.1(ii).*

(iii) Bounds and rate. *For every $w > 0$,*

$$A^*(w) < \min\Big\{A_0^*,\ \frac{\Phi^*}{\sigma_F^2\, w}\Big\},$$

where

$$\Phi^* = \max_{D\in[m,1]} \frac{(1-D)(D-r)(D-\bar{a})}{D} \le \frac{(1-\bar{a})(1-r)^2}{4},$$

and $w\,A^*(w) \to \Phi^*/\sigma_F^2$ as $w \to \infty$. In terms of the feedback intensity $G(1) = w/(1-r)$, strong feedback gives $A^* \approx \Phi^*/(\sigma_F^2(1-r)G(1))$, and always $A^* < (1-\bar{a})(1-r)/(4\sigma_F^2 G(1))$.

(iv) Several groups. *Let all weights be positive.*

- *The positive-impact equilibria with $D > \max_i r_i$ are exactly the pairs $((1-D)/G(D), D)$ with $D \in (\max(\bar{a}, \max_i r_i), 1)$ at which the analogue of Ψ_w, with $\kappa = (1-D)/G(D)$, equals A.*
- *Every such equilibrium has*

$$A < \min\Big\{A_0^*,\ \frac{(1-\sqrt{\bar{a}})^2}{\sigma_F^2\, G(1)}\Big\},$$

whatever the number of groups and their rates.

- *Every such equilibrium is stable.*

In words. With positive feedback, dealers can make a market only if their dislike of risk is below a threshold, and the threshold falls as feedback grows: for strong feedback it is inversely proportional to the feedback intensity. Past the threshold, no price impact is consistent with dealers' optimization. Below it, every equilibrium has stable prices. Liquidity does not erode through growing instability; it disappears when the two equilibria merge, at a point that stable prices give no warning of. At the calibration of Supplement S9 ($\bar{a} = 0.6$, $r = 0.7$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$), A^* is 0.40 without feedback, 0.25 at $G(1) = \frac{1}{15}$ and 0.034 at $G(1) = \frac{2}{3}$, and $w\,A^*$ approaches $\Phi^* = 0.00688$. The necessary conditions behind parts (i) and (iii) are machine-checked (`Paper1EquilibriumReducedForm.lean`). With several groups, part (iv) gives a threshold that depends on the population only through its total feedback intensity $G(1)$. At the same calibration, numerically, the exact thresholds for two and three groups lie between 28% and 86% of that bound, and the equilibria on the threshold curve are stable. Every such equilibrium is stable (part (iv)), as the scan also shows.

Theorem S9.3 (A strategic insider in the inventory equilibrium)**.** *Add three things to the model of Theorem S3.1:*

- News. *The fundamental moves by $F_{t+1} = F_t + \theta_t + \varepsilon_{t+1}$, with $\theta_t \sim N(0, \sigma_\theta^2)$ and ε_t i.i.d. of variance σ_ε^2.*
- An insider. *At date t a short-lived insider privately observes θ_t, and nothing else, submits d_t, and values the position at F_{t+1}.*
- What groups and dealers see. *The groups trade on the observed deviation $s_t = p_t - F_t$. The dealers price from total flow.*

Let $\nu_t = \xi_t + d_t$ be the unexpected flow.

(i) Characterization. *The strategy $d_t = b\theta_t$ and the price $p_t = F_t + \lambda\nu_t + x_t$, with x_t the price law of Theorem S3.1, form an equilibrium exactly when the induced dynamics are stable and*

$$D = 1 - \kappa G(D), \quad \kappa(D - \bar{a}) = AVD, \quad \lambda = \frac{b\sigma_\theta^2}{\sigma_\nu^2}, \quad b = \frac{D}{2(\kappa + \lambda)},$$

$$\sigma_\nu^2 = \sigma_\xi^2 + b^2\sigma_\theta^2, \qquad V = \sigma_\varepsilon^2 + \frac{\sigma_\theta^2\sigma_\xi^2}{\sigma_\nu^2} + \Big(\frac{\kappa + \lambda}{D}\Big)^2\sigma_\nu^2.$$

One unit of unexpected flow moves the price by $(\kappa + \lambda)/D$, and the insider's expected profit per date is $b\sigma_\theta^2/2$.

(ii) Kyle limit. *Without inventory risk ($A = 0$), $\kappa = 0$, $D = 1$, $\lambda = \sigma_\theta/(2\sigma_\xi)$ and $b = \sigma_\xi/\sigma_\theta$. These are Kyle's one-period values at every date, and the feedback groups do not move prices.*

(iii) Invariance. *The coefficients depend on the groups only through G, as Theorem S2.1 requires.*

(iv) Observer. *The observed deviation follows*

$$s_t = a^* s_{t-1} + \kappa^* \sum_i \tilde{u}_{i,t} + c_0\nu_t - c_1\nu_{t-1}, \qquad c_0 = \frac{\kappa + \lambda}{D}, \quad c_1 = \frac{\bar{a}\lambda}{D},$$

with $(a^*, \kappa^*, \tilde{u})$ as in Theorem S3.2. This is the closed loop of Section S1 driven through a known moving-average filter $c_0 - c_1 z^{-1}$, which is invertible because $c_1 < c_0$.

- *The memory rates remain the transmission zeros of the feedback part.*
- *The lower-bound pair of Theorem S4.1 is realized by two such equilibria with identical primitives.*
- *So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present.*

In words. A short-lived insider who knows the next piece of news trades against noise traders and against the feedback traders' predictable flow. Dealers net the predictable flow out. So the insider's trades move prices through two channels: an information part λ and an inventory part, both amplified by feedback through D. Positive feedback raises the total impact. The insider then trades less and earns less, and prices reveal less of her news. Negative feedback does the opposite.

- *An example from Supplement S9.* One positive-feedback group lowers the share of the insider's information impounded in prices from 0.372 to 0.366. One negative-feedback group raises it to 0.415.
- *What prices cannot identify.* It is still worth nothing to anyone trading on public information.
- *Machine check.* `Paper1EquilibriumInsider.lean` checks the observer's pathwise reduced form, the insider's best response, the Kyle limit and the posterior variance.
- *Scope.* The insider's information is short-lived, and she conditions only on her news. Two cases are not solved: an insider who also saw the public state, who would trade against the predictable mispricing; and a long-lived Kyle insider, who would also time her trades against the groups' memory.

A misjudged dealer and a targeted halt. The next two results use the simplest linear market inside Theorem S2.1. They are summarized in Section S3.

- Dates are $t = 0, \dots, T-1$ and the price change is $s_t = p_t - p_{t-1}$, with $p_{-1} = 0$.
- The dealer moves the price by $s_t = \lambda(y_t - \hat{f}_t)$, with $\lambda > 0$, where $\hat{f}_t = \sum_{m<t} \hat{k}_m s_{t-1-m}$ is the feedback flow it predicts from its kernel $\hat{k}$.
- The insider, with value v of mean zero and variance Σ_0 independent of the noise, commits to a schedule $x = bv$, $b \in \mathbb{R}^T$. Schedules may change sign, so pump-and-dump patterns are allowed.
- The kernel error is $\Delta k = k - \hat{k}$, with $\varepsilon = \sum_{m \le T-2} |\Delta k_m|$.

Let C be the lower-triangular matrix of ones and L the strictly lower Toeplitz matrix with $L_{t,t-1-m} = \Delta k_m$.

Theorem S9.4 (Rents from a misjudged aggregate response)**.** *(i) The insider's expected profit is $J(b) = \Sigma_0(\mathbf{1}^\top b - b^\top A b)$, with $A = \lambda C(I - \lambda L)^{-1}$. If $Q = \frac{1}{2}(A + A^\top) \succ 0$, the optimal schedule is $b^* = \frac{1}{2}Q^{-1}\mathbf{1}$ and the value is $V(\Delta k) = \frac{\Sigma_0}{4}\mathbf{1}^\top Q^{-1}\mathbf{1}$.*

(ii) If $\lambda T\varepsilon \le \frac{1}{4}(1 - \lambda\varepsilon)$, then $Q \succ 0$ and

$$|V(\Delta k) - V(0)| \le \frac{2\Sigma_0 T^2 \varepsilon}{1 - \lambda\varepsilon}.$$

(iii) To first order, $V(\Delta k) = V(0) - \Sigma_0 \sum_m \omega_m \Delta k_m + O(\varepsilon^2)$, with $\omega_m = \frac{(T-m)(T-m-1)}{2(T+1)^2} > 0$ and $V(0) = \frac{\Sigma_0 T}{2\lambda(T+1)}$. The rent is therefore of exact order ε. The insider gains when the dealer overstates the feedback ($\Delta k < 0$) and loses when it understates it.

(iv) If the dealer estimates the aggregate kernel from n observations with error $\varepsilon_n = O_p(n^{-1/2})$, the rent is $O_p(n^{-1/2})$. Separating the groups behind the same kernel still needs order $(\Delta\eta)^{-(4K-2)}$ experiments (Theorem S4.1).

Theorem S9.5 (Targeted intervention in equilibrium)**.** *Take the Gauss and Radau pairs of Theorem S2.1(iv), with total weight W, on rates $c+ds$ with $\rho = |c|+|d| < 1$. Place them in the market above, with the same dealer rule $(\lambda, \hat{k})$, the same insider schedule and the same draws.*

(i) The kernels agree for $m \le 2$, and $|k^G_m - k^R_m| \le 2W\binom{m}{3}|d|^3\rho^{m-3}$ for $m \ge 3$. Hence $\|k^G - k^R\|_1 \le 2W|d|^3/(1-\rho)^4$.

(ii) Before any intervention, the price paths differ by at most $\lambda^2 T\|k^G - k^R\|_1\|x+z\|$ divided by $(1-\lambda\|k^G-\hat{k}\|_1)(1-\lambda\|k^R-\hat{k}\|_1)$, which is $O(|d|^3)$. The insider's values differ by $O(|d|^3)$.

(iii) Halting the largest group leaves kernels $\frac{W}{2}(c-d\sigma)^m$ (the Gauss weights tie; we remove the group at $c+d\sigma$) and $\frac{W}{4}(c-d)^m$, which differ by $\frac{W}{4}$ at $m=0$ for every spacing. The post-halt responses therefore differ at order one, and so do the post-halt feedback budgets $\sum_m k_m$.

Proof of Theorem S2.1. Part (i). By induction, $m_{i,t} = \sum_{m<t} r_i^m s_{t-1-m}$. Multiplying by w_i and summing over i gives $f_t = \sum_{m<t} k_m s_{t-1-m}$.

Part (ii). If $k_m = k'_m$ for $m < T$, then by (i) the two aggregate flows agree on $t \le T$ for every signal. Conversely, the impulse signal sequence $s_j = \mathbf{1}\{j=0\}$ gives $f_{m+1} = k_m$, so equal flows on $t \le T$ for every signal sequence force $k_m = k'_m$ for $m < T$. This direction quantifies over signal sequences, not over price paths under a fixed rule S: a rule such as $S \equiv 0$ never produces the impulse.

On an unbounded horizon, equality of all $k_m = \int r^m\, dM(r)$ is equality of all moments of two finite signed measures on $[0,1]$. Polynomials are dense in $C[0,1]$, so the measures coincide. In terms of η this is the condition (9) of Theorem S2.3.

Part (iii). Fix a strategy profile and a realization. We argue by strong induction on $t \le T$. If prices and flows agree before t:

- the signals agree before t, because S_j depends on prices up to j;
- the aggregate flows at t agree, by part (i) and the equal kernels, since f_t uses signals before t only;
- the insider's orders at t agree, by causality;
- hence the total flows y_t agree, and then the prices p_t agree, by causality of the dealer's rule.

The insider's profit $\sum_{t<T}(v-p_t)x_t$ and the dealer's inventory $-\sum_{t<T} y_t$ are functions of these paths.

Strategy sets are the same in the two games, because the rules are defined on the same histories. Every payoff is an expectation of the same function of (v,z), so best responses coincide and so do the equilibria.

Part (iv). The Gauss rule has weights $\frac{1}{2}, \frac{1}{2}$ and nodes $\pm\sigma$ with $\sigma^2 = \frac{1}{3}$. The Radau rule has weights $\frac{1}{4}, \frac{3}{4}$ and nodes $-1, \frac{1}{3}$. Both integrate $1, s, s^2$ exactly, so on rates $c+ds$ the kernels agree for $m \le 2$. The largest weights are $\frac{1}{2}$ and $\frac{3}{4}$. □

Machine check. `Paper1EquilibriumInvariance.lean` proves the following:

- part (i): `p1_eqAggFlow_kernel`;
- both directions of part (ii) on a finite horizon: `p1_eqAggFlow_of_kernel` and `p1_kernel_of_eqAggFlow`;

- the pathwise statement of part (iii) for arbitrary causal insider, dealer and signal rules: `p1_eq_market_invariance` and `p1_eq_insider_profit_invariance`;
- part (iv): `p1_eq_group_level_differs`.

The unbounded-horizon moment statement and the passage from pathwise equality to equal equilibria are on paper.

Group-specific demand shocks. Suppose group i also trades an idiosyncratic shock $\epsilon_{i,t}$ through its memory, so that f_t gains the term $\sum_i w_i \sum_{m<t} r_i^m \epsilon_{i,t-1-m}$. The same induction shows that the equilibrium depends on the groups only through the kernel and the law of this aggregate shock process. For independent Gaussian shocks with variances σ_i^2, that law is determined by the spectral density $\sum_i w_i^2 \sigma_i^2 |1 - r_i e^{i\omega}|^{-2}$. This is a second merged object, and the invariance class shrinks accordingly. The paper's groups have no idiosyncratic shocks.

The textbook benchmark. Take a risk-neutral, competitive dealer that knows the kernel. It can compute f_t from public prices and price only $y_t - f_t$, so predictable feedback flow then has no price impact, and the closed loop of Section S1 is absent in that equilibrium. The theorem holds in that benchmark as well. Feedback reaches prices in equilibrium when the dealer bears inventory risk, or when it does not know the kernel and must learn it. These are the cases studied in the equilibrium extension.

Numerical illustration. `equilibrium/e1_demo.py` runs one market. The insider submits $x_t = 0.3(v - p_{t-1})$. The dealer sets $p_t = p_{t-1} + 0.3\, y_t - 0.05\, (I_t - I_{t-1})$. The signal is $s_t = p_t - p_{t-1}$. Two group configurations are compared, the Gauss and Radau pairs of part (iv) on rates $0.6 + 0.2s$, with total weight 0.5, under the same (v, z) draws. Over 200 draws:

- price paths coincide to machine precision on the first four dates ($t \le 3$), with largest gap 2×10^{-16};
- from date five, where k_3 first differs (by $\frac{2}{9}d^3$ times the total weight), they separate, with largest gap 1.8×10^{-4} at date five and 1.9×10^{-2} by date twenty;
- the largest group's share of feedback positions stays at medians of 0.59 and 0.79.

The result file is `equilibrium/e1_demo_results.json`.

Proof of Theorem S9.4. *Part (i).* Total flow is $y = x + z + f$. The dealer subtracts $\hat{f}$, so $s = \lambda(x + z + Ls)$. The matrix L is strictly lower triangular, so $I - \lambda L$ is invertible and $s = \lambda(I - \lambda L)^{-1}(x + z)$. Prices are $p = Cs$. With $x = bv$, realized profit is $\sum_t (v - p_t)x_t = v^2 \mathbf{1}^\top b - v\, b^\top C s$. Taking expectations with $E[v] = 0$ and v independent of z gives $J(b)$. When $Q \succ 0$, J is strictly concave, and its maximizer and value are as stated.

Part (ii). A Toeplitz operator is bounded by the ℓ_1 norm of its symbol, so $\|L\| \le \varepsilon$. For $\lambda\varepsilon < 1$ this gives $\|(I - \lambda L)^{-1}\| \le 1/(1 - \lambda\varepsilon)$ and $(I - \lambda L)^{-1} - I = \lambda L(I - \lambda L)^{-1}$. Since $\|C\| \le T$, we get $\|Q - Q_0\| \le \|A - A_0\| \le \lambda^2 T\varepsilon/(1 - \lambda\varepsilon)$. Moreover $Q_0 = \frac{\lambda}{2}(I + \mathbf{1}\mathbf{1}^\top) \succeq \frac{\lambda}{2} I$. Under the stated condition $\|Q - Q_0\| \le \lambda/4$, so $Q \succeq \frac{\lambda}{4} I$. Then

$$|V - V_0| = \tfrac{\Sigma_0}{4} |\mathbf{1}^\top Q^{-1}(Q_0 - Q)Q_0^{-1}\mathbf{1}| \le \tfrac{\Sigma_0}{4}\, T \cdot \tfrac{4}{\lambda} \cdot \tfrac{2}{\lambda} \cdot \tfrac{\lambda^2 T\varepsilon}{1 - \lambda\varepsilon}.$$

Part (iii). $Q_0^{-1}\mathbf{1} = \frac{2}{\lambda(T+1)}\mathbf{1}$, which gives $V(0)$. Since $A = A_0 + \lambda^2 CD + O(\varepsilon^2)$, the derivative of $\mathbf{1}^\top Q^{-1}\mathbf{1}$ is $-u^\top(\lambda^2 CD)u$ with $u = Q_0^{-1}\mathbf{1}$. Using $\mathbf{1}^\top C = (T, T-1, \ldots, 1)$ and $(L\mathbf{1})_t = \sum_{m<t} \Delta k_m$,

we get $\mathbf{1}^\top CL\mathbf{1} = \sum_m \Delta k_m \frac{(T-m)(T-m-1)}{2}$. Multiplying by $\frac{\Sigma_0}{4} \cdot \frac{4}{\lambda^2(T+1)^2} \cdot \lambda^2$ gives ω_m.

Part (iv). This is part (ii) with $\varepsilon = \varepsilon_n$. The rate for ε_n is an assumption on the dealer's estimator, for example least squares of order flow on lagged price changes in a stationary market. It is not proved here.

Proof of Theorem S9.5. *Part (i).* Write μ_j^G, μ_j^R for the jth moments of the two rules, which are at most one in absolute value. Expanding $(c+ds)^m$ gives

$$k_m^G - k_m^R = W \sum_{j \geq 3} \binom{m}{j} c^{m-j} d^j (\mu_j^G - \mu_j^R).$$

The terms with $j \leq 2$ vanish because both rules are exact for those degrees. Since $\binom{m}{j} \leq \binom{m}{3}\binom{m-3}{j-3}$, the sum is at most $2W\binom{m}{3}|d|^3\rho^{m-3}$. Finally, $\sum_{m\geq 3}\binom{m}{3}\rho^{m-3} = (1-\rho)^{-4}$.

Part (ii). With residual matrices L_G, L_R built from $k^G - \hat{k}$ and $k^R - \hat{k}$,

$$\begin{aligned} p^G - p^R &= \lambda C\big[(I-\lambda L_G)^{-1} - (I-\lambda L_R)^{-1}\big](x+z) \\ &= \lambda^2 C(I-\lambda L_G)^{-1}(L_G - L_R)(I-\lambda L_R)^{-1}(x+z). \end{aligned}$$

Bound each factor as in the proof of Theorem S9.4. The value comparison follows from the same perturbation argument, with base point L_R instead of 0.

Part (iii). Removing one of the two Gauss groups leaves weight $W/2$. Removing the Radau group with weight $3/4$ leaves $W/4$. At $m = 0$ the kernel is the remaining weight.

Proof of Theorem S3.1. *Part (i).* Conjecture $x_t = -\kappa H_t + \sum_i \beta_i u_{i,t+1}$ and set $D = 1 - \sum_i \beta_i w_i$. The next-period expectation follows from three facts: $E_t\xi_{t+1} = 0$, $E_t H_{t+1} = \bar{a}H_t - \sum_i u_{i,t+1}$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Together they give

$$D\, E_t x_{t+1} = -\kappa\bar{a}H_t + \sum_i (\kappa + \beta_i r_i) u_{i,t+1}.$$

The dealers' condition $x_t = E_t x_{t+1} - AVH_t$ must hold for all states. Matching the coefficient on $u_{i,t+1}$ gives $\beta_i = (\kappa + \beta_i r_i)/D$, that is, $\beta_i = \kappa/(D - r_i)$. The definition of D then becomes $D = 1 - \kappa G(D)$, and matching on H_t gives $\kappa(D - \bar{a}) = AVD$.

For the variance, x_{t+1} depends on ξ_{t+1} through H_{t+1} and through its own effect on $u_{i,t+2}$. Its innovation δ therefore satisfies $\delta = \kappa\xi_{t+1} + \sum_i \beta_i w_i \delta$, so $\delta = \kappa\xi_{t+1}/D$ and $V = \sigma_F^2 + \kappa^2\sigma_\xi^2/D^2$.

Conversely, take a solution of (11). Then the conjectured law satisfies the dealers' condition in every state (`p1_inventory_foc`), and V is consistent with it (`p1_inventory_innovation`). If the induced dynamics are stable, the process is stationary and the solution is an equilibrium.

Part (ii). With no groups, β is empty and $D = 1$. Then $x_t = -\kappa H_t$ and $H_t = \bar{a}H_{t-1} - y_t$ give $x_t = \bar{a}x_{t-1} + \kappa y_t$. The quadratic $A\sigma_\xi^2\kappa^2 - (1-\bar{a})\kappa + A\sigma_F^2 = 0$ has discriminant Δ (`p1_inventory_benchmark`).

Part (iii). The system (11) involves the groups only through G. The price process is a function of the history of flows, and by Theorem S2.1(i) it depends on the groups only through the kernel $k_m = \int r^m\, dM$. A finite measure on $[0,1]$ is determined by G off its support, and also by its moments.

Part (iv). Write $\Phi(\kappa, D; M) = \big(D + \kappa G(D) - 1,\ \kappa(D-\bar{a}) - A(\sigma_F^2 D + \kappa^2\sigma_\xi^2/D)\big)$. At $M = 0$ the zero is $(\kappa_-, 1)$. There the Jacobian with respect to (κ, D) is triangular with determinant $\sqrt{\Delta} \neq 0$:

$\partial_D \Phi_1 = 1$, $\partial_\kappa \Phi_1 = G = 0$, and $\partial_\kappa \Phi_2 = (1-\bar{a}) - 2A\kappa_- \sigma_\xi^2 = \sqrt{\Delta}$. The implicit function theorem, applied in the norm of the statement, gives a unique zero near $(\kappa_-, 1)$ that is continuous in M.

Linearizing, $dD = -\kappa_- G(1)$. Differentiating the second equation and using $A(\sigma_F^2 + \kappa_-^2 \sigma_\xi^2) = \kappa_-(1-\bar{a})$ gives $d\kappa = -\kappa_-(\bar{a} + 2A\kappa_- \sigma_\xi^2)\, dD/\sqrt{\Delta}$, and so the stated formula. Finally, $G(1) = \sum_i w_i/(1-r_i) = \sum_i q_i$.

At $M = 0$ the transition matrix has eigenvalues $\bar{a}$ and the r_i, all inside the unit disk, so the dynamics remain stable for small M.

Proof of Theorem S3.2. *Part (i).* Write the implicit form of the price at t, $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t}$, and the price law at $t-1$, $x_{t-1} = -\kappa H_{t-1} + \sum_i \beta_i u_{i,t}$. Use $H_t = \bar{a} H_{t-1} - \sum_i u_{i,t} - \xi_t$. Then

$$Dx_t - \bar{a} x_{t-1} = \kappa \xi_t + \sum_i (\kappa + \beta_i (r_i - \bar{a})) u_{i,t}.$$

Since $\kappa = \beta_i (D - r_i)$, the coefficient is $\beta_i (D - \bar{a}) = \kappa (D - \bar{a})/(D - r_i)$. Divide by D.

The scaled flows inherit the group recursion, because the scaling is constant (`p1_eq_scaled_cohort`). The states $(H_t, u_{\cdot,t+1})$ and $(x_t, \tilde{u}_{\cdot,t+1})$ are related by an invertible linear map, since $\kappa \neq 0$ and every scale factor is nonzero. So the two dynamics are conjugate and have the same eigenvalues.

Part (ii). The effective weights are nonzero because $D \neq \bar{a}$. Apply Theorem S4.3(i) to the closed loop of part (i).

Part (iii). Since $G(D) = \sum_i w_i/(D - r_i) = \sum_i \tilde{w}_i/(D - \bar{a})$, the equation $D = 1 - \kappa G(D)$ reads $D = 1 - \kappa^* D\, \tilde{M}([0,1])/(D(1 - a^*))$. Solve for D; then $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. Conversely, $(\bar{a}, \kappa, D, M)$ determine the effective objects. So two equilibria give the same price response if and only if their effective objects agree, and then their merged measures agree. This is consistent with Theorem S2.1.

Part (iv).

- *D exceeds every rate.* The hypothesis of Theorem S4.1 is $a^* + \kappa^* W/(1 - r_c - \Delta\eta) < 1$. It gives $\kappa^* W/(1 - a^*) < 1 - r_c - \Delta\eta$, so $D > r_c + \Delta\eta$, which is at least every rate.
- *Each economy is an equilibrium.* With the stated choices, $D + \kappa G(D) = 1$ holds (`p1_eq_realize_D`), and so does $\kappa(D - \bar{a}) = AVD$ with $V = \sigma_F^2 + \kappa^{*2} \sigma_\xi^2$ (`p1_eq_realize_dealer`). By part (i) the equilibrium price response is the given closed loop, which is stable. So each economy is an equilibrium by Theorem S3.1(i).
- *The two economies share everything but the groups.* They have the same total effective weight W, so they share D, κ and the primitives, and their price responses are those of Theorem S4.1.
- *Upper bound.* On a class where the effective parameters are known or estimated at the parametric rate (Corollary S8.2), Theorem S4.2 recovers $\tilde{w}$. Then D follows from part (iii), and $w_i = \tilde{w}_i (D - r_i)/(D - \bar{a})$ is Lipschitz on the class.

Proof of Theorem S9.2. *Part (i).*

- *D lies in $(m, 1)$.* Let $\kappa > 0$ and $D > 0$. The dealer equation $\kappa(D - \bar{a}) = AVD > 0$ gives $D > \bar{a}$. The group equation is $(D-1)(D-r) + \kappa w = 0$. If $D < r$, it forces $D > 1$, which is impossible; and $D = r$ is excluded. So $D \in (r, 1)$, and $\kappa = \kappa_w(D)$.
- *The equilibria are the solutions of $\Psi_w(D) = A$.* Multiplying the dealer equation by D gives $\kappa(D - \bar{a})D = A(\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2)$, which is $\Psi_w(D) = A$. Conversely, every such D solves the system.

- *Stability.* By Theorem S3.2(i) the dynamics are those of the closed loop with $c = \kappa^* \tilde{w} = (1 - D)(D - \bar{a})/D \in (0, 1)$. Its characteristic polynomial is $p(\lambda) = \lambda^2 - (a^* + r + c)\lambda + a^* r$. The roots are real and nonnegative, because the discriminant is at least $(a^* - r)^2$ and the coefficients alternate in sign. Next, $p(1) = (1 - a^*)(1 - r) - c = (D - \bar{a})(D - r)/D > 0$, so 1 lies outside the interval between the roots. Their product $a^* r$ is below 1, so both roots are below 1.
- *Existence and count.* Ψ_w is continuous and positive on $(m, 1)$ and tends to 0 at both ends: as $D \to 1$, $\kappa_w \to 0$; as $D \to m$, either $D - \bar{a} \to 0$ or $\kappa_w \to 0$. So its maximum A^* is attained. Every $A \in (0, A^*)$ is taken at least twice, once on each side of a maximizer, and no $A > A^*$ is taken.

Part (ii).

- *A parametrization by κ.* For fixed $\kappa \in (0, (1 - r)^2/(4w)]$, the solutions of $\kappa_w(D) = \kappa$ are $D_\pm(\kappa, w) = \big((1 + r) \pm \sqrt{(1 - r)^2 - 4\kappa w}\big)/2$.
- *The larger root gives the maximum.* $\tilde{\Psi}(\kappa, D) = \kappa(D - \bar{a})D/(\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2)$ is strictly increasing in D for $D > \bar{a}$. The numerator of its D-derivative is $\bar{a}\sigma_F^2 D^2 + (2D - \bar{a})\kappa^2 \sigma_\xi^2 > 0$. So $A^*(w) = \max_\kappa \tilde{\Psi}(\kappa, D_+(\kappa, w))$.
- *Strict decrease.* Let $w_1 < w_2$, and let κ° attain $A^*(w_2)$. Then κ° is feasible at w_1, and $D_+(\kappa^\circ, w_1) > D_+(\kappa^\circ, w_2)$. So $A^*(w_1) \geq \tilde{\Psi}(\kappa^\circ, D_+(\kappa^\circ, w_1)) > A^*(w_2)$.
- *The limit as $w \downarrow 0$.* The bound $A^* < A_0^*$ of part (iii) is one half. For the other, fix $\kappa_0 = \sigma_F/\sigma_\xi$. As $w \downarrow 0$, $D_+(\kappa_0, w) \to 1$, and $\tilde{\Psi}(\kappa_0, D_+) \to (1 - \bar{a})/(2\sigma_F \sigma_\xi)$.

Part (iii).

- *The bound A_0^*.* By the arithmetic–geometric mean inequality, $\sigma_F^2 D^2 + \kappa^2 \sigma_\xi^2 \geq 2\sigma_F \sigma_\xi \kappa D$. So $\Psi_w(D) \leq (D - \bar{a})/(2\sigma_F \sigma_\xi) < A_0^*$. Machine-checked in squared form as $4A^2 \sigma_F^2 \sigma_\xi^2 \leq (D - \bar{a})^2$ (`p1_fragility_benchmark_bound`).
- *The bound $\Phi^*/(\sigma_F^2 w)$.* Dropping $\kappa^2 \sigma_\xi^2 > 0$ from the denominator gives $\Psi_w(D) < (1 - D)(D - r)(D - \bar{a})/(\sigma_F^2 w D) \leq \Phi^*/(\sigma_F^2 w)$. Machine-checked as $A\sigma_F^2 w D \leq (1 - D)(D - r)(D - \bar{a})$ (`p1_fragility_feedback_bound`).
- *The bound on Φ^*.* It uses $(1 - D)(D - r) \leq (1 - r)^2/4$ and $(D - \bar{a})/D \leq 1 - \bar{a}$.
- *The limit as $w \to \infty$.* We have $\kappa_w = O(1/w)$ uniformly on $[m, 1]$. So $w\Psi_w(D) \to (1 - D)(D - r)(D - \bar{a})/(\sigma_F^2 D)$ uniformly, and the maxima converge.

Part (iv).

- *κ is determined by D.* Let $D > \max_i r_i$. All weights are positive, so $G(D) > 0$. Then $D = 1 - \kappa G(D)$ with $\kappa > 0$ gives $D < 1$ and $\kappa = (1 - D)/G(D)$. The dealer equation then reads as in part (i), and $D > \bar{a}$ as there.
- *Comparison with $G(1)$.* For $D \in (\max_i r_i, 1)$, each $D - r_i < 1 - r_i$, so $G(D) > G(1)$.
- *The bound.* The dealer equation gives $A\sigma_F^2 D < \kappa(D - \bar{a}) < (1 - D)(D - \bar{a})/G(1)$. Moreover

$$(1 - D)(D - \bar{a}) = D(1 - \sqrt{\bar{a}})^2 - (D - \sqrt{\bar{a}})^2 \leq D(1 - \sqrt{\bar{a}})^2.$$

  Machine-checked as `p1_fragility_K_bound`. The bound A_0^* follows as in part (iii).
- *Stability.* By Theorem S3.2(i), the dynamics are those of the closed loop with $(a^*, \kappa^*, \tilde{w})$. Here $a^* \geq 0$, $\kappa^* > 0$, $\tilde{w}_i > 0$ and $r_i \geq 0$, so its transition matrix in the state $(x_t, y_{1,t}, \ldots, y_{K,t})$ is nonnegative.

- *The spectral radius is the root above the rates.* By Perron–Frobenius, the spectral radius ρ is an eigenvalue. Every eigenvalue $z \notin \{0, r_i\}$ solves $F(z) = 1$, where $F(z) = a^*/z + \kappa^* \sum_i \tilde{w}_i/(z - r_i)$. No r_i is an eigenvalue, because $\tilde{w}_i \neq 0$ (Theorem S4.3(i)). F decreases strictly from $+\infty$ to 0 on $(\max_i r_i, \infty)$, so it has exactly one root z^* there, and $\rho \geq z^*$. Being real and at least z^*, ρ equals z^*.
- *The criterion.* Hence $\rho < 1$ if and only if $F(1) < 1$, that is, $a^* + \kappa^* \sum_i \tilde{w}_i/(1 - r_i) < 1$. With $g_i = w_i/(D - r_i)$ this reads $\kappa \sum_i g_i/(1 - r_i) < 1$ (`p1_eq_effective_stability_iff`).
- *It holds.* Each $1 - r_i > 1 - D > 0$ and $\kappa \sum_i g_i = 1 - D$, so $\kappa \sum_i g_i/(1 - r_i) < 1$ (`p1_eq_stability_K`).

Proof of Corollary S9.1. *Part (i).*

- The group recursion is part of the model.
- By Theorem S3.2(i), the price is the closed loop with $a^* \geq 0$, $\kappa^* > 0$ and $\tilde{w}_i > 0$, driven by the i.i.d. flow ξ.
- It is stable by Theorem S9.2(iv).
- These are the hypotheses of Proposition S5.2. Its regression involves only group i's own flow and the price, so it delivers the true (r_i, w_i).

Part (ii).

- *Price-only side.* By Theorem S3.2, the price-only experiment is the reduced form at the effective parameters. The realization of Theorem S3.2(iv) puts the Gauss and Gauss–Radau pair inside the equilibrium class, so the price-only side of Theorem S5.3 holds with the effective weights.
- *Tagged side.* Each group's flow response to a unit impulse is $f_{i,t} = w_i \sum_{s \leq t} r_i^{t-s} h_s$, where h is the equilibrium response and $h_0 = \kappa^* > 0$. That is the tagged record of Theorem S5.3, and its argument uses only that h is bounded with $h_0 > 0$.
- On the stated classes the true weights are a Lipschitz function of the effective ones, and conversely, so the orders are unchanged.

Numerical checks of Theorems S3.2 and S9.2. The script `equilibrium/e4_reduced_form.py` runs three checks.

- *Reduced form.* For four solved equilibria, the equilibrium price path and the reduced-form path with $(a^*, \kappa^*, \tilde{w})$ agree to 4.4×10^{-16} over 5,000 simulated periods. The four are: one group with positive feedback, one with negative feedback, two groups, and the Gauss pair. Their impulse responses agree to 10^{-16}. For one group of weight 0.02 at $r = 0.7$, for example, $a^* = 0.629$, $\kappa^* = 0.621$ and $\tilde{w} = 0.0279$.
- *Realization.* Take the Gauss and Gauss–Radau pair with $K = 2$, $a^* = 0.5$, $\kappa^* = 0.5$, $W = 0.2$, $r_c = 0.6$ and $\Delta\eta = 0.1$, so that the stability index of Theorem S4.1 is 0.833. Both economies are equilibria with $D = 0.8$, $\bar{a} = 0.4$, $\kappa = 0.4$ and $A = 0.188$, and both equilibrium equations hold to machine precision.
  - Their true group weights are $(0.064, 0.036)$ and $(0.0375, 0.0625)$.
  - Their equilibrium impulse responses differ in ℓ_2 by $1.22\, d^3$, $1.25\, d^3$ and $1.29\, d^3$ at $d = 0.1$, 0.05 and 0.025. That is the order $(\Delta\eta)^{2K-1}$.

- *Fragility.* The maximum of Ψ_w reproduces the bisection thresholds of the solver to 10^{-12} ($A^* = 0.2547, 0.1297, 0.0677, 0.0342$ at $w = 0.02, 0.05, 0.1, 0.2$).
  - $A^* = 0.3967$ at $w = 5 \times 10^{-4}$, against the limit 0.4.
  - $w\,A^* = 0.006875$ at $w = 1$ and 0.0068764 at $w = 10$, against $\Phi^* = 0.0068764$.

Several groups. `equilibrium/e4_fragility_K.py` scans D for three populations at weights 0.005 to 1 per group. The populations are two groups at rates 0.5 and 0.85, three at 0.3, 0.6 and 0.8, and the Gauss pair.

- The exact threshold A^* lies below the part (iv) bound in all 15 cases, at 28% to 86% of it. For example, for two groups with $G(1) = 1.73$, $A^* = 0.0084$ against a bound of 0.0293.
- Every equilibrium on the threshold curve is stable. At the threshold the spectral radius is between 0.79 and 0.93; it approaches 1 only at the end of the curve where $D \to \max_i r_i$ and $A \to 0$.

The result files are `equilibrium/e4_reduced_form_results.json` and `equilibrium/e4_fragility_K_results.json`.

Proof of Theorem S9.3. *Part (i).*

- *Dealers' information.* The groups' flow is predictable. So the dealers' information about θ_t is ν_t, and $E[\theta_t \mid \nu_t] = \lambda\nu_t$ with $\lambda = \mathrm{Cov}(\theta, \nu)/\mathrm{Var}(\nu) = b\sigma_\theta^2/\sigma_\nu^2$.
- *The price decomposes.* $E_t F_{t+1} = F_t + \lambda\nu_t$ and $E_t\nu_{t+1} = 0$, so $E_t p_{t+1} - p_t = E_t x_{t+1} - x_t$. The dealers' condition is therefore the one of Theorem S3.1. Its state is $(H_t, u_{\cdot,t+1})$, and $E_t u_{i,t+2} = r_i u_{i,t+1} + w_i E_t x_{t+1}$. Coefficient matching gives the same equations for β, D and κ (`p1_inventory_necessity`).
- *The price's response to ν_t.* Since the groups trade on $s_t = x_t + \lambda\nu_t$ and $\sum_i \beta_i w_i = 1 - D$, the implicit form is $Dx_t = -\kappa H_t + \sum_i \beta_i r_i u_{i,t} + (1-D)\lambda\nu_t$. The response of s_t to ν_t is therefore $(\kappa + (1-D)\lambda)/D + \lambda = (\kappa + \lambda)/D$.
- *The insider.* $F_{t+1} - p_t = \theta_t + \varepsilon_{t+1} - m_t - c\,\xi_t - c\,d_t$, where $c = (\kappa + \lambda)/D$ and m_t is the predictable part of s_t. The insider does not observe the public state, and m_t has mean zero and is independent of θ_t. So she maximizes $(\theta - c\,d)\,d$. So $d = \theta/(2c)$ and the profit is $\theta^2/(4c)$ (`p1_insider_best_response`). Taking expectations, $b\sigma_\theta^2/2$.
- *The variance.* The innovation of p_{t+1} given dealers' information is $(\theta_t - \lambda\nu_t) + \varepsilon_{t+1} + c\,\nu_{t+1}$, which gives V (`p1_insider_posterior_variance`).
- *Converse.* A solution of the system satisfies every condition just used.

Part (ii). With $A = 0$, $\kappa(D - \bar{a}) = 0$ forces $\kappa = 0$, and then $D = 1$. Then $b = 1/(2\lambda)$ and $\lambda\sigma_\nu^2 = b\sigma_\theta^2$ give $4\lambda^2\sigma_\xi^2 = \sigma_\theta^2$ (`p1_insider_kyle_limit`).

Part (iii). The groups enter only through $D = 1 - \kappa G(D)$.

Part (iv).

- *The reduced form.* Combine the implicit form at t, the price law at $t-1$ and $H_t = \bar{a}H_{t-1} - \sum_i u_{i,t} - \nu_t$, as in Theorem S3.2. This gives $Ds_t - \bar{a}s_{t-1} = \sum_i \beta_i(D - \bar{a})u_{i,t} + (\kappa + \lambda)\nu_t - \bar{a}\lambda\nu_{t-1}$ (`p1_insider_reduced_form`).
- *The filter is invertible.* $c_1 < c_0$ because $\bar{a} < 1$, $\lambda \geq 0$ and $\kappa \geq 0$. So the observer, knowing the primitives, inverts the filter, and is left with the closed loop of Theorem S3.2.
- *Realization.* Proceed as in Theorem S3.2(iv):

– set $D = 1 - \kappa^* W/(1-a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$;
– solve the insider's pair (λ, b) given (κ, D);
– set $A = \kappa(D - \bar{a})/(VD)$.

The two economies then share every primitive.

Numerical check of Theorem S9.3. The script `equilibrium/e6_insider.py` solves the system at $A = 0.05$, $\sigma_\varepsilon^2 = 1$, $\sigma_\theta^2 = \sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$, and simulates 2×10^5 dates for each solution.

Groups	κ	D	λ	b	$(\kappa+\lambda)/D$	profit	inform.
none	0.1656	1	0.4835	0.7704	0.6490	0.0963	0.372
+0.02 at 0.7	0.1692	0.9883	0.4816	0.7592	0.6586	0.0949	0.366
−0.2 at 0.7	0.1463	1.0775	0.4928	0.8430	0.5931	0.1054	0.415
two groups, or split†	0.1679	0.9924	0.4823	0.7632	0.6552	0.0954	0.368

Informativeness (inform.) is $b^2\sigma_\theta^2/\sigma_\nu^2$, the share of $\mathrm{Var}(\theta)$ resolved by the price. †Weights $0.005, 0.005$ at rates $0.5, 0.85$, or the same merged measure split into three groups; both give identical coefficients.

- In every case, the dealers' first-order residual is below 5×10^{-17}, and the observer's reduced form holds to 4.4×10^{-16}.
- The regression of θ on ν reproduces λ to the third decimal.
- The simulated profit at $b' = 0.6b, \ldots, 1.4b$ peaks at b, as does the analytic profit.
- As $A \downarrow 0$ (from 0.05 to 0.001), (λ, b) moves from $(0.482, 0.759)$ to $(0.49999, 0.994)$, toward Kyle's $(0.5, 1)$.

Numerical solution of Theorem S3.1. The script `equilibrium/e4_inventory.py` solves the system (11) at $A = 0.2$, $\sigma_F^2 = 1$, $\sigma_\xi^2 = 0.25$ and $\bar{a} = 0.6$.

How the solver finds equilibria.

- For one group with $w > 0$, every equilibrium with $\kappa > 0$ and $D > 0$ has $D \in (\max(\bar{a}, r), 1)$ and $\kappa = (1-D)(D-r)/w$. A scan over D therefore finds all of them.
- For several groups, the solver follows the branch of $D(\kappa)$ that is continuous with $D = 1$.
- Each solution is checked by simulating 2×10^5 periods of the closed loop. The check computes the dealers' first-order residual and the innovation variance of the price.

Benchmark. Here $\Delta = 0.12$, and the two equilibria are $\kappa_- = 0.536$ and $\kappa_+ = 7.46$, as in part (ii).

Results.

Groups (weights at memories)	κ	D	κ/D	spectral radius
none	0.5359	1	0.5359	0.600
0.02 at 0.7	0.5916	0.9533	0.6206	0.787
−0.2 at 0.7	0.4212	1.1767	0.3579	0.597
$0.005, 0.005$ at $0.5, 0.85$	0.569147	0.970292	0.586573	0.881
$0.0025, 0.0025, 0.005$ at $0.5, 0.5, 0.85$	0.569147	0.970292	0.586573	0.881
Gauss pair, $0.01, 0.01$	0.5759	0.9649	0.5969	0.766
Radau pair, $0.005, 0.015$	0.5743	0.9662	0.5945	0.738
0.2 at 0.7; or $0.1, 0.1$ at $0.5, 0.85$	no equilibrium			

- *The simulation checks hold in every case.* The relative first-order residual is below 4×10^{-16}. The simulated innovation variance matches $\kappa^2 \sigma_\xi^2 / D^2$ to within 0.3%.
- *Invariance (part (iii)).* The split configuration has the same merged measure as the two-group one, and its coefficients agree to all printed digits.
- *Gauss and Radau pairs.* They agree on the first three kernel moments, and their impacts differ by 0.0016.
- *Negative feedback* lowers impact.

First-order formula of part (iv), for one group at $r = 0.7$:

- at $w = 0.005$, $\kappa = 0.5457$ against the first-order value 0.5449, and $D = 0.9906$ against 0.9911;
- at $w = 0.01$, $\kappa = 0.5574$ against 0.5540.

Doubling w multiplies the error by about 4.5, as a second-order remainder should.

Existence region. For one group at $r = 0.7$, an equilibrium exists for dealer risk aversion below a threshold A^*. This is computed by bisection, with the full scan over D at each step:

w	0	0.02	0.05	0.1	0.2
$G(1) = w/(1-r)$	0	0.067	0.167	0.333	0.667
A^*	0.400	0.255	0.130	0.068	0.034

- Without feedback, $A^* = (1 - \bar{a})/(2\sigma_F \sigma_\xi) = 0.4$, which is the closed form of part (ii).
- For larger $G(1)$, A^* falls roughly in proportion to $1/G(1)$.
- At the threshold the two equilibria merge and disappear.

The results file is `equilibrium/e4_inventory_results.json`.

Numerical checks.

- `equilibrium/e2_rents.py`, with $T = 10$, $\lambda = 0.5$ and $\Delta k_m = \pm 0.4\,\epsilon\,(0.7)^m$:
  - the exact rent is -0.322ϵ when the dealer understates feedback and $+0.322\epsilon$ when it overstates it, as $\epsilon \to 0$;
  - this matches the first-order formula $\mp \Sigma_0 \sum_m \omega_m |\Delta k_m|$ to three digits at $\epsilon = 0.0125$, and to within 6% at $\epsilon = 0.4$.
- `equilibrium/e3_halt.py`, with the market of `e1_demo.py`, $c = 0.6$, $W = 0.5$, 200 draws and 40 dates:
  - the largest pre-halt price gap is 3.8×10^{-4}, 2.9×10^{-3} and 2.3×10^{-2} for $d = 0.05$, 0.1 and 0.2, a ratio to d^3 of 3.0, 2.9 and 2.8;
  - the largest post-halt gap is 0.196, 0.190 and 0.177.

The result files are `e2_rents_results.json` and `e3_halt_results.json`. Parts of both theorems are machine-checked.

- *Theorem S9.4(i):* the symmetrization, the optimal schedule and its value (`Paper1EquilibriumExtras.lean`).
- *Theorem S9.5(i):* the kernel bound $|k_m^G - k_m^R| \le 2W \binom{m}{3} |d|^3 (|c| + |d|)^{m-3}$ for every $m \ge 3$ (`Paper1HaltKernelBound.lean`).
- *Theorem S9.5(iii):* the post-halt gap $W/4$.

The remaining parts are proved on paper.

S10 Execution cost and capital

Theorem S10.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ and $\eta \in (0,1)$. Compare a single group of weight q and rate η with two groups of weight $q/2$ and rates $\eta \pm \delta$, both in $[\epsilon, 1-\epsilon]$. All parameters are known. The price receives executed quantity through κv_t, starts from zero, has no process noise, and is observed with independent $N(0,\sigma^2)$ noise. Probes are deterministic and common to both alternatives. Let $d_T = (h^\delta_j - h^0_j)_{j<T}$ and $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.*

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{14}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e^*_T(V) = \Phi\Big(-\frac{V\|d_T\|_2}{2\sigma}\Big), \tag{15}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, so it cannot make the error arbitrarily small at fixed V. Error at most $\alpha < 1/2$ requires $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ independently of T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q\delta^2$. A single pulse of size

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q \delta^2} \tag{16}$$

and the third observation achieve error exactly α. The necessary and sufficient quantity order is δ^{-2}.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$, the necessary horizon order is δ^{-4}, attained by the pulse whenever $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \ne 0$, additive reserve L and cost envelope $c\sum|v_t|$, the sharp robust quantity allowance under budget $B > L$ is $V_B = (B-L)/(|\lambda|\kappa + c)$.

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma S10.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{17}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

Positivity here means that the response is a positive mixture of decaying exponentials (Lemma S8.1), whose spectral density is positive. Positive, decreasing taps alone do not suffice: for $h = (1, 0.9, 0.1)$ the 3×3 matrix T_h has determinant -0.468, so some round trip earns money. The conclusion concerns modelled execution cost only.

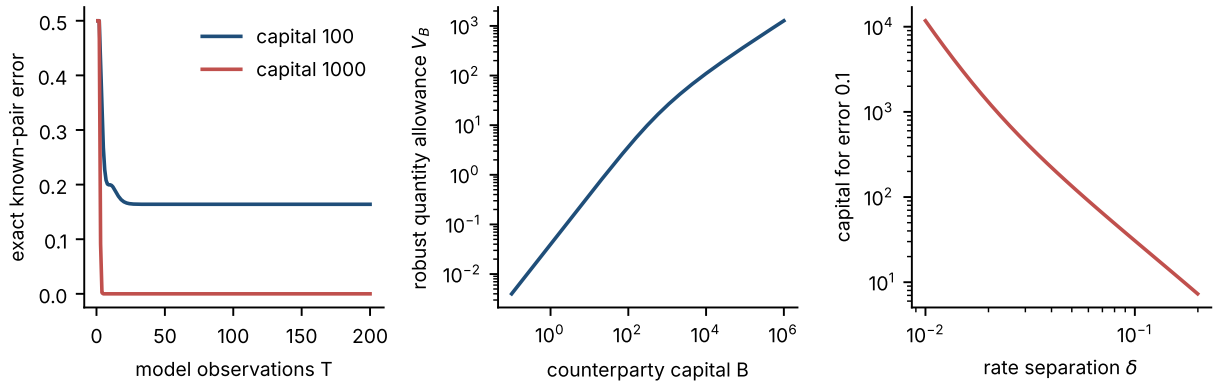


Figure S3: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (18). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

We verify the ledger identity (17) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

S10.1 Capital supplied by a counterparty

Theorem S10.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{18}$$

equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive group representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure S3 illustrates the experiment.

Proposition S10.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \leq V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (see the disclosure contracts of the full working paper).

S10.2 Capital paid by the probe trader

Lemma S10.5 (Time-domain Toeplitz coercivity: capital order for funded probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{19}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (16) achieves error α and remains margin-solvent at every later mark with capital $B_A = \frac{\kappa}{2} A^2 + (c + \mu P_*) A$. Necessary and sufficient capital orders are δ^{-4}.

Lemma S10.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of the account-level analysis of the full working paper. Its explicit constant is very conservative, because it scales as $m/K_*^2 \propto \epsilon^7 (1 - S)^5$: at the calibrated parameters it is 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any feasible design. We therefore state it as a lemma; the account-level analysis of the full working paper computes the sharp constant, within 5% of a feasible design at every separation.

Theorem S10.1. In either positive economy $J \ge 0$ and $\|J\|_\infty \le 1 - \epsilon(1 - S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1 - S)$. From the display above, $|G_\delta - G_0| \le 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \le K_* \delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \le K_* \delta^2 \|v\|_1$, and Gaussian KL with Pinsker gives (14). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \le V \|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q \delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t |\lambda x_t| \le |\lambda| \kappa V$, attained by a single pulse.

Lemma S10.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i} v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2} v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \le \lambda < 1$, and a nonnegative

mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem S10.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \le x_t \le \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2} \sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (18). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition S10.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2 \kappa^2 q \delta^2 > 0$.

Lemma S10.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2} v^\top T_h v - c \sum |v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1 - \lambda^2)/(1 - 2\lambda \cos\omega + \lambda^2) \ge (1 - r)/(1 + r)$, so $v^\top T_h v \ge m \|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2 \delta^4 \|v\|_2^2$, hence the KL bound; Pinsker gives (19). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu |I_t| P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

S11 Two unknown groups: pole fitting and the global two-point bound

Because the memory rates are zeros of H (Theorem S2.3), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A three-step inversion repairs this: (1) compute a minimal realization of H itself from the Hankel matrix of $h_1, h_2, \ldots$ (with h_0 as the direct term) by Kung's eigensystem realization; (2) compute its transmission zeros, which are the memory rates and the timing zero at 0 (transforming the sequence first, for example to $h_{t+1} - a h_t$, cancels these zeros); (3) project reconstructed flow onto the recovered memory filters to obtain the weights (full working paper, internet appendix). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-group systems exactly; with noise, per-group weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

S11.1 A global two-point bound for any estimator

A Cramér–Rao benchmark (full working paper) applies only to estimators that are unbiased near the true parameter. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem S11.1 (Global two-point bound and matching upper bound, $K = 2$)**.** *In this theorem w_k denotes group k's trading intensity q_k, not the merged weight $q_k \eta_k$ of Section S1, and $\Delta\eta$ is the actual spacing of the two rates. Fix total intensity $q = 0.20$, mean rate $\bar\eta = 0.40$, target separation*

$\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$ (so that $\bar{\eta} - p_+\Delta\eta > 0$). Let $p_\mp = \frac{1}{2} \mp \tau/q$ and

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{20}$$

(i) Label groups in increasing order of rate, so that weight vectors are compared rate by rate (as unordered sets the two weight vectors coincide). Both economies have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$ in this labelling, and equal rate moments $M_m = \sum_k w_k \eta_k^m$ for $m \in \{0, 1, 2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{21}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP) that reports weights in the same rate order, the supremum being over the class of two-group economies with total weight q, mean rate $\bar{\eta}$ and spacing $\Delta\eta$,

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{22}$$

so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2 / \|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$. For $\Delta\eta \geq 2\tau q/(q + 2\tau)$ (≈ 0.018 at the stated values) the same bound holds without any labelling convention, for the permutation-invariant loss

$$d_\infty(\hat{\theta}, \theta) = \min_{\pi \in S_2} \max_k (|\hat{w}_{\pi(k)} - w_k| + |\hat{\eta}_{\pi(k)} - \eta_k|),$$

because then $d_\infty(\theta_0, \theta_1) = \min\{2\tau(1 + \Delta\eta/q), (1 + 2\tau/q)\Delta\eta\} \geq 2\tau$. For smaller spacings the relevant permutation-invariant statement is the Wasserstein bound of Theorem S4.1.

(iii) Conversely, the matching upper bound $N = O((\Delta\eta)^{-6})$ on rich separated two-group classes is Theorem S4.2 with $K = 2$. On the separated class with weights and rates bounded away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ for the deconvolved kernel $g_m = \sum_i w_i \eta_i r_i^m$, and perturbation analysis gives the same order.

We recomputed the two-point quantities exactly (Table S1): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table S1).

Remark S11.2 (The hard directions carry no risk). The two economies of Theorem S11.1 have the same total weight $\sum_i q_i = q$, so their summed price response $\sum_t h_t = \kappa/(1 - a - \kappa \sum_i q_i)$ is identical, 2.353 at the calibration of Table S1: the directions in which prices are least informative about the groups are directions in which cumulative price risk does not move. The summed response is a linear functional of the response and is estimated at the parametric rate whatever the spacing of the memories; the companion paper turns this into a certification theorem, a minimax-optimal pre-trade guard and a decision value of disclosure (Niroula et al., 2026). Identification and safety ask different questions of the same data.

S12 The supervisory decision in detail

This section completes Section S5.

Table S1: Two-point sample complexity for two-group weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N_{CR}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$, $N = 50$, *1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Estimators: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

Proof of Proposition S5.1.

Proof of Proposition S5.1. Part (i) follows by scaling the intensities. For (ii), expand $(1 - r_c - \Delta\eta s)^{-1}$: the first three moments agree, so aggregate budgets differ by order $(\Delta\eta)^3$. Each intensity converges to $W\omega_i/(1 - r_c)$, giving the stated dominant-weight gap. This exceeds the aggregate discrepancy at sufficiently small spacing. A ceiling between the two post-halt budgets reduces the action problem to testing two Gaussian mean vectors. Their common-covariance testing error is $\Phi(-\sqrt{N}\|h^G - h^R\|_2/(2\sigma))$; the response bound in Theorem S4.1 gives the lower bound. Part (iii) follows from continuity of $w/(1 - r)$ on the declared compact parameter set and the fixed margin. □

General-K scope. The same reduction gives exponent $4K - 2$ for a pair whose price responses have that moment-matching order *and* whose dominant intensities have a fixed gap. Moment matching alone does not supply the gap: the symmetric odd/even binomial pair has equal maximal normalized weights. Lemma S12.3 and Proposition S12.4 below supply a pair with a gap for every K. The $K = 3$ rows below are numerical examples for the Gauss and Gauss–Radau pair.

Explicit robustness condition. Smooth best responses alone do not preserve separation: the constant smooth map $\Phi(M) = 0$ sends both post-halt budgets to a. Instead assume that, over the stated action and holding horizon, $|\widetilde{R}_j - R_j| \le \varepsilon$ for $j \in \{G, R\}$, with $R_R < R_G$. Then

$$R_R + \varepsilon \le \bar{s} < R_G - \varepsilon \quad \Longrightarrow \quad \widetilde{R}_R \le \bar{s} < \widetilde{R}_G. \tag{23}$$

Indeed, $\widetilde{R}_R \le R_R + \varepsilon$ and $\widetilde{R}_G \ge R_G - \varepsilon$. A nonempty robust window exists if $2\varepsilon < R_G - R_R$. This is a maintained response bound, not an equilibrium result. The same allowance must cover the alternative uniform action. At the $K = 2$, $\Delta\eta = 0.10$ calibration, $R_R = 0.532$, $R_G = 0.569909$, and $\bar{s} = 0.55$ has margin 0.018; an allowance 0.005 fits within it.

A complete conditional cost-and-error model. Let j denote the economy, π_j its declared prior, H the holding horizon, b the surplus per active class per period, C_h the implementation cost

of halting and L the penalty per period for violating the ceiling. These are assumed accounting primitives, not estimated social welfare. An envelope $S_{\max}$ supports the robust uniform action

$$\theta_\varepsilon = \min\{1, \max\{0, (\bar{s} - \varepsilon - a)/(S_{\max} - a)\}\}, \qquad L_T = HbK(1 - \theta_\varepsilon).$$

This action is feasible only if its worst-case budget meets the ceiling (in particular $a + \varepsilon \le \bar{s}$). For a halt,

$$L_{h,j} = Hb + C_h + HL\,\mathbf{1}\{R_j + d_j > \bar{s}\}, \quad \mathcal{L} = F + Nc + \sum_j \pi_j\{p_j L_{h,j} + (1 - p_j)L_T\}.$$

Here $F + Nc$ is the information bill, p_j the probability of halting, and $|d_j| \le \varepsilon$ the action response. Require $\Pr_j(\text{halt and violate}) \le \alpha$ separately in each world. Compare this loss with the best feasible no-purchase lottery over the *same two actions*, and allow refusing to buy information. Neither arbitrary alternative policies nor an optimum over sample size are claimed. Active probing additionally incurs its execution loss and capital requirement; these are not automatically the per-record fee c and must not be counted twice.

Existing runs and the remaining synthetic check. The archived `R_P1_targeted_intervention.json` supplies six response/decision rows. Recomputing them independently from the stated state recursion agrees to 2.1×10^{-17} in the response discrepancy. Those rows supply S_j, R_j and Gaussian testing distances; they contain no measured disclosure fees, intervention payoffs or strategic response bounds. The existing GPU and historical diagnostics cannot fill these quantities by relabelling their outputs. The new CPU study `studies/supervisory_release_20261105/run_study.py` therefore evaluates only this missing accounting layer, with exact Gaussian probabilities rather than a new large simulation.

For this check alone, assume independent scalar disclosures $Z_n \sim N(R_j, \sigma_t^2)$ and halt when $\bar{Z} + z_{1-\alpha}\sigma_t/\sqrt{N} + \varepsilon \le \bar{s}$. This is an illustrative measurement contract, *not* a claim that class positions or tagged flow directly observe post-halt budgets. If a world is unsafe under the response allowance, this gate falsely halts with probability at most α under the stated unbiased Gaussian law. The price comparator observes the full 400-tap Gaussian reset record of the same known pair, with a likelihood-ratio rule constrained by the same per-world error budget. Both information channels have the same assumed monetary fee here; neither represents free passive observation.

With $K = 2$, $N = 32$, $\sigma_t = 0.02$, $\varepsilon = 0.005$, $\alpha = 0.05$, $\pi_R = 1/2$, $H = b = 1$, $C_h = 0.02$, $L = 10$, $F = 0.05$, and $c = 0.0005$, the uniform loss is 1.449219, the information bill is 0.066, and disclosure loss is 1.305132. Its conditional net advantage is 0.144087 surplus units. Halt probabilities are approximately zero in G and 0.978928 in R. Buying the price record yields loss 1.515219 at this sample size; its optimal acquisition decision is to decline and retain the no-data action. An informed oracle before information fees has loss 1.234609.

Across 2,304 deterministic configurations varying fees, noise, horizon, prior, sample size and allowance, paid disclosure improves on the no-data benchmark in 1,207, worsens it in 1,079, and ties in 18. These counts have no population or statistical-significance interpretation. Negative cases remain in the output. A downward measurement bias of 0.03 produces false-halt probability 0.418776; a future response of 0.025 despite an allowance of 0.005 makes halting unsafe in R with probability 0.978928. The latter also invalidates the uniform-action guarantee, so payoff comparisons under that misspecification are not presented as a fair safety comparison. The experiment establishes conditional accounting and failure modes, not real-market gains, a validated equilibrium response, or the general-K theorem.

Proposition S12.1 (A robust scalar disclosure gate)**.** *Let $Z_1, \dots, Z_N$ be independent $N(R, v)$ readings of the fixed post-halt feedback budget, $N > 0$, and let the realized action response d satisfy $d \le \varepsilon$. For $t > 0$, halt only if $\bar{Z} + t + \varepsilon \le \bar{s}$. Then the probability of halting while the realized budget exceeds the ceiling is at most*

$$\Pr\{\bar{Z} + t + \varepsilon \le \bar{s},\ R + d > \bar{s}\} \le \frac{v}{Nt^2}.$$

Consequently $v \le \alpha N t^2$ gives a per-world error bound α.

Proof. On an unsafe-halt event, $\bar{Z} + t + \varepsilon \le \bar{s} < R + d \le R + \varepsilon$, so $t \le |\bar{Z} - R|$. Chebyshev's inequality for the actual Gaussian sample mean gives the bound. Lean checks the product Gaussian law, the sample statistic, this event inclusion and the probability inequality in `p1_supervisory_gaussian_unsafe_halt`; it is not merely a check of a scalar named "probability bound." The response allowance and the interpretation of Z remain explicit assumptions. □

This conservative gate uses $t = \sigma_t/\sqrt{N\alpha}$. It differs from the sharper Gaussian-quantile gate above, whose exact normal probabilities are numerically evaluated but whose quantile identity is not formalized here. Under the same calibration with $N = 128$, the conservative gate has radius 0.007906, information bill 0.114, loss 1.349034 and conditional net advantage 0.100185. Its halt probabilities are approximately zero in G and 0.998023 in R. Across the same 2,304 sensitivity configurations it improves on the fixed-menu no-data benchmark in 781, worsens it in 1,499 and ties in 24. Both gates' rows remain in the result file: 4,608 configurations in total. The welfare and error interpretation remains conditional on the declared observation and response contracts.

Operational bridge from tagged flows to a halt decision. The scalar-reading contract above can be replaced, within the reset model, by five observed channels: two first class-flow responses, two second responses, and the first price response. For a unit pulse write $f_{i,0} = \kappa w_i$, $f_{i,1} = w_i(\kappa r_i + h_1)$, and $h_1 = \kappa a + \kappa^2 \sum_i w_i$. Thus

$$w_i = f_{i,0}/\kappa, \qquad r_i = f_{i,1}/f_{i,0} - h_1/\kappa, \qquad q_i = w_i/(1 - r_i).$$

Assume known impact, verified stable class membership, $0 \le r_i \le 0.95$, independent unbiased Gaussian reset readings with per-channel variance σ_f^2, and N independent records. A simultaneous radius $e = \sigma_f\sqrt{5/(N\alpha)}$ covers the five sample means with probability at least $1 - \alpha$ by Chebyshev and a union bound. Interval division, only when the first-flow lower endpoint is positive, gives bounds on w_i, r_i, q_i; intersect rate bounds with the declared class. Halt only if the remaining upper budget plus the response allowance meets the ceiling. Otherwise retain the same robust uniform fallback. On simultaneous coverage, every selected halt is safe. This supplies a conditional observation-to-action argument without feeding the unknown R, w or r to the rule.

The separate Lean module `Paper1FlowDecision.lean` covers exact flow inversion, ratio and clipped intervals, the intensity interval, selected-halt inequality, actual product-Gaussian joint mean bound and a probability-composition theorem with an explicit rule-soundness premise. Their deterministic composition, clipping implementation and executable simulation still require separate correspondence checks; no claim of a fully formalized executable decision algorithm is made. The pinned build receipt records the checked declarations.

The exploratory script `studies/paper1_uplift_20261105/flow_decision.py` runs 20,000 independent replications per world in each of 60 configurations: $N \in \{32, 128, 512, 2048, 8192\}$, $\sigma_f \in \{0.0005, 0.002, 0.005\}$, $H \in \{1, 10\}$ and $F \in \{0.05, 0.25\}$. Other quantities use the two-group calibration above, a per-channel-vector record fee 0.0005, and the same class-halt or throttle

action menu. Fourteen configurations have positive estimated net information value; the others are retained. The largest observed gain is exploratory, selected from the grid: at $N = 128$, $\sigma_f = 0.0005$, $H = 10$ and $F = 0.05$, no-data loss is 14.492188, information-inclusive loss 12.370094 and net gain 2.122094 surplus units. A simultaneous grid-adjusted Hoeffding interval is reported in the result file; zero observed unsafe halts is not a proof of zero probability. Costs, horizons and response bounds are synthetic, and no real-market gain or calibrated welfare conclusion follows. Six additional fixed stress cases retain both smaller biases with no observed unsafe halt and failures: a downward second-flow bias of 0.010 produces unsafe halts in G in all 20,000 replications, while a response 0.030 exceeding the allowance 0.005 produces unsafe halts in R in all replications. Swapped second-flow labels lead to abstention in both worlds. These are synthetic contract falsifications, not guarantees under misspecification.

Proposition S12.2 (No-data action and information purchase)**.** *For the fixed halt/throttle menu, let* $\overline{L}_h = \sum_j \pi_j L_{h,j}$ *and let* $p_{\max} = \alpha$ *if a halt is unsafe in at least one world, otherwise* $p_{\max} = 1$*. Assuming the fallback meets the ceiling, the best constant no-data halt probability is* $p = 0$ *when* $\overline{L}_h \geq L_T$*, and* $p = p_{\max}$ *when* $\overline{L}_h < L_T$*. A specified information rule with expected action loss* L_I *is strictly preferred if and only if*

$$F + Nc < \mathcal{L}_0 - L_I.$$

Allowing refusal to purchase gives loss $\min\{\mathcal{L}_0, F + Nc + L_I\} \leq \mathcal{L}_0$*.*

Proof. The no-data loss is affine in its feasible halt probability: $L_T + p(\overline{L}_h - L_T)$. Its slope determines the minimizing endpoint. Subtracting the information rule's total loss gives the break-even inequality, and the minimum inequality proves the optional-purchase statement. These algebraic decisions are machine checked in `Paper1SupervisoryDecision.lean`. They do not establish an optimum over other actions, sample sizes, or information contracts. □

S12.1 The targeted halt for every K

Proposition S5.1(ii) is stated for two groups. Its argument needs two populations whose kernels agree to order $(\Delta\eta)^{2K-2}$ and whose largest groups differ at order one. This subsection supplies such a pair for every K.

Let μ be the uniform probability measure on $[-1, 1]$, let $p_0, p_1, \ldots$ be its orthonormal polynomials, and put $\lambda_K(s) = 1/\sum_{j<K} p_j(s)^2$. Call a rule *admissible* if it has K distinct nodes in $[-1, 1]$ and integrates every polynomial of degree at most $2K - 2$ exactly against μ. The Gauss rule and the Gauss–Radau rule of Theorem S4.1 are admissible.

Lemma S12.3 (Largest weights in the family)**.** *Fix* $K \geq 2$*.*

(i) An admissible rule with nodes $t_1, \ldots, t_K$ *has weights* $\omega_i = \lambda_K(t_i)$*. In particular its weights are positive.*

(ii) There is $c_0 > 0$ *such that, for every* $|c| \leq c_0$*, the zeros of* $p_K - c\,p_{K-1}$ *are the nodes of an admissible rule* R_c*. A point is a node of* R_c *for at most one* c*.*

(iii) There are $c_1 \neq c_2$ *in* $[-c_0, c_0]$ *for which* R_{c_1} *and* R_{c_2} *have different largest weights. Their moments of degree* $2K - 1$ *differ.*

Proof. (i) Let ℓ_i be the Lagrange polynomials of the nodes. Each product $\ell_i\ell_j$ has degree $2K - 2$, so the rule integrates it exactly: $\int \ell_i\ell_j\, d\mu = \sum_m \omega_m \ell_i(t_m)\ell_j(t_m) = \omega_i \mathbf{1}\{i = j\}$. Taking $i = j$ gives $\omega_i = \int \ell_i^2 d\mu > 0$. So $\ell_i/\sqrt{\omega_i}$, $i = 1, \ldots, K$, is an orthonormal basis of the polynomials of degree at

most $K-1$, as is $p_0, \dots, p_{K-1}$. The sum of squares of an orthonormal basis at a point does not depend on the basis, so $\sum_{j<K} p_j(s)^2 = \sum_i \ell_i(s)^2/\omega_i$. At $s = t_i$ the right side is $1/\omega_i$.

(ii) The zeros of p_K are real, simple and inside $(-1,1)$. Simple real zeros move continuously with the coefficients, so for $|c| \le c_0$, with c_0 small, $q_c = p_K - c\,p_{K-1}$ has K simple zeros in $(-1,1)$. Give these nodes the interpolatory weights. Let f have degree at most $2K-2$ and write $f = q_c g + \rho$ with $\deg g \le K-2$ and $\deg \rho \le K-1$. Both p_K and p_{K-1} are orthogonal to polynomials of degree at most $K-2$, so $\int q_c g \, d\mu = 0$, and the rule integrates ρ exactly and gives $q_c g$ the value zero. Hence the rule is exact for f. If s is a zero of q_c and of $q_{c'}$ with $c \neq c'$, then $p_{K-1}(s) = 0$ and so $p_K(s) = 0$. By the three-term recurrence a common zero of p_K and p_{K-1} is a zero of p_{K-2}, and so on down to $p_0 = 1$, which has none.

(iii) The reciprocal $1/\lambda_K$ is a polynomial of degree $2K-2 \ge 2$, so every level set of λ_K is finite. Suppose all the rules R_c, $|c| \le c_0$, had the same largest weight v. By (i) each R_c would have a node in the finite set $\{\lambda_K = v\}$, and by (ii) different values of c would give different nodes. There are infinitely many c, so this is impossible. For the last statement, let ν be the difference of the two rules. It is supported on at most $2K$ points. If the two rules also agreed at degree $2K-1$, ν would vanish on every polynomial of degree at most $2K-1$, including the one that equals 1 at one support point and 0 at the others. So ν would be zero, which contradicts (ii) and the positivity of the weights. □

Proposition S12.4 (Supervisory separation for every K)**.** *Fix $K \ge 2$ and let R_{c_1} and R_{c_2} be the rules of Lemma S12.3(iii), with nodes s_i^j, weights ω_i^j and largest weights $v_1 < v_2$. Under the hypotheses of Theorem S4.1, let economy $j \in \{1,2\}$ have K groups with rates $r_c + \Delta\eta\, s_i^j$ and merged weights $W\omega_i^j$, and write $q_i^j = W\omega_i^j/(1 - r_c - \Delta\eta\, s_i^j)$.*

(i) The response bounds of Theorem S4.1 hold for this pair: $\|h^1 - h^2\|_2 \le C_U(\Delta\eta)^{2K-1}$ for every horizon, and $\|h^1 - h^2\|_2 \ge \kappa^2 W|g|(\Delta\eta)^{2K-1}$ when at least $2K+1$ taps are observed, where $g \neq 0$ is the difference of the two rules at degree $2K-1$.

(ii) The feedback budgets satisfy $|S^1 - S^2| = O((\Delta\eta)^{2K-1})$, and the largest groups satisfy $\max_i q_i^2 - \max_i q_i^1 = W(v_2 - v_1)/(1 - r_c) + O(\Delta\eta)$.

(iii) For sufficiently small spacing there is a nonempty interval of ceilings at which halting the largest group meets the ceiling in economy 2 and not in economy 1. For N independent reset records with Gaussian noise $\sigma^2 I$, a rule that makes this decision with both errors at most $1/4$ needs

$$N \;\ge\; \left(2\Phi^{-1}(0.75)\,\sigma/\|h^1 - h^2\|_2\right)^2 \;\ge\; c\,\sigma^2(\Delta\eta)^{-(4K-2)}.$$

Proof. (i) The proof of Theorem S4.1 uses four properties of its pair: nodes in $[-1,1]$, positive weights of unit mass, equal moments up to degree $2K-2$, and a nonzero difference at degree $2K-1$. Lemma S12.3 gives all four. (ii) Expanding $(1 - r_c - \Delta\eta\, s)^{-1}$ in powers of $\Delta\eta$, the sum $\sum_i q_i^j$ is $W\sum_{m\ge 0}(\Delta\eta)^m \mu_m^j/(1-r_c)^{m+1}$, where μ_m^j is the moment of degree m of rule j. The moments agree for $m \le 2K-2$, which gives the first statement. Each q_i^j equals $W\omega_i^j/(1-r_c) + O(\Delta\eta)$, and the maximum is continuous, which gives the second. (iii) The post-halt budgets are $R^j = S^j - \kappa \max_i q_i^j$, so $R^1 - R^2 = \kappa W(v_2 - v_1)/(1-r_c) + O(\Delta\eta) > 0$ for small spacing, and any ceiling strictly between R^2 and R^1 separates the two decisions. The decision is then a test between two Gaussian mean vectors, as in the proof of Proposition S5.1, and (i) gives the bound. □

What is and is not shown. The pair in Proposition S12.4 is two members of the family R_c, not the Gauss and Gauss–Radau pair itself. For that pair the largest weights differ in every case we

computed ($K = 2, \dots, 200$), and a proof of that comparison for every K is open. Lemma S12.3(ii) proves admissibility only for c near zero. Numerically, for $K = 2, \dots, 14$ the rules R_c are admissible on the whole range between the two Gauss–Radau rules, their weights equal λ_K at the nodes to 7×10^{-13}, and on a grid of 41 values of c the largest weight is smallest at the Gauss rule and largest at the Gauss–Radau rules (`equilibrium/every_k_halt_check.py`). The counting step in part (iii) is machine-checked (`p1_largest_weight_not_constant` in `Paper1EveryKHalt.lean`), with parts (i) and (ii) and the finiteness of the level sets as hypotheses. The rest is proved on paper.

S12.2 Proof of Proposition S4.5

Write $D_X(z) = z - a_X - \kappa G_X(z)$ for economy X, so that $H_X = \kappa z / D_X$ and the stationary price spectrum is $f_X(\omega) = \sigma^2 \kappa^2 / |D_X(e^{i\omega})|^2$. Because both economies have budget $1 - \epsilon$, $D_A(1) = D_B(1) = \epsilon$.

Step 1: the difference vanishes at zero frequency. $D_B - D_A = -(a_B - a_A) - \kappa(G_B - G_A)$, and the choice of persistences gives $a_B - a_A = -\kappa(G_B(1) - G_A(1))$. With $E = G_B - G_A$ this is $\Delta(z) \equiv D_B(z) - D_A(z) = -\kappa\,(E(z) - E(1))$, which does not depend on ϵ. Write $E(z) = \sum_m e_m z^{-m}$ with $e_m = k^B_m - k^A_m$. Then $|E(e^{i\omega}) - E(1)| \le |\omega| \sum_m m|e_m|$. The rates are $r_c + \Delta\eta\, s$, and expanding $(r_c + \Delta\eta\, s)^m$ in powers of $\Delta\eta$, the moments of order up to $2K-2$ of the two normalized rules cancel, so $|e_m| \le 2W \sum_{j \ge 2K-1} \binom{m}{j} r_c^{m-j} (\Delta\eta)^j$. Summing, $\sum_m m|e_m| \le C_1 (\Delta\eta)^{2K-1}$, with C_1 depending on K, W, r_c and a bound on $\Delta\eta$. Hence $|\Delta(e^{i\omega})| \le \kappa C_1 (\Delta\eta)^{2K-1} |\omega|$.

Step 2: the denominator is bounded below by $\epsilon + |\omega|$. Let $a^* = \lim_{\epsilon \to 0} a_A(\epsilon)$ and $D^* = D_A - \epsilon$, the denominator of economy A at budget one; $D_A = D^* + \epsilon$. At budget one the closed-loop matrix J of the price and the groups is entrywise nonnegative with a positive diagonal and positive first row and column (since $a^* > 0$ and all weights are positive), so it is primitive. Its Perron root is 1, it is simple, and no other eigenvalue lies on the unit circle. The zeros of D^* in $|z| \ge 1$ are eigenvalues of J, so D^* has no zero on the unit circle except $z = 1$, and there $D^{*\prime}(1) = 1 + \kappa \sum_i w_i r_i / (1 - r_i)^2 > 0$. Hence there are $\delta, c_1, C_2, m > 0$ with $|\operatorname{Im} D^*(e^{i\omega})| \ge c_1 |\omega|$ and $\operatorname{Re} D^*(e^{i\omega}) \ge -C_2 \omega^2$ for $|\omega| \le \delta$, and $|D^*(e^{i\omega})| \ge m$ for $\delta \le |\omega| \le \pi$. For $|\omega| \le \delta$: if $\epsilon \ge 2C_2\omega^2$ then $\operatorname{Re} D_A \ge \epsilon/2$ and $|D_A| \ge \max(\epsilon/2, c_1|\omega|)$; if $\epsilon < 2C_2\omega^2$ then $|D_A| \ge c_1|\omega|$ and $\epsilon < 2C_2\delta|\omega|$. In both cases $|D_A(e^{i\omega})| \ge c_2(\epsilon + |\omega|)$. For $|\omega| \ge \delta$ and $\epsilon \le m/2$, $|D_A| \ge m/2$. So $|D_A(e^{i\omega})| \ge c_3(\epsilon + |\omega|)$ on $[-\pi, \pi]$ for all $\epsilon \le \epsilon_1$.

Step 3: the bound. By Steps 1 and 2, $\rho(\omega) = |\Delta / D_A| \le \kappa C_1 (\Delta\eta)^{2K-1} |\omega| / (c_3(\epsilon + |\omega|)) \le C_3 (\Delta\eta)^{2K-1}$, uniformly in ϵ and ω. The ratio of spectra is $u = f_B / f_A = |D_A / D_B|^2 = |1 + \Delta/D_A|^{-2}$, so for $\rho \le 1/2$, $|u - 1| \le C_4 \rho$. The divergence rate of stationary Gaussian processes is $D_\epsilon(B\|A) = \frac{1}{4\pi} \int_{-\pi}^{\pi} (u - 1 - \log u)\, d\omega$, and $u - 1 - \log u \le (u-1)^2$ for $|u - 1| \le 1/2$. Hence $D_\epsilon(B\|A) \le \frac{1}{2} C_4^2 C_3^2 (\Delta\eta)^{4K-2}$ for all $\epsilon \le \min(\epsilon_0, \epsilon_1)$ and all $\Delta\eta$ small enough that $C_3(\Delta\eta)^{2K-1} \le 1/2$. For $\epsilon_1 < \epsilon \le \epsilon_0$ the family is uniformly stable and the bound is that of Theorem S4.4. □

Numerical check. For $K = 2, 3$, $\Delta\eta \in \{0.1, 0.05, 0.025\}$ and margins from 10^{-1} to 10^{-6}, the ratio $D_\epsilon(B\|A) / (\Delta\eta)^{4K-2}$ increases to a finite limit as $\epsilon \to 0$ (0.41 to 0.48 for $K = 2$, 0.65 to 0.71 for $K = 3$; `fragility_uniform_check.py`). With a common persistence instead of a common budget, the divergence rate grows like ϵ^{-1}, which is the budget difference becoming visible. This proposition is not machine-checked.

S13 Saturating traders: proofs and checks

Theorem S6.1.

Proof. $f_t = \sum_i c_i g_t(r_i, \ell_i)$ with $g_t(r, \ell) = \psi(\ell \sum_{j<t} r^{t-1-j} s_j)$, so equal measures give equal flows for every signal. The rest is the induction of Theorem S2.1(iii), which uses only that the two flow maps agree and depend on signals before t. The last sentence follows from the example below Theorem S6.2. □

Theorem S6.2.

Proof. (i) By strong induction on $t < \tau$. If the paths agree before t, the signals agree before t, so every trader of either population has the same state in both markets. All are inside their bands, so both aggregate flows equal the linear flow $\sum_{j<t} k_{t-1-j} s_j$, which is the same for P and P' because the kernels agree. The insider's order and the dealer's price are causal, so they agree at t. The same argument with P' replaced by the linear market (the band-linear shape $\psi'(u) = u$) gives the second statement. (ii) Build both markets, and the record noise, on one probability space. By (i) the records coincide on $\{\tau \geq T\}$, so the total-variation distance is at most $\Pr(\tau < T)$, and τ is computed on the linear market. (iii) In the linear market from rest, $x = g * \varepsilon$ with $g = h/\kappa$, and $m_t(r) = (\varphi_r * x)_t$ with $\varphi_r(j) = r^j$ shifted by one date. So $m_t(r)$ is centred Gaussian with variance at most $s_r^2 \leq \sigma_\varepsilon^2 \|\varphi_r\|_1^2 \|g\|_1^2$. Here $\|\varphi_r\|_1 = 1/(1-r)$. Also $|h_t|$ is dominated by the response of the positive economy with persistence a and kernel $|k_m|$, whose total mass is $\kappa/(1 - S_{\mathrm{abs}})$, so $\|g\|_1 \leq 1/(1 - S_{\mathrm{abs}})$. For a standard normal Z and $z \geq 0$, $\Pr(|Z| > z) \leq e^{-z^2/2}$. A union bound over the T dates and $n + n'$ traders gives the bound. A test with both errors at most α needs $\mathrm{TV} \geq 1 - 2\alpha$. (iv) Without process noise, the linear path is deterministic, so $\tau \geq T$ surely and (ii) gives distance zero. For a pulse, the linear state is $m_t(r) = A \sum_{j<t} r^{t-1-j} h_j$. □

Proposition S6.3.

Proof. For $u \geq 0$, $\phi(u) = u - \tanh u$ has $\phi(0) = 0$ and $\phi'(u) = \tanh^2 u \in [0, u^2]$. So $0 \leq u - \tanh u \leq u^3/3$, and by oddness $|\tanh u| \leq |u|$ and $|\tanh u - u| \leq |u|^3/3$ for all u. (i) *Size of each path.* Since $|\tanh u| \leq |u|$, $|x_t| \leq a|x_{t-1}| + \kappa \sum_i c_i |\ell_i| \, (\varphi_{r_i} * |x|)_t$ for $t \geq 1$, with $|x_0| = \kappa |A|$. So $|x|$ is dominated by the pulse response of the positive economy with weights $c_i|\ell_i|$. That response has peak $\kappa|A|$ (Lemma S8.1) and total mass $\kappa|A|/(1 - \rho_P)$. Hence $\|x\|_\infty \leq \kappa|A|$ and $\|x\|_2 \leq \|x\|_1 \leq \kappa|A|/(1 - \rho_P)$, and the same holds for the memory states, with an extra factor $1/(1 - r_i)$. *Difference.* Write $c_i \tanh(\ell_i m_{i,t}) = c_i \ell_i m_{i,t} + R_{i,t}$ with $|R_{i,t}| \leq c_i |\ell_i|^3 |m_{i,t}|^3/3$. The linear parts give $(k * x)_t$ with the common kernel. So $e = x^P - x^{P'}$ solves the linear closed loop driven by $u_t = \sum_i R_{i,t} - \sum_j R'_{j,t}$ in place of the probe, and $e = h * u$. By Young's inequality, $\|e\|_2 \leq \|h\|_1 \|u\|_2$, and $\|h\|_1 \leq \kappa/(1 - S_{\mathrm{abs}}) \leq \kappa/(1 - \hat{S})$ as in Theorem S6.2(iii). Also $\| \, |m_i|^3 \|_2 \leq \|m_i\|_\infty^2 \|m_i\|_2 \leq \kappa^3 |A|^3/((1 - r_i)^3 (1 - \rho_P))$, so $\| \sum_i R_i \|_2 \leq \Lambda_P \kappa^3 |A|^3/(1 - \rho_P)$. Combining gives the bound on $\|e\|_2$. For N experiments, the Kullback–Leibler divergence is $N\|e\|_2^2/(2\sigma^2)$. Pinsker's inequality and $\mathrm{TV} \geq 1 - 2\alpha$ give the requirement. (ii) Both markets have $x_0 = \kappa A$ and $m_{i,1} = x_0$, so $x_1^P - x_1^{P'} = \kappa(\sum_i c_i \tanh(\ell_i \kappa A) - \sum_j c'_j \tanh(\ell'_j \kappa A))$. The linear terms cancel because k_0 is common, and $\tanh u = u - u^3/3 + O(u^5)$ gives the expansion. For $|A|$ small, $|x_1^P - x_1^{P'}| \geq \kappa^4 |\Delta M_3| |A|^3/6$. The test that compares the average of N first taps with the midpoint has error $\Phi(-\sqrt{N} \, |x_1^P - x_1^{P'}|/(2\sigma))$, which gives the stated N. □

Theorem S6.4.

Proof. Step 1: probes steer the price. Write F_t for aggregate feedback flow, a function of $x_0, \ldots, x_{t-1}$. Since $x_t = a x_{t-1} + \kappa(F_t + v_t)$ with $\kappa \neq 0$, the probe $v_t = (x_t^* - a x_{t-1}^*)/\kappa - F_t(x_{<t}^*)$ realizes any target path x^*, and $F_t = (x_t - a x_{t-1})/\kappa - v_t$ is read from the response. So two populations give identical responses to all probes if and only if their flow maps agree on all paths.

Step 2: sufficiency of $\tilde{\mu}$. The shape is odd, so $c\,\psi(\ell m) = c\,\mathrm{sgn}(\ell)\,\psi(|\ell| m)$ and $F_t = \int \psi(\lambda\, m_t(r))\,\tilde{\mu}(dr, d\lambda)$. Equal $\tilde{\mu}$ give equal flow maps, hence identical responses.

Step 3: identification. Let $D = \tilde{\mu}_P - \tilde{\mu}_{P'} = \sum_k d_k \delta_{(r_k,\lambda_k)}$ with distinct atoms and $\lambda_k > 0$. Take the path with $x_0 = \theta u$ and $x_1 = u$. Then $m_2(r) = (r\theta + 1)u$, and equal flow maps at date 2 give $g(u) = \sum_k d_k \psi(s_k u) = 0$ for all u, with $s_k = \lambda_k(r_k\theta + 1)$. For two distinct atoms, $\lambda_k(r_k\theta + 1) = \lambda_j(r_j\theta + 1)$ is an affine equation in θ that is not an identity, so it has at most one root. Fix $\theta > 0$ outside these finitely many roots. Then the s_k are positive, because $r_k \geq 0$, and distinct. Independent dilations give $d_k = 0$ for every k, so $D = 0$. □

Theorem S6.4, parts (i)–(iii).

Proof. (i) Let $g(u) = \sum_k w_k \,\mathrm{clip}(s_k u) = 0$ for all u, with distinct $s_k > 0$, and let s_1 be the largest scale. At $u = 1/s_1$ every copy is inside its band or at its edge, so $g(1/s_1) = \sum_k w_k s_k/s_1 = 0$. If there is a second-largest scale s_2, then at $u = 1/s_2$ the first copy is saturated and the others are not, so $w_1 + \sum_{k\neq 1} w_k s_k/s_2 = 0$. The first equation gives $\sum_{k\neq 1} w_k s_k = -w_1 s_1$, so $w_1(1 - s_1/s_2) = 0$ and $w_1 = 0$. If there is no other scale, the first equation is $w_1 = 0$. Remove s_1 and repeat. The statement on probes inside the bands is Theorem S6.2(iv).

(ii) For every y, $1 - \tanh y = 2e^{-2y}/(1 + e^{-2y})$. Let $g(u) = \sum_k w_k \tanh(s_k u) = 0$ for all u, with distinct $s_k > 0$, and let s_1 be the smallest scale. Letting $u \to \infty$ gives $\sum_k w_k = 0$, so $\sum_k w_k(1 - \tanh(s_k u)) = 0$ for all u. Multiply by $e^{2s_1 u}$ and let $u \to \infty$: the first term tends to $2w_1$, and every other term is at most $2|w_k|e^{-2(s_k - s_1)u}$, which tends to 0. So $w_1 = 0$. Remove s_1 and repeat.

Small probes. g is real-analytic, so if it vanishes near 0 it vanishes everywhere, and the argument above applies. A direct route: the odd Taylor coefficients of tanh are $\tau_{2n-1} = 2^{2n}(2^{2n} - 1)B_{2n}/(2n)!$, which are nonzero because the Bernoulli numbers B_{2n} are nonzero. If g vanishes near 0, then $\sum_k (w_k s_k)(s_k^2)^{n-1} = 0$ for $n = 1, \ldots, K$, a Vandermonde system in the distinct numbers s_k^2, so every $w_k = 0$. In Step 1 of the theorem, $|F_t| \leq \sum_i c_i |\ell_i| |m_{i,t}|$, so small target paths need only small probes.

(iii) $\sum_k w_k s_k = 5 - 8 + 3 = 0$ and $\sum_k w_k s_k^3 = 5 - 32 + 27 = 0$, so $\sum_k w_k \psi(s_k u) = u \sum_k w_k s_k - \gamma u^3 \sum_k w_k s_k^3 = 0$. For a cubic shape the date-t flow map is $\int \lambda\, m_t(r)\, d\tilde{\mu} - \gamma \int \lambda^3 m_t(r)^3\, d\tilde{\mu}$, which depends on $\tilde{\mu}$ only through the measures $\lambda\,\tilde{\mu}$ and $\lambda^3\tilde{\mu}$ in r. □

Proposition S6.6.

Proof. (i) The proofs of these results use the dealer's rule only through causality and through its being the same map in both markets. (ii) The linear parts of the two expected flows agree because the kernels agree, so the merged weight at each memory is the same. For each trader, $|\mathbb{E}[c\,\mathrm{clip}(Y)] - c\,\mathbb{E}Y| \leq c\,\mathbb{E}[(|Y| - 1)_+]$ with $Y = \ell m^+ \sim N(\ell\bar{m}, \sigma^2)$. Since $|Y| \leq |\ell\bar{m}| + \sigma|Z|$, this is at most $c\,\mathbb{E}[(\sigma|Z| - \delta)_+] = c(2\sigma\phi(\delta/\sigma) - 2\delta(1 - \Phi(\delta/\sigma))) \leq c\sqrt{2/\pi}\,\sigma e^{-\delta^2/(2\sigma^2)}$. □

Numerical checks. The scripts are `research_extension/nonlinear_scoping/identification_and_figure.py`, `sim_wiener.py`, `sim_passive_clip.py` and `dealer_anticipation.py`. Two populations with the same $\tilde{\mu}$ (a trend follower of capacity 2 and a contrarian of capacity 1, against one trend follower of capacity 1, all at $r = 0.7$ and $|\ell| = 0.1$) give identical responses to 50 random probe sequences of standard deviation 10, with gap exactly 0, under both shapes. The constructive recovery in the main text locates the kinks at 3.702 and 14.813 (true 3.704 and 14.815) and returns net capacities 0.24989 and 0.99988.

S14 Machine checks in detail

Theorem ~S3.1. `Paper1InventoryEquilibrium.lean` checks the algebraic core of part (i):

- the dealers' first-order condition holds in every state;
- the next price loads on the next flow shock with coefficient κ/D;
- the no-feedback existence condition.

Part (iii) is proved on paper. For part (iv), `Paper1InventoryIFT.lean` applies Mathlib's inverse function theorem along a one-parameter family of feedback $G = \epsilon g$, with g strictly differentiable at $D = 1$. For all small ϵ it gives a zero near $(\kappa_-, 1)$, its uniqueness in a fixed neighbourhood, and the two first-order derivatives in the statement (`p1_inventory_small_feedback`). The version that is uniform over all small group measures, in the norm of the statement, is proved on paper.

Theorem S4.1. Lean proves the bound for every K and every horizon T, with a constant that does not depend on T. It uses the theorem's stability hypothesis $|a| + |\kappa|W/(1 - r_c - \Delta\eta) \le \varsigma < 1$, and holds for all error probabilities with $\alpha_0 + \beta_0 \le 1$ (`p1_thm_kcohort_lower_bound_uniform`): $4\sigma^2(1-\alpha_0-\beta_0)^2 \le N\, C_U^2 (\Delta\eta)^{4K-2}$. The checked pair is a different moment-matched pair with the same order: binomial weights on $2K$ equally spaced nodes, split into odd and even halves. For the Gauss and Gauss–Radau rules themselves, Lean checks the case $K = 3$ (positive weights, matched moments and the fifth-moment gap). Lean also checks both estimator statements for this pair, for every horizon (`p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1`). The first covers an estimator of the weights, listed in order of memory rate, that is accurate to $LW/4^L$ in each coordinate with probability at least 3/4 under both economies. The second covers an estimator of the merged measure that integrates every 1-Lipschitz function vanishing at zero to within $2\Delta\eta W/(\pi(2K-1))$. For measures of equal mass, a Wasserstein error below that accuracy implies this. Both force $\sigma^2 \le N\, C_U^2 (\Delta\eta)^{4K-2}$, here with $L = K - 1$. The matching lower order $\|h^A - h^B\|_2 \gtrsim (\Delta\eta)^{2K-1}$ is proved on paper.

Theorem S4.2. Lean verifies this theorem for every $K \ge 1$, using only Lean's standard axioms. The formal proof follows the one above: a Lipschitz bound for the closed-loop deconvolution that does not depend on $\Delta\eta$, the $(\Delta\eta)^{-m}$ rescaling, a sub-Gaussian tail bound over the $2K$ taps, injectivity of the Jacobian of the moment map (proved by Hermite interpolation, without the determinant formula), and a uniform inverse function theorem on the compact class. The formal estimator is the representation with increasing nodes and nonzero weights that matches the $2K$ estimated moments; Lean shows that it is unique and that Prony's linear system has a unique solution whose polynomial has exactly its nodes as roots, so the roots are real. The formal statement allows sub-Gaussian noise and assumes $|r_c| + \Delta\eta \le 1$, so that every memory rate has modulus at most one.

Theorem S4.3. For every K, Lean checks part (i) and part (ii) starting from equal spectra at every frequency (`p1_thm_minphase_zeros`, `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged`). The check uses H in the rational form $\kappa z D(z)/((z-a)D(z) - \kappa N(z))$ of Theorem S2.3, and *stable* in the sense that the poles of H lie in the open unit disk. Both are derived from the market recursion. The eigenvalues of the transition of the state $(x_t, y_{1,t}, \dots, y_{K,t})$ are exactly the zeros of the denominator of H, so Schur stability of the recursion is the pole condition (`p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`). The generating function of the recursion's price response h_t to a unit trade satisfies $\left(\sum_t h_t x^t\right) x^{K+1} B(1/x) =$

$x^{K+1}\kappa x^{-1}D(1/x)$ as formal power series, where B is the denominator of H; with $x = 1/z$ this is $\sum_t h_t z^{-t} = H(z)$ (`p1_impulse_generating`, `p1_tfBottomRev_eval`, `p1_tfTopRev_eval`).

Theorem S4.4. Lean checks the finite-sample consequence of parts (i) and (iii), not the divergence-rate statements. In the checked record the market starts from rest and runs for any number M of warm-up periods before the T recorded prices. The bound does not depend on M, but the passage to the stationary law ($M \to \infty$) is not checked. Part (iii) is checked for every pair of populations with nonnegative merged weights of total at most W, normalized rates in $[-1, 1]$ and equal moments below order $2K-1$ (`p1_thm_passive_cost_matched`). This covers the K-group Gauss and Gauss–Radau economies of Theorem S4.1: both rules are positive with nodes in $[-1, 1]$ and integrate polynomials of degree $2K-2$ exactly. Lean proves these properties of the two rules for $K = 2$ and $K = 3$ (`p1_thm_passive_cost_gauss_radau_two`, `p1_thm_passive_cost_gauss_radau_three`), so for these K the check covers the paper's economies with no extra hypothesis. For general K, the exactness of the two quadrature rules, a classical property, enters the check as the moment hypothesis. It also covers the binomial pair used in the numerical checks (`p1_thm_passive_cost_every_K`).

S15 Machine-checked declarations

The table lists, for each main result, the Lean 4 declarations that check it and what is proved only on paper. The project builds against a pinned Mathlib and has 1,137 named declarations in 126 modules, with no `sorry`. An axiom audit of every named declaration finds only Lean's three standard axioms (`propext`, `Classical.choice` and `Quot.sound`). A machine check confirms the formal statement, not that it matches the prose. The prose statements above say where the two differ.

Result	Declarations (module)	On paper only
Lemma S1.1	`tracking_gain_explicit_root`, `tracking_prior_riccati_equation` (`Microfoundations.lean`); the third example's recursion, `p1_proportional_trade_recursion` (`Paper1StressIdentification.lean`)	Convergence of sample covariances. The third example's first-order condition is assumed in the checked statement, not derived
Theorem S2.1 (sufficiency)	`p1_eqAggFlow_kernel`, `p1_eqAggFlow_of_kernel`, `p1_kernel_of_eqAggFlow`, `p1_eq_market_invariance`, `p1_eq_insider_profit_invariance`, `p1_eq_group_level_differs` (`Paper1EquilibriumInvariance.lean`)	The unbounded-horizon moment statement; the passage from pathwise equality to equal equilibria
Proposition S2.2	`p1_anylinear_flow_kernel`, `p1_anylinear_iff`, `p1_anylinear_invariance` (`Paper1CoreExtensions.lean`)	The passage from pathwise equality to equal equilibria

Result	Declarations (module)	On paper only
Theorem S3.1	`p1_inventory_foc`, `p1_inventory_innovation`, `p1_inventory_benchmark` (`Paper1InventoryEquilibrium.lean`); `p1_inventory_small_feedback` (`Paper1InventoryIFT.lean`)	Part (iii); the version of part (iv) uniform over small group measures
Theorem S3.2	`p1_eq_realize_D`, `p1_eq_realize_dealer`, `p1_eq_scaled_cohort` (`Paper1EquilibriumReducedForm.lean`)	Parts (ii)–(iii), and part (iv) beyond the realization
Theorem S4.1	`p1_thm_kcohort_lower_bound_uniform` (`Paper1KCohortUniform.lean`); `p1_thm_kcohort_lower_bound_weights`, `p1_thm_kcohort_lower_bound_W1` (`Paper1KCohortEstimators.lean`)	The matching lower order of the response gap; the checked pair is a binomial moment-matched pair of the same order
Theorem S4.2	`p1_thm_prony_upper_bound` (`Paper1PronyUpper.lean`)	—
Theorem S4.3	`p1_thm_minphase_zeros` (`Paper1MinPhase.lean`); `p1_thm_minphase_spectrum_H`, `p1_thm_minphase_spectrum_merged` (`Paper1SpectralH.lean`); model link `p1_eigen_iff_bottom_root`, `p1_schur_stable_iff_poles_in_disk`, `p1_impulse_generating` (`Paper1ModelLink.lean`)	—
Theorem S4.4	`p1_thm_passive_cost_matched`, `p1_thm_passive_cost_gauss_radau_two` (`Paper1PassiveMatched.lean`); `p1_thm_passive_cost_gauss_radau_three` (`Paper1GaussRadau3.lean`); `p1_thm_passive_cost_every_K` (`Paper1PassiveKCohort.lean`)	The divergence-rate statements; the stationary limit; Gauss–Radau exactness for general K enters as a hypothesis
Theorem S9.3	`p1_insider_reduced_form`, `p1_insider_best_response`, `p1_insider_kyle_limit`, `p1_insider_posterior_variance` (`Paper1EquilibriumInsider.lean`)	The fixed-point characterization in part (i), part (iii), and the numerical comparative statics

Result	Declarations (module)	On paper only
Proposition S5.1	`p1_prop_target_two_cohorts`, `p1_prop_target_every_K` (`Paper1SupervisorySeparation.lean`)	The checked two-group pair is a different moment-matched pair from the Gauss and Gauss–Radau pair, with the same orders; the every-K version assumes a dominant-group margin; for the quadrature pairs the largest-weight comparison is checked numerically for $K \leq 200$, not proved
Proposition S12.4 and Lemma S12.3	The counting step of Lemma S12.3(iii) only: `p1_largest_value_not_constant`, `p1_largest_weight_not_constant` (`Paper1EveryKHalt.lean`)	Lemma S12.3(i)–(ii), the finiteness of the level sets, and all of Proposition S12.4
Proposition S4.5	none	All of it; checked numerically
Proposition S5.4	none; it rests on Theorem S2.1	All of it
Proposition S5.2	`p1_tagged_iv_determinant_bound` (`Paper1TaggedIV.lean`); `p1_iv_boundary_decomposition` (`Paper1IVBoundary.lean`)	The stationary IV central limit theorem
Theorem S6.1	`p1_sat_sufficiency`, `p1_flow_market_invariance` (`Paper1SaturatingSufficiency.lean`)	The passage from pathwise equality to equal equilibria
Theorem S6.2	`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`, `p1_sat_capacity_differs` (`Paper1SaturatingSufficiency.lean`)	Parts (ii)–(iv): the total-variation, Gaussian and probe bounds
Proposition S6.3	For a cubic shape $u - \gamma u^3$ only: `p1_cubic_pulse_tap1_kernel_cancel`, `p1_cubic_pulse_tap1_price_gap` (`Paper1StressIdentification.lean`)	All of the proposition as stated for tanh: part (i), and the tanh expansion and test in part (ii). The checked statements show the linear cancellation and the exact A^3 gap for the cubic shape

Result	Declarations (module)	On paper only
Theorem S6.4	General statement, any number of traders: `p1_thm_stress_identification`; hard limit: `p1_clip_indep_dilations`, `p1_thm_stress_identification_hard_limit`; cubic counterexample: `p1_cubic_not_indep_dilations` (`Paper1StressIdentificationGeneral.lean`); tanh: `p1_tanh_indep_dilations`, `p1_thm_stress_identification_tanh` (`Paper1StressIdentificationTanh.lean`); steps: `p1_probe_steer_target`, `p1_probe_flow_recovery_eq`, `p1_sat_signed_capacity_sufficiency` (`Paper1StressIdentification.lean`)	The small-probe statement in part (ii); the last sentence of part (iii); the statement in part (i) on probes inside the bands, which is Theorem S6.2(iv)
Proposition S6.5	`p1_delever_stress_invariance` (`Paper1CoreExtensions.lean`)	Parts (ii)–(iv), as for Theorem S6.2
Proposition S6.6	none	All of it

S16 Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \le a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	group weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i\eta_i$	merged weight
$M = \sum_i w_i \delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa \sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t \rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown groups
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices